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AI Requirements Framework

for Safety-Critical Systems

Requirements and Verification Standards for Artificial Intelligence in Safety-Critical Applications

Safety Critical Labs
AI CERTIFICATION AUTHORITY

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Section 1: Introduction

1.0 DEVELOPMENT AND GOVERNANCE

This framework is developed and maintained by Safety Critical Labs under its documented Framework Development Methodology. The SCL Principal approves every normative sentence before a version is called built. Drafting, build execution and validation are assisted by AI drafting tools operated under Safety Critical Labs' direction, and no normative sentence is adopted without that reading and approval. No external technical committee governs the text as of this version.

The requirements are derived from a structured analysis of the failure modes of learning from data, documented in the change logs and in the applied demonstrations they cite. The derivation's completeness can be questioned and is

not proven. Readers who find a failure mode without a requirement, or a requirement without a failure mode, are invited to report it to Safety Critical Labs.

1.1 PURPOSE

This framework establishes the AI-specific requirements and verifications against which Safety Critical Labs conducts certification assessments for systems incorporating artificial intelligence or machine learning capabilities. It is built for safety-critical and mission-critical applications; the classification assigned under Section 1.2.2 determines which of its requirements apply to a given system and at what stringency.

This framework:

- Establishes requirements addressing AI and ML-specific failure modes not covered by traditional software verification practices
- Defines verification methods and evidence standards for data-driven, probabilistic systems
- Provides AI-specific requirements for bias, drift, hallucination, explainability, and human-AI interaction
- Applies across aerospace & defense, aviation, medical, automotive, industrial & manufacturing, and other safety-critical domains

Rationale: *Traditional software assurance practices assume deterministic, logic-based systems with stable behavior after deployment. AI and ML systems challenge this paradigm through probabilistic outputs, emergent behavior from training data, and potential performance degradation over time. These characteristics require verification approaches designed specifically for data-driven systems.*

Two conditional requirement sets carry handling obligations rather than safety requirements: AI-10 (personal data regimes) and AI-15 (information classification and export control regimes). They apply where the regime applies, and they are named here so that the core sets of Section 2 can be read as what attaches to every learned component.

1.2 SCOPE AND APPLICABILITY

This framework applies to any system, subsystem, or component incorporating artificial intelligence or machine learning capabilities, as determined under Section 1.2.1. The classification assigned under Section 1.2.2 (Safety-Critical, Mission-Critical, or Operational Support) determines which requirements apply and at what

stringency; classification selects requirements, it does not gate entry. The framework is built for safety-critical and mission-critical operations, is domain-agnostic, and is applicable to aerospace & defense, aviation, medical, automotive, industrial & manufacturing, and other domains where AI outputs carry operational consequence.

The framework structures requirements into two tiers: Core Requirements (Section 2) apply to every AI system the framework certifies, regardless of underlying technology, with AI-5 applying where the system produces generative or free-form output; Architecture and Paradigm Requirements (Section 3) apply conditionally based on the system's design choices and classification, including multi-model coordination (AI-11), neural network use (AI-12), continuous learning (AI-13), the runtime assurance architecture required at Safety-Critical (AI-14), and information classification regimes (AI-15). Section 4 establishes verification methods, evidence classes, outcomes, the verification matrix and the shared monitoring, alerting, response and logging clause (Section 4.6). Section 5 provides informative implementation patterns. Appendix A contains the glossary. The classification determination, the verification evidence templates and the revision record are Part 2 of this standard (SCL-ARF-002), issued at the same version.

Table 1-1: Applicability Determination Process

Step	Action	Outcome
1	Identify AI/ML presence using Section 1.2.1 criteria	Yes → Step 2; No → standard software assurance practices apply
2	Classify AI system (Safety-Critical, Mission-Critical, Operational Support)	Determines applicable requirements
3	Document applicability determination	Retained as verification evidence
4	Establish tailoring if applicable	Tailoring rationale documented

Should tailoring of listed requirements be considered, reference Section 1.2.6 on how to assess tailoring approaches.

1.2.1 Defining AI

Traditional software behavior is explicitly defined by code. Verification confirms the code implements the required logic, and once verified, behavior remains stable unless the code is changed. AI/ML systems operate differently: behavior emerges from patterns learned during training rather than from explicitly programmed instructions.

This fundamental difference means code inspection alone cannot verify AI behavior, validated performance may not persist over time, and outputs may be non-deterministic. These characteristics do not make AI systems unsuitable for safety-critical operations, but they do require verification approaches designed for data-driven, emergent behavior. The requirements in this framework provide those approaches.

Table 1-2: Applicability Criteria

Criterion	Indicator	Verification Impact
Learned behavior (anchor criterion)	System behavior is specified by an artifact induced from data (through training, fine-tuning, continuous learning, or retrieval augmentation, for example) rather than by an explicitly written specification against which correctness can be fully verified by established methods	Behavior cannot be verified by code inspection alone; if met, this framework applies
Probabilistic outputs (supporting indicator)	System outputs are non-deterministic or include confidence/uncertainty	Identical inputs may not produce identical outputs; verification emphasizes confidence indication and calibration (AI-8.3)
Potential drift (supporting indicator)	System behavior may change or degrade over time without code modifications	Validated performance may not persist without ongoing monitoring (AI-4); triggers the Mission-Critical classification floor per Section 1.2.2

The Learned behavior criterion is the anchor criterion and is necessary for applicability. The determination is made by inspection of how the system's correctness can be established. If correctness can be fully verified against explicitly written specification by established methods, with no residual reliance on fidelity to any learned artifact (Appendix A), the system is not AI for the purposes of this framework and standard software assurance practices apply. If correctness can only be established empirically, against data or against fidelity to a learned artifact, the anchor criterion is met and the system is AI for the purposes of this framework; proceed to Section 1.2.2 to determine classification and applicable requirements. The mechanisms named in Table 1-2 (training, fine-tuning, continuous learning,

retrieval augmentation) illustrate how learned behavior commonly arises; the determination turns on the availability of an explicitly written specification, not on the presence of any particular mechanism. The supporting indicators do not establish applicability on their own; a system that does not meet the anchor criterion is outside the scope of this framework even where its outputs are probabilistic or its behavior varies over time. Where the anchor criterion is met, the supporting indicators sharpen verification emphasis as noted in Table 1-2. Decision impact, listed as an applicability criterion in prior versions, is a criticality attribute rather than an AI-presence attribute; it is applied during classification in Section 1.2.2.

The applicability determination is performed on a declared determination boundary: a defined and bounded collection of hardware and software items, together with its declared interfaces, containing the candidate learned component or components. This determination level is adapted from the AI/ML constituent concept of the EASA Artificial Intelligence Concept Paper Issue 2, which introduces an intermediate assurance level between the system level and the item level for learning-based components; the same concept appears in the SAE ARP6983/EUROCAE ED-324 working material (informative cross-reference per Section 1.3.3), and comparable software-item granularity is established practice under NPR 7150.2. Performing the determination at the boundary level prevents over-scoping, in which a predominantly conventional system is swept into AI applicability by the presence of a single learned component, and under-scoping, in which a learned component escapes determination because the surrounding system is conventional. Boundary declarations are subject to a coverage condition: the declaration shall name the encompassing system and enumerate every candidate learned component within it; every candidate learned component shall lie within a declared determination boundary, and a negative determination discharges only the boundary for which it was made.

This framework attaches to each component within the determination boundary whose behavior is induced from data, and to any artifact whose correctness criterion is fidelity to such a component, including distilled, compiled, quantized, or otherwise transformed derivatives (Table 1-2b; AI-4.7). Framework obligations end at the declared interfaces of the determination boundary. Obligations placed on the integrating system, including independent output validation (AI-5.4), fallback response (AI-6.3), and human decision authority (AI-9.1), attach where those requirements specify, in the system that consumes the boundary's outputs. The determination boundary and its interfaces are declared in the Applicability Determination Record (Part 2 C.1).

For a Safety-Critical classification, the determination boundary shall either include the runtime assurance monitor and fallback required by AI-14, or declare them in the integrating system with their conventional assurance evidenced by reference (AI-14.2, AI-14.5) before a Safety-Critical determination can issue. A Safety-Critical claim about a learned component alone, with no monitor and fallback inside or declared beside the boundary, is not a claim this framework can attest.

For systems containing data-fitted parameters, the determination applies the deletion test: delete every value derived from data, or designed to be derived from data during operation, and inspect what remains. If a behavioral specification remains, the written structure bounding behavior with values re-derivable by a documented derivation that constitutes a correctness argument, the anchor criterion is not met; if no behavioral specification remains, the learned artifact (Appendix A) is the specification and the anchor criterion is met. Parameters generated at runtime by an explicitly specified procedure are treated as specified behavior only where established analysis of the procedure bounds the correctness of the resulting behavior. Parameter determination or improvement by machine learning (ISO/IEC 22989:2022 3.3.5), including retraining and continuous learning, does not satisfy this condition regardless of when the optimization runs; such systems meet the anchor criterion, and where learning continues during operation, the AI-13 conditional requirements apply. The deletion test is a synthesis of the structure-versus-parameters distinction from system identification practice and the correctness-criterion logic of Table 1-2b; Part 2 B.1 states the test as a determination step.

The following system types are not AI for the purposes of this framework, because their behavior is explicitly specified and verifiable by established methods, notwithstanding that several produce probabilistic outputs or exhibit time-varying behavior. Table 1-2b provides worked examples with the reason for exclusion.

Table 1-2b: Negative Criteria (Not AI for Framework Purposes)

System Type	Why Excluded
Deterministic control laws (e.g., PID and gain-scheduled controllers)	Behavior is fully specified by programmed control equations and gains; verifiable by code inspection and classical control analysis
Classical state estimators (e.g., Kalman filters, fixed-model particle filters)	Outputs are probabilistic, but dynamics, noise models, and covariances are explicitly specified and analytically verifiable; no behavior is induced from data. For adaptive variants with runtime covariance estimation, see the data-fitted parameters row

System Type	Why Excluded
Fixed rule-based systems (hand-authored expert systems, decision tables)	Logic is explicit and inspectable. Rule sets or rule weights induced from data meet the anchor criterion and are in scope
Statistical process control	Fixed statistical tests against programmed control limits; no learned behavior
Monte Carlo simulation and dispersion analysis tools	Sampling from explicitly specified models and probability distributions; outputs are statistical ensembles, but every generating model and distribution is written specification, verifiable by established methods; no behavior is induced from data
Lookup tables and interpolation from engineering data	Static, explicitly specified mappings verifiable by standard methods. Artifacts derived from a learned model (distilled tables, compiled or quantized models) inherit AI applicability, with transformation validation per AI-4.7
Data-fitted parameters within explicitly written model structures (calibrated aerodynamic tables, fitted noise covariances, regression coefficients with documented derivation)	Deleting every data-derived value leaves a behavioral specification intact: the written structure bounds behavior, and the values are re-derivable by a documented derivation that constitutes a correctness argument. Parameters generated at runtime by an explicitly specified procedure whose established analysis bounds the correctness of the resulting behavior (adaptive filter covariance estimation) are excluded on the same basis; parameter updating by machine learning, including continuous learning, is not excluded (AI-13 applies where learning continues during operation). Values produced by machine learning and frozen into tables or parameter sets have no such derivation and remain in scope per the lookup table row above

Part 2 B.1 operationalizes this determination as an evidence-cited decision sequence, and the determination, its declared boundary, and the evidence inspected are recorded in the Applicability Determination Record (Part 2 C.1).

1.2.2 AI System Classification

Once a system has been identified as meeting the applicability criteria in Section 1.2.1, it shall be classified according to Table 1-3. Classification determines which requirements apply and the level of tailoring authority available.

Table 1-3: AI System Classification

Classification	Description	Applicable Requirements
Safety-Critical AI	AI outputs directly affect human safety or system survivability	All AI-1 through AI-10 (Core Requirements), AI-5 where the system produces generative or free-form output and AI-10 where personal data is processed; AI-14 (Runtime Assurance) mandatory; plus AI-11 if multi-model, AI-12 if neural network (AI-12.2 and AI-12.4 also apply to other statistical machine learning models), AI-13 if continuous learning (shadow mode only at this tier), and AI-15 where an information classification regime applies
Mission-Critical AI	AI outputs affect operational success but not human safety	AI-1 through AI-4, AI-6, AI-8, AI-9 (Core Requirements); AI-5 where generative or free-form output; AI-10 where personal data is processed; AI-7 tailored, with the threat-model reason stated in the tailoring record; AI-14 available by tailoring-in; plus AI-11, AI-12, AI-13 and AI-15 conditionally per system design and regime
Operational Support AI	AI supports operations but does not drive critical decisions	AI-1 through AI-4 and AI-6 (Core Requirements) minimum; AI-5 where generative or free-form output; AI-10 where personal data is processed; AI-7, AI-8, AI-9 as tailored; plus AI-11, AI-12, AI-13 and AI-15 conditionally per system design and regime

Decision impact is applied at this step. AI outputs that directly affect human safety or system survivability classify the system as Safety-Critical; outputs that affect operational success but not human safety classify it as Mission-Critical; all remaining AI systems are Operational Support. Part 2 B.2 and B.3 operationalize this determination. Classification shall be informed by the severity of the worst credible

consequence of AI system malfunction, drawing on the hazard analysis (AI-1.5); where checklist outcomes and severity assessment conflict, the more severe classification shall apply.

The classification shall be decided against four questions, recorded with the evidence they rest on in the Applicability Determination Record: what happens if the AI output is wrong; whether a human can detect and correct the error before harm occurs; whether the AI is in the control path, the advisory path, or neither; and the latency between the AI output and its operational consequence. Part 2 B.2 and B.3 operationalize these questions. The applicant proposes the classification; the Lead Assessor's determination governs where it differs, and the difference is recorded.

The classification determination shall be documented in the Applicability Determination Record (Part 2 C.1) and approved at the authority level specified for the system's classification in Table 1-4 (Project Safety Review Board for Safety-Critical AI, Chief Engineer for Mission-Critical AI, Software Assurance authority for Operational Support AI).

The approval authorities named in Tables 1-4 and 1-8 are defined functionally so that any organization can map its roles to them. Project Safety Review Board: a body independent of the development team, accountable for system safety, with a documented charter. Chief Engineer: the individual accountable for the technical integrity of the system, independent of the component's development team. Software Assurance authority: the function independent of development that is accountable for software assurance. An applicant whose organization uses other names records the mapping in the Applicability Determination Record; a role that is not independent of the development team does not satisfy the definition, whatever it is called.

Systems meeting the Potential drift supporting indicator in Table 1-2 shall be classified no lower than Mission-Critical and require tailoring approval per Section 1.2.6.

Classification also triggers the ODD declaration rigor specified in Section 1.2.3 Table 1-4, and AI-10 (Privacy and Data Protection) applies where the system processes personal data under any applicable privacy framework; where personal data is not processed, AI-10 sub-requirements may be documented as Not Applicable with rationale.

1.2.3 Operational Design Domain

Following the procedural pattern of Section 1.2.1 (Defining AI) and Section 1.2.2 (AI System Classification), this section establishes ODD declaration as a process step that runs alongside classification. The formal requirement is established in AI-1.0.

The AI system shall have a declared Operational Design Domain (ODD) prior to verification. The ODD defines the operational conditions, inputs, users, subjects, regulatory context, and constraints under which the system is designed and validated to operate and serves as the authoritative scope statement for the certification.

Declaration rigor scales with classification per Table 1-4. The formal requirement and verification for ODD declaration are established in AI-1.0. Tailoring of ODD declaration follows the tailoring authority defined in Section 1.2.6.

Table 1-4: ODD Declaration Rigor by Classification

Classification	Required Declaration Rigor
Safety-Critical AI	Operational Claim summary statement plus structured ODD using the full attribute taxonomy (all eight categories) with quantitative bounds for all measurable attributes. Exclusions explicitly enumerated using all five exclusion sub-categories. ODD declaration formally reviewed and approved by the Project Safety Review Board.
Mission-Critical AI	Operational Claim summary statement plus structured ODD addressing all eight attribute categories. Quantitative bounds where measurable, qualitative descriptors where appropriate. Exclusions explicitly enumerated using applicable sub-categories. ODD declaration approved by the Chief Engineer.
Operational Support AI	Operational Claim summary statement plus documented ODD addressing Operational Environment, Operational Phases, Regulatory and Operational Authorization, Subject Population (where applicable), and Exclusions at minimum. Other categories addressed as applicable to the AI function. ODD declaration approved by the Software Assurance authority.

1.2.4 Relationship to Traditional Software Standards

Traditional software engineering standards were developed for deterministic, logic-based systems. These standards implicitly assume characteristics that do not hold for AI/ML systems. This section documents the gaps that necessitate AI-specific requirements.

1.2.4.1 Gap Analysis

Table 1-5: Traditional Software vs. AI/ML Gap Analysis

Traditional Software	AI/ML Reality	Risk If Unaddressed
Fault = bug in code	Fault = bias, drift, hallucination	Wrong failure mode analysis
V&V at delivery	Requires continuous validation	Post-deployment failures undetected
Requirements to Code traceability	Behavior emerges from training data	Loss of traceability
Deterministic outputs	Probabilistic outputs	Unpredictable behavior in edge cases
Code coverage metrics	No equivalent for trained models	Inadequate test coverage
Explicit decision logic	Black-box decision making	Operators cannot verify AI reasoning
Scope implicit from documentation	Validated envelope varies by training data	Unbounded certification claim without explicit operational scope
Privacy handled procedurally outside the system	Personal data can be reconstructed from models, membership inferred, decisions evade data subject rights	Privacy regulation violation; model inversion and membership inference risk

1.2.4.2 Gap-to-Requirement Mapping

Each AI-specific requirement set in Section 2 addresses one or more gaps identified above. Table 1-6 provides traceability from identified gaps to the corresponding requirements.

Table 1-6: Gap-to-Requirement Mapping

Gap / Risk	Requirement Set
Unbounded operational scope	AI-1.0: Operational Design Domain
Training data integrity and traceability	AI-1: Operational and Data Foundations

Gap / Risk	Requirement Set
Bias and fairness issues	AI-2: Addressing AI Bias
No test coverage equivalent	AI-3: ML Test Coverage
Static after deployment / drift	AI-4: Continuous Validation
Unpredictable outputs	AI-5: Hallucination Prevention
Vulnerable to OOD inputs	AI-6: Out-of-Distribution Detection
Vulnerable to adversarial attacks	AI-7: Adversarial Robustness
Black-box decision making	AI-8: Explainability
Operators cannot verify reasoning	AI-9: Human-AI Teaming
Personal data protection and data subject rights	AI-10: Privacy and Data Protection
Multi-model coordination failure modes	AI-11: Multi-Model Systems Requirements
Neural network architectural concerns (calibration, adversarial vulnerability, training integrity, generative hallucination)	AI-12: Neural Network Requirements
Continuous learning paradigm risks (catastrophic forgetting, learning instability, ODD evolution)	AI-13: Continuous Learning and Adaptation Requirements
Statistical and bounded evidence cannot close a Safety-Critical claim on its own	AI-14: Runtime Assurance Requirements
Classification markings lost through training, fine-tuning and retrieval	AI-15: Regulated Information Handling Requirements (AI-1.5 to AI-1.8 in versions 3.x)

1.2.5 Integration with Software Lifecycle

All AI software shall also meet existing software engineering requirements. AI systems are software systems, and the foundational practices of software planning, configuration management, verification, and safety assurance remain applicable. This framework does not create a parallel lifecycle; rather, it identifies AI-specific activities that shall be performed within each phase of the existing software lifecycle. One requirement is deliberately an exception: AI-1.5 addresses the system-level hazard analysis, which is conducted under the project's system safety

process (see Appendix A) rather than within the software lifecycle; AI-1.5 is this framework's interface to that upstream process.

Table 1-7: AI Integration with Software Lifecycle

Lifecycle Phase	Title	Key Artifacts
Requirements	Define AI function criticality; identify protected classes	AI applicability determination
Design	Define model architecture; establish explainability approach	AI design documentation
Implementation	Partition datasets; document provenance	Dataset partition records
Test	Execute AI-specific test matrix	Test reports, coverage reports
Verification	Complete verification matrix; compile evidence	Verification matrix, evidence package
Operations	Monitor drift, performance, and bias; revalidate per AI-4	Drift logs, monitoring reports, revalidation records
Retirement	Decommission the AI system; disposition models, datasets, retrieval corpora, and logs per their classification and retention requirements (AI-15.2, AI-15.4)	Disposition records, destruction or archival certification

1.2.6 Tailoring Guidance

Should tailoring of listed requirements be considered, Table 1-8 identifies the tailoring authority for each AI system classification. All tailoring shall be documented with rationale and retained as verification evidence. Changes to ODD declarations (AI-1.0) follow the same tailoring authority as other requirements in the corresponding classification.

Table 1-8: Tailoring Authority by Classification

Classification	Tailoring Authority
Safety-Critical AI	No tailoring without Project Safety Review Board approval
Mission-Critical AI	Tailoring permitted with Chief Engineer approval and documented rationale

Classification	Tailoring Authority
Operational Support AI	Tailoring permitted with Software Assurance authority approval

1.3 NORMATIVE REFERENCES

This framework distinguishes normative references from informative ones. Table 1-9 lists the normative references: the documents on which conformity with this framework depends, each to the stated extent of its invocation and no further. Table 1-10 lists informative references: documents the framework draws on for context, provenance, or crosswalk, which carry no conformity burden. The domain standards listed under each requirement are likewise informative (Section 1.3.1). The conformity basis is stated in Section 1.6.

Table 1-9: Normative References

Document	Title	Extent of Invocation
ISO/IEC 22989:2022	AI Concepts and Terminology	Terminology imported in Appendix A at the clauses cited there; the machine learning definition at 3.3.5, invoked by the deletion test in Section 1.2.1 and Part 2 B.1
ISO/IEC 5338:2023	AI System Lifecycle Processes	Terminology imported in Appendix A at the clauses cited there
IEEE 1012-2024	System, Software, and Hardware Verification and Validation	Verification method definitions (Inspection, Analysis, Demonstration, Test) adopted in Section 4.1
SCL-ARF-002, Part 2 of this standard, same version	AI Requirements Framework for Safety-Critical Systems, Part 2: Classification Determination, Evidence Templates and Revision Record	Normative for the classification determination (B.1 to B.6) and the required content of the evidence records (C.1 to C.8) to the extent the sections of this document that cite them state; the revision record is informative

Table 1-10: Informative References

Document	Title	Relevance
NIST AI RMF 1.0	AI Risk Management	Risk taxonomy

Document	Title	Relevance
	Framework	
DO-178C	Software Considerations in Airborne Systems	Aviation software
ISO 26262	Road Vehicles - Functional Safety	Automotive safety
ISO/IEC 27701:2019	Privacy Information Management	Privacy controls
ISO/IEC 29100:2011	Privacy Framework	Privacy principles
NIST Privacy Framework v1.0	Enterprise Privacy Risk Management	Privacy risk taxonomy
EU AI Act	Regulation on Artificial Intelligence	AI risk tiering (EU)
NPR 7150.2D	NASA Software Engineering Requirements	Software lifecycle reference
NASA-STD-8739.8	Software Assurance and Safety Standard	Safety analysis reference
NPR 8000.4C	NASA Agency Risk Management Procedural Requirements	Consequence classification lineage (Section 1.2.2); the dimension structure of the evidence-strength profile in the assessment methodology
ISO/IEC 23053:2022	Framework for AI Systems Using ML	ML framework concepts
ISO/IEC TR 24027:2021	Bias in AI Systems and Decision Making	Bias assessment guidance
NIST AI 600-1	AI RMF: Generative AI Profile	Generative AI risks
FAA AI Safety Roadmap	Roadmap for AI Safety Assurance	Aviation AI certification concepts
GDPR (Regulation 2016/679)	EU General Data Protection Regulation	Personal data processing (EU)
HIPAA Privacy and Security Rules	45 CFR 164 Subparts C and E	Protected health information (US)
CCPA/CPRA	Cal. Civ. Code §§ 1798.100 et	Consumer privacy

Document	Title	Relevance
	seq.	(California)
NIST SP 800-122	Guide to Protecting the Confidentiality of PII	PII handling guidance
ISO 34503	Road Vehicles - Taxonomy for ODD	ODD attribute taxonomy (automotive)
UL 4600	Standard for Safety for the Evaluation of Autonomous Products	Autonomous product safety case reference
EASA AI Concept Paper Issue 2	EASA Artificial Intelligence Concept Paper Issue 2: Guidance for Level 1 and 2 machine learning applications	AI/ML constituent concept; determination boundary source (Section 1.2.1)

1.3.1 Domain Standards Citation Convention

The framework requirements in Sections 2 and 3 are algorithm-agnostic and domain-agnostic. Safety Critical Labs certification assessments are frequently conducted in the context of domain-specific regulatory regimes that impose analogous or complementary requirements. To make that relationship visible at the point of reading, each requirement lists related domain standards inline. The list is a crosswalk: it maps the requirement to obligations the applicant's domain regime may already impose, and it levies nothing.

Domain standards citations are organized in a "Related Domain Standards" subsection at the end of each parent requirement and of selected sub-requirements, structured by domain:

- **General:** Domain-agnostic standards (ISO/IEC, NIST, IEEE)
- **Aviation:** FAA regulations, advisory circulars, and guidance
- **Automotive:** SAE standards, ISO automotive safety standards, NHTSA guidance, UNECE regulations
- **Medical:** FDA guidance documents and regulations, HIPAA where applicable
- **EU:** EU AI Act articles, GDPR articles, and related Union regulations
- **Cross-Domain:** NIST AI RMF function and category references, NIST SP 800 series

A domain block is included only where the requirement has relevant citations in that domain. Absence of a domain block indicates that no specific domain standard currently addresses the requirement at the cited level of detail; this does not imply the requirement is inapplicable in that domain.

Related Domain Standards are informative, not normative. No external document is levied by this framework unless a requirement statement names it and states the extent of the invocation; the normative references are listed in Table 1-9, and the conformity basis is stated in Section 1.6. Certification is conducted against the requirements in Sections 2 and 3, and assessment checklists derive only from the [V.AI-x.y] verification statements, never from the crosswalk. During assessment, Safety Critical Labs performs the crosswalk between those requirements and the applicant's domain regime: the applicant submits the evidence its domain regime already requires, as produced; Safety Critical Labs maps that evidence to the framework requirements it satisfies; and the assessment addresses only what the AI-specific failure modes leave uncovered. The applicant does not demonstrate conformance twice. Safety Critical Labs maps and assesses; it does not produce, design, or remediate the applicant's evidence, and it does not certify conformance to any domain standard, which is not a determination this framework issues.

1.3.2 Source Traceability Typology

This framework operates in a domain where no single canonical standard yet covers all relevant requirements. The framework therefore extends, combines, and in some cases originates content. To preserve the rigor expected of certification frameworks, every addition is categorized by its relationship to existing sources, and citations follow the convention below.

Direct Adoption: Verbatim use of a concept from an existing standard. Cited as: "Per [Standard] [clause]." Example: the IEEE 1012-2024 verification methods used directly in Section 4.1.

Adaptation: Modification of an existing concept for AI/ML context. Cited as: "Adapted from [Standard] [clause]; modified to [reason]." Example: the Operational Design Domain concept adapted from SAE J3016 §3.16 (automotive) to a domain-agnostic AI ODD declaration in AI-1.0.

Synthesis: Combination of concepts from multiple sources where no single source covers the whole. Cited as: "Synthesis of [Standard A], [Standard B], [Standard C]." Example: the consequence-based classification tier system in Section 1.2.2 (Safety-Critical AI, Mission-Critical AI, Operational Support AI) draws on NPR 8000.4C (consequence classification), ISO 26262 (ASIL), and DO-178C (DAL). Additional example: AI-13 Continuous Learning and Adaptation Requirements (Section 3.3) is a synthesis of ISO/IEC 22989:2022 5.11.9.2 continuous learning provisions, ISO/IEC

23894:2023 AI risk management for evolving systems, and the framework's own ODD evolvability concept.

Extension: Addition of new content to an existing concept where the source recognizes the territory but does not address the AI-specific case. Cited as: "Extends [Standard] [clause] to address [AI-specific concern]." Example: the Operations row in Table 1-7 extends NPR 7150.2D §4 (Software Lifecycle, including Operations, Maintenance, and Retirement) to host AI-specific continuous activities (AI-2.4 bias monitoring, AI-4 family continuous validation, AI-6.2 runtime out-of-distribution detection, AI-9.6 operator role verification).

Framework-defined: Novel content where no upstream standard covers the territory. Cited as: "Framework-defined to address [specific gap], drawing analogy from [related concept]." Example: the Subject Population versus User Population distinction in AI-1.0, framework-defined to clarify bias concerns (Subject Population, entities the AI acts upon) versus operator concerns (User Population, humans who direct, monitor, or oversee the AI).

This typology is applied to all framework content and is expected to be maintained in future revisions. Where a concept blends categories (most adaptations also synthesize), the dominant relationship is named.

1.3.3 Verification Status of Cited References

Inline domain-standard citations throughout this framework have been verified at clause level against the source documents listed in Tables 1-9 and 1-10 to the extent those documents are available in the certification project library. The following commercially-distributed standards are cited in this framework but were not available for clause-level verification at the time of this revision; their citations are retained as informative cross-references and shall be re-verified when source documents become accessible:

- RTCA DO-326A (Airworthiness Security Process Specification); cited in R.AI-7 Aviation rows
- RTCA DO-330 (Software Tool Qualification Considerations); cited in R.AI-1 Aviation row
- RTCA DO-355 (Information Security Guidance for Continuing Airworthiness); cited in R.AI-7.5 Aviation row
- RTCA DO-356A (Airworthiness Security Methods and Considerations); cited in R.AI-7 Aviation row

- AAMI HE75 (Human Factors Engineering - Design of Medical Devices); cited in R.AI-9.8 Medical row
- SAE ARP6983/EUROCAE ED-324 (Process Standard for Development and Certification/Approval of Aeronautical Safety-Related Products Implementing AI); cited in the Section 1.2.1 determination boundary discussion and in the R.AI-11 and R.AI-12 Aviation rows

Per §1.3.1 and Section 1.6, inline domain citations are informative rather than normative; the framework's normative references are listed in Table 1-9. The unverified citations above are retained because they represent the recognized authoritative standards in their respective domains and support cross-regulatory recognition for clients operating under those regimes. Their inclusion does not constitute a normative dependency of this framework.

1.4 VERB APPLICATION AND LEXICON

Requirements are defined in Sections 2 and 3 and are denoted by "shall" in the requirements statement. Associated verifications are co-located with their requirements in Sections 2 and 3 and are summarized in the Section 4.2 Verification Matrix. The certification determination rests on these statements as set out in Section 1.6.

Outcome and commitment statements at the parent requirement level are denoted by "will." "Will" statements describe the aggregate outcome that is expected to be met by the sum of the supporting "shall" sub-requirements. "Will" statements do not carry an independent verification; verification is established at the sub-requirement level.

Best practices are denoted "should" and documented in Section 5. "Should" statements do not carry a formal verification requirement.

Each verification statement in Sections 2 and 3 names the evidence class its governing success criterion belongs to (Section 4.1: deterministic, statistical, bounded, surveillance) and resolves to one of four verification outcomes defined there: pass, fail, not applicable with rationale, or insufficient evidence. The procedural clauses of Section 4.6 are verified by the evidence they name.

AI-specific terminology is defined in Appendix A: Glossary, aligned with ISO/IEC 22989:2022 and ISO/IEC 5338:2023 where applicable.

1.5 THRESHOLD CATEGORY DEFINITIONS

The following threshold categories shall be documented for all AI systems regardless of classification. These categories are referenced throughout the

requirements in Section 2. Threshold values are project-defined; Safety Critical Labs sets none. Every declared threshold value shall carry a derivation basis with the following elements: the hazard it mitigates, by reference to the AI-1.5 analysis; the worst credible consequence of that hazard; the acceptable failure probability or rate the project adopts for it, with justification; the threshold value; the test sample size and confidence that support the value under the statistical evidence class of Section 4.1; and the sensitivity of the determination to the value across its plausible range. Threshold values shall be derived from and bounded by the declared Operational Design Domain (AI-1.0); thresholds that admit conditions outside the declared ODD are inconsistent with the certification claim.

- **Minimum performance threshold:** Minimum acceptable accuracy, precision, recall, or domain-appropriate metric below which system response is required
- **Confidence threshold:** Minimum output confidence level for autonomous action versus human review escalation
- **Data drift threshold:** Maximum deviation from training data distribution (using KL divergence, PSI, or equivalent metric) before revalidation is triggered
- **OOD detection threshold:** Boundary conditions that classify inputs as out-of-distribution requiring fallback behavior
- **Human escalation threshold:** Conditions under which AI outputs require mandatory human review before action

Rationale: *Threshold values are project-defined because the acceptable failure probability for a hazard belongs to the system safety process and the applicant's domain regime, not to this framework, and because no sourced basis exists for framework-set values across domains. What the framework requires is that every value be derived, traceable to a hazard, supported by the sample behind it, and declared together with the population it is measured on (Section 1.7). Section 5.1 gives selection considerations and illustrative values; the illustrative values are unsourced and are not criteria.*

Adequacy criteria. The assessor judges the derivation basis and rejects the declared value itself, not only its rationale, where any of the following holds: the value admits conditions outside the declared ODD; the value is not traceable to a hazard in the AI-1.5 analysis; the stated sample size cannot support the claimed rate at the stated confidence; the sensitivity analysis shows the determination changes within the plausible range of the value; or the declared population the value is measured on is narrow relative to the ODD it represents or omits a stratum

Section 1.7 requires. These criteria are stated in the published standard so that the determination rests on it; the assessment methodology applies them and adds nothing to them.

1.6 CONFORMITY BASIS

The certification determination under this framework rests on the requirement statements of this document, together with the instruments of Part 2 (SCL-ARF-002) at the same version where this document names them, and on nothing else. For the requirements of Sections 2 and 3, the determination rests on each [R.AI-x.y] requirement statement verified per its paired [V.AI-x.y] verification statement, which comprises the verification method, the evidence class, the confirmations the method shall establish, the success criterion, the declared population or set, and the evidence artifact, applied by the methods of Section 4.1 at the cadence and independence Section 4 specifies. The procedural requirements stated with “shall” in Sections 1 and 4, namely the applicability determination and its record (Section 1.2.1, Part 2 B.1, Part 2 C.1), classification and its approval (Section 1.2.2), the Operational Design Domain declaration (Section 1.2.3, AI-1.0), threshold documentation and its derivation basis (Section 1.5), the declared evaluation populations and sets (Section 1.7), the sufficiency statements and declarations required by this section, the monitoring, alerting, response and logging clauses (Section 4.6), and verification independence (Section 4.5), form part of the conformity basis and are verified by the evidence those sections name. Rationale paragraphs, the implementation patterns of Section 5, and the worked examples in Part 2 are informative. The residual list of Section 4.1 accompanies the determination on the certificate; the evidence-strength profile the assessment methodology defines is reported in the Technical Assessment Report and does not decide the determination.

Evidence classes and outcomes. Each verification produces evidence of a declared class (Section 4.1) and resolves to pass, fail, not applicable, or insufficient evidence. Statistical and bounded evidence supports a requirement and does not close it; the residual it leaves is recorded and carried. At Safety-Critical, an insufficient-evidence outcome on any applicable requirement blocks the determination. At Mission-Critical and Operational Support it may be carried as a declared residual on the certificate with an obligation to resolve it by the first surveillance, except on a requirement that mitigates a hazard at the tier’s worst credible consequence, where it blocks.

Sufficiency statements. For each applicable requirement set the applicant shall supply a sufficiency statement in the structure of Part 2 C.7: the parent outcome claimed for the declared Operational Design Domain, the sub-requirement

evidence listed by class, the residuals (insufficient-evidence outcomes, bounded budgets, surveillance obligations) and the compensation for each. The statement is verified by inspection against the structure and forms part of the conformity basis. This framework is the evidence layer beneath whatever system-level assurance case the applicant's domain regime requires; it does not replace that case, and the sufficiency statements are written to be consumed by it.

The Related Domain Standards listed under each requirement, the informative references in Table 1-10, and the citations retained under Section 1.3.3 carry no conformity burden. No external document forms part of the conformity basis unless a requirement statement names it and states the extent of the invocation; the documents so invoked are the normative references of Table 1-9. Assessment checklists derive only from the [V.AI-x.y] verification statements, never from the crosswalk. Where a requirement refers to the regulatory regime applicable to the applicant (for example AI-15, AI-8.5, and AI-10), that regime is the one the applicant identifies in its Operational Design Domain declaration under Regulatory and Operational Authorization (AI-1.0); the framework does not select it and does not levy it.

The controls of AI-7 assume a foundational cybersecurity posture for the AI system and for the environment in which it is developed, built, and deployed. That posture is a prerequisite of certification, not assessed scope. The foundational cybersecurity baseline shall be declared in the Applicability Determination Record (Part 2 C.1), either by attestation or by reference to existing conformity evidence such as an ISO/IEC 27001 certificate, a SOC 2 report, or a compliance record under DO-326A or ISO/SAE 21434. Safety Critical Labs reviews the declaration for completeness and records it; it does not assess the baseline. The declared baseline is carried on the certificate as an element of the certified configuration, in the same manner as the declared Operational Design Domain (AI-1.0) and the declared threshold values (Section 1.5), so that the certificate states the security baseline on which the certified AI-specific controls rest.

The certified configuration comprises the elements the applicant declares and Safety Critical Labs records on the certificate: the determination boundary and classification (Section 1.2), the Operational Design Domain (AI-1.0), the threshold values with their derivation bases (Section 1.5), the declared evaluation populations and sets (Section 1.7), the threat model, perturbation budget and attack pattern sets declared under AI-7.1 and AI-7.2, the foundational cybersecurity baseline declared under this section, and, for Safety-Critical systems, the runtime assurance monitor and fallback of AI-14 with the conventional standard and level recorded under AI-14.2. Certification attaches to that configuration. A change to any of its

elements is a change to the certified configuration and is subject to the change and surveillance provisions of the assessment methodology.

1.7 DECLARED EVALUATION POPULATIONS AND SETS

Every rate, score or coverage figure a verification statement in Sections 2 or 3 asks for is measured on a population, partition or set. For each such figure the applicant shall declare the population, partition or set it is measured on, with its derivation basis and its size, before the verification is executed. The declarations are recorded in the Declared Evaluation Population Record (Part 2 C.8), form part of the certified configuration (Section 1.6), and are printed in the Certification Determination Document. A rate is reported only together with the declared population it was measured on; a rate without its population is not evidence under this framework.

Where a failure mode has a known easy case and a known hard case, the declaration shall separate them into strata that are measured and reported separately, and the verification statement names which stratum governs. The required strata are: for out-of-distribution test populations, a far-from-envelope stratum and a near-boundary stratum, each with its boundary validated before use (AI-6.1); for hallucination test sets, injected cases and errors the model itself produced on held-out inputs (AI-5.2); for disparity, the Subject Population strata and User Population roles declared under AI-1.0 (AI-2); for coverage, the cells of the declared Operational Design Domain attribute grid (AI-3.1); for robustness, the threat model, perturbation budget and attack pattern set (AI-7.2); and for every monitored indicator, the monitoring window and the sample floor (Section 4.6.1). Declared sets also include the injected-failure, injected-scenario, seeded and qualification sets the verification statements name.

The adequacy of a declared population is judged under the criteria of Section 1.5: it is rejected where it is narrow relative to the Operational Design Domain it represents, where it omits a required stratum, or where its size cannot support the rate claimed on it at the stated confidence. Verified by inspection of the Declared Evaluation Population Record against the verification statements: every rate in the evidence package resolves to a declared population. Evidence class: Deterministic. A rate without a declared population is a fail of the verification that reports it.

Section 2: Requirements

The requirements in this section apply to AI systems as defined in Section 1.2.1. Each requirement set (AI-1 through AI-10) addresses a specific AI failure mode identified in Section 1.2.4; AI-5 applies where the system produces generative or free-form output, and AI-10 where personal data is processed. Threshold categories

referenced throughout these requirements are defined in Section 1.5; declared populations in Section 1.7; the monitoring, alerting, response and logging cycle that several sets rely on is stated once in Section 4.6. Section 2 requirements are algorithm-agnostic and do not prescribe specific implementation approaches; architecture-specific, paradigm-specific and classification-specific requirements are established in Section 3.

Verb application in this section follows Section 1.4.

Note: Verifications are co-located with requirements in Section 2 for ease of review. They are summarized in the verification matrix in Section 4.2.

2.1 AI-1: OPERATIONAL AND DATA FOUNDATIONS

[R.AI-1] OPERATIONAL AND DATA FOUNDATIONS

The AI system will establish operational scope and data integrity sufficient to support reliable model behavior and a defensible certification claim throughout the model lifecycle.

Traditional software has direct traceability from requirements to code; an engineer can inspect the code to verify it implements the requirement. AI systems break this traceability because behavior emerges from training data within a declared operational envelope. AI-1 establishes the foundations on which all downstream verification depends: the operational envelope within which the system is designed to function (AI-1.0), the integrity of the data partitions that produced the model (AI-1.1 through AI-1.4), the integration of AI failure modes into the project's hazard analysis and the derivation chain for every threshold (AI-1.5, numbered AI-1.9 in versions 3.x), and the controls governing retrieval corpora where retrieval augmentation is used (AI-1.6, numbered AI-1.10 in versions 3.x). The information classification controls that were AI-1.5 through AI-1.8 in versions 3.x are carried by AI-15; the identifier map in the Document Revision History maps every identifier of version 3.7 to this version. AI-1.0 is numbered as a foundational scoping requirement that precedes the data controls; the Operational Design Domain declared under AI-1.0 establishes the bounds within which all subsequent AI-1.x sub-requirements (and the broader AI-2 through AI-10 requirement sets) apply.

Applicability: the information classification controls of versions 3.x (AI-1.5 through AI-1.8 in 3.x) are carried by AI-15 (Section 3.5), which applies where the AI system or its data pipeline is subject to an information classification or export control regime. AI-1.6 (AI-1.10 in versions 3.x) applies where the system employs retrieval-augmented generation; where it does not, AI-1.6 may be documented as Not Applicable with rationale.

Related Domain Standards

- General: ISO/IEC 22989:2022 5; ISO/IEC 5338:2023 §6.4
- Aviation: DO-178C §5 (Software Development); DO-330 §5 (Tool Qualification); FAA AI Safety Roadmap
- Automotive: ISO 26262-6:2018 §5 (Software development); ISO 21448:2022 §6; ISO 34503 (ODD attributes)
- Medical: FDA AI/ML SaMD Action Plan (2021); FDA PCCP Guidance (2023); IEC 62304:2006+A1:2015 §5.1
- EU: EU AI Act Article 10 (Data and data governance); EU AI Act Article 17 (Quality management)
- Government/Defense: 32 CFR Part 2002 (CUI); 22 CFR 120.17 (ITAR deemed exports); 15 CFR 762 (EAR recordkeeping)
- Cross-Domain: NIST AI RMF MAP-1 through MAP-5; NIST SP 800-53 Rev. 5

2.1.0 Operational Design Domain

[R.AI-1.0] OPERATIONAL DESIGN DOMAIN

The AI system shall declare an Operational Design Domain (ODD) that defines the operational conditions, inputs, users, subjects, regulatory context, and constraints under which the system is designed and validated to operate. The ODD declaration shall begin with an Operational Claim summary statement and shall address the eight attribute categories defined in Table 2.1.0-A at the declaration rigor established for the system's classification in Table 1-4; Operational Support systems shall address, at minimum, the reduced category set specified there.

Rationale: *AI systems trained against representative data perform reliably within the conditions represented in that data and become unreliable outside those conditions. Without an explicit declaration of operational scope, the certification claim is unbounded; the certificate appears to attest to system behavior under all conditions when verification evidence supports only a subset. ODD declaration establishes the operational envelope as a contractual element of the certification, scoping the certification claim to declared conditions and providing the basis from which downstream requirements are derived: dataset partitioning (AI-1.1), bias evaluation across subject populations (AI-2), operational input distribution testing (AI-3.3), output validity boundaries (AI-4.4), training distribution characterization (AI-6.1), and operational role verification (AI-9.6). ODD is the operational*

complement to system classification: classification establishes what the system is; ODD establishes where, for whom, and how it is intended to operate.

Operational Claim: The ODD declaration shall begin with a single-sentence Operational Claim summarizing the scope of the certification: *"This AI system is designed and validated to [function] for [subject population] under [operational conditions], operated by [user role] within [regulatory regime]."* The Operational Claim is the elevator-summary of the ODD; the attribute categories below specify it in detail.

The ODD declaration shall address the attribute categories defined in Table 2.1.0-A, at the coverage and rigor established for the system's classification in Table 1-4. Exclusions shall be declared using the five exclusion sub-categories defined in Table 2.1.0-B. Threshold categories (Section 1.5) and operational input distributions (AI-3.3) shall be derived from and consistent with the declared ODD.

Table 2.1.0-A: Required ODD Attribute Categories

Category	Description	Examples	Downstream Requirements
Operational Environment	Physical, spatial, atmospheric, and temporal conditions of operation	Geographic scope, weather and atmospheric conditions, illumination and time-of-day ranges, terrain or surface types, ambient temperature ranges, care setting (medical), facility class	AI-3.3, AI-6.1
System Inputs	Sensor and data source characteristics, supported health states, input quality envelopes, and defined system behavior at degradation thresholds	Sensor types and configurations, sensor health states under which operation is supported, data source provenance, input quality bounds, defined behavior (degraded mode, disengage, fallback, non-diagnostic message) when degradation thresholds are reached	AI-3.3, AI-6.1, AI-9.4
User Population	Operator qualifications, training, currency,	Required operator certifications, system-specific training, currency requirements, role-based	AI-9.1, AI-9.3

Category	Description	Examples	Downstream Requirements
	authorized roles, and role within operational workflow	authorization levels, supported team configurations, user role in the decision workflow (autonomous, decision support, advisory)	
Subject Population	Populations or entities the AI's decisions act upon, with validated subject characteristics and excluded subject conditions	Validated patient demographics (medical), vehicle/pedestrian/cyclist populations the perception system handles (automotive), aircraft and obstacle classes (aviation), entity classes for surveillance or targeting systems	AI-2, AI-3.3, AI-4.4
Operational Phases	Intended use cases, operational phase transitions, and phase-level constraints	Supported workflow stages or mission segments, mode transition conditions, supported task categories, maximum phase or session duration, operational tempo (frequency, duty cycle, throughput per operator)	AI-9.4, AI-9.6
Performance Envelope	Per-decision performance limits and timing requirements	Detection ranges, sensitivity and specificity, accuracy thresholds, decision latency, throughput bounds, confidence calibration error, computational resource availability	AI-3.3, AI-4.2, AI-4.4, AI-8.3
Regulatory and Operational Authorization	Applicable regulatory regime, required certifications, waivers, authorization conditions, and	Governing regulatory framework (FAA Part 107, FDA SaMD class, ISO 26262 ASIL, EU AI Act risk class), required waivers or special authorizations, jurisdictional limitations, applicable	All of Section 2 (foundational scoping)

Category	Description	Examples	Downstream Requirements
	ODD evolvability under change control	industry standards, ODD evolvability declaration (locked at certification, or evolvable under defined change control mechanism such as FDA PCCP)	
Exclusions	Explicit out-of-scope conditions under which the system shall not engage	See Table 2.1.0-B for required exclusion sub-categories	AI-4.4, AI-6.3, AI-9.1

Note on Operational Phases vs. Performance Envelope: Operational Phases captures phase-level or mission-level constraints (duration, sequence, transitions, tempo). Performance Envelope captures per-decision performance (latency, throughput, accuracy at the inference level). Time-bounded attributes that describe the workflow or mission belong in Operational Phases; time-bounded attributes that describe individual inferences belong in Performance Envelope.

Note on Subject Population: Not all systems have a distinct subject population from their user population. Where the AI acts on the user themselves (e.g., a personal recommendation system with no third-party subject), the Subject Population entry may state "Subject = User; see User Population." Where subjects are abstract objects rather than people (e.g., obstacle detection in drones), Subject Population still applies and should declare the validated object classes.

Table 2.1.0-B: Required Exclusion Sub-Categories

[V.AI-1.0] OPERATIONAL DESIGN DOMAIN

The AI system shall verify operational design domain by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the ODD declaration opens with a single-sentence Operational Claim; that it addresses every attribute category of Table 2.1.0-A at the rigor Table 1-4 sets for the classification, with a quantitative bound for every attribute that is measurable; that every exclusion sub-category of Table 2.1.0-B is either populated or marked none with a rationale; that the declaration is approved at the authority level Table 1-4 names, as defined functionally in Section 1.2.2; and

that the declaration records the ODD evolvability status under Regulatory and Operational Authorization.

The inspection shall further confirm that every downstream artifact that depends on the ODD is traceable to a named ODD attribute: the operational input distribution and grid of AI-3.3, the training distribution characterization of AI-6.1, the output validity boundaries of AI-4.4, the Subject Population strata used by AI-2, and every declared evaluation population recorded under Section 1.7.

Success Criteria: Every measurable attribute carries a bound; no exclusion sub-category is blank without rationale; every downstream artifact named above resolves to an ODD attribute by identifier; and the approval record is present. One unresolved trace is a fail, not a partial.

Declared population or set: Not applicable to the declaration itself. The strata, bounds and exclusions declared here are the source of every population declared under Section 1.7.

Evidence artifact: ODD Declaration Record (Part 2 C.4) with the trace table to downstream artifacts. Cadence: One-time; reviewed at the AI-9.6 interval.

Sub-Category	Description	Example
Excluded Environments	Environmental conditions outside the operational envelope	Night operations, heavy precipitation, visibility below threshold, temperature extremes, terrain types not supported, deployment outside cleared jurisdiction
Excluded Use Cases	Mission types or operational scenarios outside intended use	Aerobatic maneuvers, formation operations, payload drop, multi-system coordination, mission durations beyond validated bounds, screening use of a diagnostic-only system
Excluded Operator Conditions	Operator states, qualifications, or staffing configurations not supported	Operators without current certification, operators below minimum currency, staffing below required minimum, non-radiologist users of a radiology decision support system
Excluded Subject Conditions	Subject populations or conditions outside validated scope	Pediatric patients in an adult-validated medical system, pedestrians in wheelchairs or carrying large objects in a perception system not validated for those cases, subject conditions

Sub-Category	Description	Example
		explicitly excluded from training data
Excluded System States	System or sensor degradation conditions that trigger automatic disengage or fallback	Sensor degradation beyond defined thresholds, communication loss conditions, computational resource exhaustion, image quality below threshold, GPS-denied conditions

ODD Maintenance: The ODD declaration shall be reviewed at the periodic intervals defined for operational role verification (AI-9.6). Identified divergence between declared ODD and actual operational use shall trigger formal ODD revision under the same authority structure as initial declaration. For systems declared as evolvable under a change control mechanism (e.g., FDA PCCP), ODD evolution shall follow the approved change control plan; ODD changes outside the approved plan shall require recertification.

2.1.1 Dataset Separation

[R.AI-1.1] DATASET SEPARATION

The AI system shall maintain separation between training, validation, and test datasets.

Rationale: *Data leakage, which occurs when information from test or validation sets contaminates training data, produces artificially inflated performance metrics. A model that has* ***"seen"*** *test data during training will perform well on that data but may fail on novel operational inputs. Strict separation with technical enforcement ensures performance estimates reflect true generalization capability.*

[V.AI-1.1] DATASET SEPARATION

The AI system shall verify dataset separation by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the training, validation and test partitions are documented with unique identifiers; that the partition key is declared at the level of the physical or logical asset whose samples are correlated (engine, patient, vehicle, session) rather than the individual sample; and that partition boundaries are enforced by technical controls (separate storage, access control, automated pipeline checks).

The analysis shall confirm, by computation on the partitions themselves, that no asset appears in more than one partition; that the near-duplicate rate across

partitions under a declared similarity measure is at or below the declared tolerance; and that the integrity check is executed and its result recorded before every training run and every evaluation run. Where the data admits it, the analysis shall include a counterfactual evaluation on a sample-level split and report the performance difference it produces.

Success Criteria: Zero assets shared across partitions; cross-partition near-duplicate rate at or below the declared tolerance, with the measure and tolerance named and the default tolerance being zero; integrity check logged before every training and evaluation run in the record. A single shared asset is a fail.

Declared population or set: All three partitions in full, under the declared asset key and similarity measure (Section 1.7).

Evidence artifact: Partition integrity report: inspection record (Part 2 C.3) and the analysis output with the counterfactual where run. Cadence: One-time; repeated before every training run.

2.1.2 Partition Use Restriction

[R.AI-1.2] PARTITION USE RESTRICTION

The AI system shall ensure each dataset partition is used only for its intended purpose.

Rationale: *Even with proper separation, using partitions incorrectly (e.g., using validation data for final testing, or repeatedly evaluating against test data during development) compromises the integrity of performance estimates. Restricting partition use to intended purposes and logging all access ensures data integrity is maintained throughout the development lifecycle.*

[V.AI-1.2] PARTITION USE RESTRICTION

The AI system shall verify partition use restriction by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the intended use of each partition is documented: training for model fitting, validation for hyperparameter tuning and model selection, test for final performance evaluation only.

The analysis shall confirm from the access logs that no training or tuning process accessed the test partition before the final evaluation; that every access to the test partition is attributable to a named evaluation run; and that any violation of intended use was alerted, investigated and dispositioned.

Success Criteria: Zero test-partition accesses by training or tuning processes before the final evaluation, established from the complete access log; every test-partition access attributable; every violation, if any, carries an investigation record. An unattributable test-partition access is a fail.

Declared population or set: The complete partition access log for the development period.

Evidence artifact: Access logs and partition audit record. Cadence: One-time.

2.1.3 Data Provenance

[R.AI-1.3] DATA PROVENANCE

The AI system shall document data provenance for all datasets used in model development.

Rationale: *AI model behavior is a direct function of training data. Without documented provenance, the source, quality, and potential biases in training data cannot be assessed. Provenance documentation enables root cause analysis when model failures occur, supports retraining decisions, and provides traceability required for safety-critical systems.*

[V.AI-1.3] DATA PROVENANCE

The AI system shall verify data provenance by inspection.

Evidence class: Deterministic.

The inspection shall confirm that every dataset named in the training configuration, and every retrieval corpus source where AI-1.6 applies, has a provenance entry identifying its source, collection method (sensors, sampling rates, time periods, environmental conditions), transformations and preprocessing (normalization, augmentation, filtering, feature extraction), and that each entry is traceable to a source record.

Success Criteria: One hundred per cent of datasets named in the training configuration have a provenance entry traceable to a source record; no dataset without an entry entered training. One dataset without provenance is a fail.

Declared population or set: All datasets named in the training configuration and, where applicable, all retrieval corpus sources.

Evidence artifact: Provenance records. Cadence: One-time.

2.1.4 Ground Truth Labeling Quality

[R.AI-1.4] GROUND TRUTH LABELING QUALITY

The AI system shall validate ground truth labeling quality for all supervised learning datasets.

Rationale: *Supervised learning models learn to predict labels. If labels are incorrect, the model learns incorrect behavior. Label errors are particularly dangerous because they directly encode wrong answers into the model. For safety-critical applications, label quality must be validated through independent verification, as label errors could cause the model to make dangerous predictions with high confidence.*

[V.AI-1.4] GROUND TRUTH LABELING QUALITY

The AI system shall verify ground truth labeling quality by inspection and analysis.

Evidence class: Statistical.

The inspection shall confirm that the labeling methodology and labeler qualifications are documented, and that the agreement threshold and the label error-rate budget are derived under Section 1.5 with the hazard they trace to.

The analysis shall confirm that inter-rater agreement on the declared audit sample meets the derived threshold, reported with its interval; that the label error rate is estimated on an independent audit sample with its interval; that ambiguous or disputed labels are identified and resolved with rationale; and that the effect of the estimated error rate on the performance claim is assessed.

Success Criteria: Agreement interval entirely above the derived threshold and error-rate interval entirely below the derived budget on the declared audit sample. Either interval containing its criterion is insufficient evidence; a point estimate on the wrong side is a fail.

Declared population or set: An audit sample drawn from each partition and each declared Subject Population stratum, with its size and the confidence it supports stated (Section 1.7).

Evidence artifact: Labeling quality report with agreement and error-rate intervals. Cadence: One-time.

2.1.5 AI Hazard Analysis Integration (AI-1.9 in versions 3.x)

[R.AI-1.5] AI HAZARD ANALYSIS INTEGRATION

The AI system shall be included in the system-level hazard analysis conducted under the project's applicable system safety process, and that hazard analysis shall address the AI-specific failure mode categories identified in Section 5.2.2: data-related, model-related, operational, adversarial, privacy, and human factors. Identified AI hazards shall trace to the framework requirements and project-defined thresholds that mitigate them, and every threshold declared under Section 1.5 shall trace back to a hazard in this analysis through the derivation chain Section 1.5 requires. Hazard analysis outputs shall inform system classification (Section 1.2.2) and shall supply, for Safety-Critical systems, the list of hazards assigned to the learned component against which the AI-14 monitor coverage statement is written.

Rationale: *The foundational principle of this framework is that requirements are defined by the failures they prevent. The hazard analysis is where a project enumerates those failures for its specific system, yet prior versions treated AI hazard analysis as informative guidance only. This requirement establishes the normative hook: it does not create a parallel hazard analysis process, but requires that the hazard analysis the project's system safety process already mandates (FMEA, FMECA, FTA, STPA, or PRA per applicable domain practice) explicitly address AI-specific failure modes and connect its outputs to classification, threshold selection, and requirement tailoring. Section 5.2 provides implementation guidance for this integration.*

[V.AI-1.5] AI HAZARD ANALYSIS INTEGRATION

The AI system shall verify ai hazard analysis integration by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that a system-level hazard analysis conducted under the project's applicable system safety process exists and includes the AI system; that it addresses each failure mode category of Section 5.2.2 or documents the category as not credible with rationale; and that it is maintained current with the system configuration under change control.

The analysis shall confirm the trace in both directions: every identified AI hazard traces to the requirement or requirements and the threshold or thresholds that mitigate it, or to a documented risk acceptance; and every threshold declared under Section 1.5 traces to a hazard, a worst credible consequence and an acceptable failure probability or rate adopted for it. For Safety-Critical systems the analysis shall confirm that the hazards assigned to the learned component are listed and that the AI-14.4 coverage statement accounts for each.

Success Criteria: A complete two-way trace: no hazard without a mitigation or acceptance, no threshold without a hazard. At Safety-Critical, no assigned hazard absent from the AI-14.4 coverage statement. One untraced item either way is a fail.

Declared population or set: The complete hazard list for the declared determination boundary.

Evidence artifact: Hazard analysis extract and two-way trace table. Cadence: One-time; re-verified on change.

Related Domain Standards

- General: ISO/IEC 23894:2023 (AI risk management); ISO 31000:2018 (Risk management principles)
- Aviation: SAE ARP4761 (Guidelines for conducting the safety assessment process on civil airborne systems)
- Automotive: ISO 26262-3:2018 Clause 6 (Hazard analysis and risk assessment); ISO 21448:2022 (Safety of the intended functionality)
- Medical: ISO 14971:2019 (Application of risk management to medical devices)
- Aerospace/Defense: NASA-STD-8739.8 (Software safety analysis); MIL-STD-882E (System safety); NPR 8000.4C (Agency risk management procedural requirements; the risk logic the evidence-strength profile of the assessment methodology descends from)
- Cross-Domain: NIST AI RMF MAP function; STPA Handbook (Leveson and Thomas, 2018)

2.1.6 Retrieval Corpus Controls (AI-1.10 in versions 3.x)

[R.AI-1.6] RETRIEVAL CORPUS CONTROLS

Where the AI system employs retrieval-augmented generation or otherwise incorporates retrieved content into model inputs or outputs at runtime, the retrieval corpus shall be subject to documented controls including: source provenance meeting the criteria of AI-1.3; content screening prior to corpus admission per AI-15.1 where AI-15 applies; integrity protection and change control for corpus additions, modifications, and removals; and protection against corpus poisoning and indirect prompt injection per AI-7.1 and AI-7.2. Where retrieval augmentation is not used, this sub-requirement may be documented as Not Applicable with rationale.

Rationale: *Retrieval augmentation moves behavior-determining content out of trained parameters and into a corpus that can change at runtime. A corpus change is functionally a model behavior change, but it bypasses the training pipeline controls of AI-1.1 through AI-1.4 unless the corpus is explicitly controlled. An uncontrolled corpus is also the entry point for indirect prompt injection, in which adversarial instructions embedded in retrieved content manipulate system outputs. This requirement extends the data foundation controls to the retrieval pathway so that corpus content receives the same provenance, screening, and integrity discipline as training data. Corpus content enters the system through loaders; the loading of corpus artifacts is subject to AI-7.6 (Artifact Load Safety).*

[V.AI-1.6] RETRIEVAL CORPUS CONTROLS

The AI system shall verify retrieval corpus controls by inspection and test.

Evidence class: Bounded.

The inspection shall confirm that corpus sources are documented with provenance per AI-1.3; that each corpus source has a recorded screening disposition per AI-15.1; that corpus additions, modifications and removals are executed under documented change control with an audit trail retained per AI-15.4; and that corpus access controls restrict modification to authorized processes and personnel.

The test shall confirm that every attempt in a declared modification-attempt set (add, modify, remove, each outside the change control process) is detected and rejected or alerted, and that every pattern in the declared indirect prompt injection set (Section 1.7, declared under AI-7.2) is either detected at screening or has its influence on system outputs bounded per the protections verified under AI-7.2, with the measured rate reported.

Success Criteria: One hundred per cent of out-of-process modification attempts detected and rejected or alerted; for the declared injection set, the detection rate and the bounded-influence rate are reported with their intervals and together cover the set. An undetected and unbounded pattern is a fail.

Declared population or set: The declared modification-attempt set and the declared indirect prompt injection pattern set (Section 1.7), with their derivation basis.

Evidence artifact: Corpus control inspection record and injection test report (Part 2 C.2). Cadence: One-time; re-tested on corpus pipeline change.

Related Domain Standards

- General: NIST AI 600-1 (Generative AI Profile); NIST SP 800-53 Rev. 5 CM-3 (Configuration change control) and SI-7 (Software, firmware, and information integrity)
- AI-Specific: OWASP Top 10 for LLM Applications LLM01 (Prompt injection); MITRE ATLAS AML.T0051 (LLM prompt injection); MITRE ATLAS AML.T0020 (Poison training data)
- EU: EU AI Act Article 10 (Data and data governance); EU AI Act Article 15(5) (Resilience against adversarial inputs)
- Cross-Domain: NIST AI RMF MAP-2.3 and MANAGE-3

2.2 AI-2: ADDRESSING AI BIAS

[R.AI-2] ADDRESSING AI BIAS

The AI system will bound and monitor performance disparity across the Subject Population strata and User Population roles declared under AI-1.0, so that no declared stratum bears a disparity the hazard analysis has not accepted.

AI systems trained on historical data can encode and amplify disparities present in that data, producing systematically different performance for different populations of the people or entities the system acts on or the operators who use it. Disparity in this set means that difference across the declared strata, measured on a population of decisions. Performance variation across operational conditions, sensor states and environments is a different question, the coverage of the declared Operational Design Domain, and is verified under AI-3.3 and AI-6. The fairness metric a project adopts is a deliberate choice: the common metrics are mutually incompatible where base rates differ across strata, so the metric is named, justified from the hazard the disparity creates, and fixed before results are seen. Evaluation strata shall be derived from the Subject Population and User Population declared under AI-1.0 and recorded under Section 1.7. The indicator, alerting, response and logging cycle for disparity is stated in Section 4.6, which carries what were AI-2.6, AI-2.7 and AI-2.9 in versions 3.x; the remaining sub-requirements of this set are renumbered, and the identifier map in the Document Revision History maps them.

Related Domain Standards

- General: ISO/IEC TR 24027:2021; ISO/IEC 22989:2022 3.5.4
- Aviation: FAA AI Safety Roadmap (equitable operation across crew demographics) `p.8 ("Focus on Safety Assurance and Safety Enhancements" principle)`

- Medical: FDA Draft Guidance on AI-Enabled Device Software Functions (2025) `Section III (TPLC Approach: General Principles) and Section VIII (Data Management)`; FDA Digital Health Policy Navigator
- EU: EU AI Act Article 10(2)(f) (bias examination and correction); EU AI Act Article 15(3) (accuracy across groups)
- Consumer/Employment: EEOC AI Technical Assistance (2023) (where employment-adjacent)
- Cross-Domain: NIST AI RMF MANAGE-2.3; NIST SP 1270 (Taxonomy of AI Bias, 2022)

2.2.1 Bias-Bounded Baseline Performance

[R.AI-2.1] BIAS-BOUNDED BASELINE PERFORMANCE

The AI system shall be baselined before deployment with performance disparities across the Subject Population strata and User Population roles declared under AI-1.0 bounded within approved disparity thresholds derived under Section 1.5. Performance variation across operational conditions, sensor states and environments is not disparity in the sense of this set and is verified under AI-3.3 and AI-6.

Rationale: *Establishing baseline disparity before deployment gives continuous monitoring a reference and makes disparity drift measurable. The disparity metric must be chosen deliberately: the common fairness metrics are mutually incompatible where base rates differ across strata, so the metric is named, its selection justified from the hazard the disparity creates, and it is not changed after the result is seen.*

[V.AI-2.1] BIAS-BOUNDED BASELINE PERFORMANCE

The AI system shall verify bias-bounded baseline performance by analysis and test.

Evidence class: Statistical.

The analysis shall confirm that the disparity metric is named, that its selection is justified from the hazard analysis, that the known incompatibility with the metrics not chosen is acknowledged, and that the disparity threshold for each stratum pair is derived under Section 1.5.

The test shall confirm, on the declared evaluation strata, that the disparity between each pair of strata is estimated with an interval and compared with the derived threshold.

Success Criteria: For every declared stratum pair the disparity interval lies entirely within the threshold. An interval that contains the threshold is insufficient evidence for that pair; a point estimate outside it is a fail.

Declared population or set: The Subject Population strata and User Population roles declared under AI-1.0 and recorded under Section 1.7, with per-stratum sample sizes stated.

Evidence artifact: Baseline disparity report with per-pair intervals. Cadence: One-time.

Related Domain Standards

- General: ISO/IEC TR 24027:2021 §6 (Bias assessment approaches); ISO/IEC 22989:2022 3.5.4 (Bias definition)
- Aviation: FAA AI Safety Roadmap (equitable performance across operational demographics) `p.8 ("Focus on Safety Assurance and Safety Enhancements" principle)`
- Automotive: UNECE WP.29 R157 (ALKS) performance consistency requirements `§5.2 (Dynamic Driving Task)`
- Medical: FDA Draft Guidance on AI-Enabled Device Software Functions (2025) `Section VIII (Data Management), Section X.A (Performance Validation), and Appendix C (Performance Validation Considerations)`; FDA Digital Health Policy Navigator
- EU: EU AI Act Article 10(2)(f) (bias examination and correction); EU AI Act Article 10(3) (training data representativeness); EU AI Act Article 15(3) (performance across demographic groups)
- Consumer/Employment (where applicable): EEOC AI Technical Assistance (2023)
- Cross-Domain: NIST AI RMF MANAGE-2.3; NIST SP 1270:2022 (Taxonomy of AI Bias)

2.2.2 Bias-Managed Training Data

[R.AI-2.2] BIAS-MANAGED TRAINING DATA

The AI system shall use training datasets that account for underrepresented groups, historical biases, and labeling errors.

Rationale: *Training data quality directly determines AI system behavior. Datasets with underrepresented groups, historical biases, or labeling errors will produce models that perpetuate those biases.*

[V.AI-2.2] BIAS-MANAGED TRAINING DATA

The AI system shall verify bias-managed training data by inspection and analysis.

Evidence class: Statistical.

The inspection shall confirm that the training data documentation names the strata it was managed against and the historical bias sources considered, with a disposition for each.

The analysis shall confirm, by count on the training partition, the representation of every declared stratum; that strata below the declared minimum representation are identified with a mitigation (augmentation, reweighting, additional collection) or are excluded from the ODD under AI-1.0; and that the labeling quality controls of AI-1.4 are applied per stratum.

Success Criteria: Every declared stratum either meets the declared minimum representation or carries a mitigation or an ODD exclusion; every listed historical bias source carries a disposition. An undispositioned stratum or source is a fail.

Declared population or set: The training partition by the declared strata (Section 1.7).

Evidence artifact: Training data representation report. Cadence: One-time.

2.2.3 Training Dataset Validation

[R.AI-2.3] TRAINING DATASET VALIDATION

The AI system shall validate, before model development, that training datasets that could impact Safety-Critical or Mission-Critical decisions meet the representation criteria of AI-2.2 and have been tested against the declared bias pattern inventory.

Rationale: *Validation of training datasets prior to model development confirms that the training data bias management requirements (AI-2.2) have been met and provides documented evidence for verification closure.*

[V.AI-2.3] TRAINING DATASET VALIDATION

The AI system shall verify training dataset validation by inspection and analysis.

Evidence class: Statistical.

The inspection shall confirm that the validation was performed before model development and that the bias pattern inventory tested against is declared under Section 1.7 with its derivation basis (protected attributes, operational contexts, historical failure cases).

The analysis shall confirm that representation per declared stratum meets AI-2.2; that every pattern in the inventory was tested with its result recorded; and that where a pattern cannot be excluded, a risk assessment and a mitigation are documented.

Success Criteria: Every inventory pattern has a recorded test result; every pattern not excluded has a risk assessment and mitigation; representation per stratum meets AI-2.2. An untested pattern is a fail.

Declared population or set: The training partition and the declared bias pattern inventory (Section 1.7).

Evidence artifact: Training dataset validation report. Cadence: One-time.

2.2.4 Continuous Bias Monitoring

[R.AI-2.4] CONTINUOUS BIAS MONITORING

The AI system shall implement continuous bias monitoring appropriate to the operational context.

Rationale: *AI system bias can emerge or amplify during operations due to data drift, changing operational contexts, or feedback loops. Continuous monitoring enables early detection of bias before it impacts safety or operational success. Monitoring frequency and methods should scale with system classification and decision criticality.*

[V.AI-2.4] CONTINUOUS BIAS MONITORING

The AI system shall verify continuous bias monitoring by test.

Evidence class: Surveillance.

The test shall confirm that the disparity indicators of AI-2.1 are computed over the declared monitoring window for every stratum whose sample in the window reaches the declared sample floor; that strata below the floor are reported as unmonitored for that window; and that threshold crossings raise the alert, response and log actions of Section 4.6.

Success Criteria: Indicators computed for every stratum at or above the floor in every window of the test period; every window reports its unmonitored strata;

every injected crossing produces the Section 4.6 actions. A window with a monitored stratum and no indicator is a fail.

Declared population or set: Operational data by the declared strata; the monitoring window and sample floor declared under Section 1.7.

Evidence artifact: Bias monitoring records. Cadence: Continuous.

2.2.5 Bias Impact Assessment

[R.AI-2.5] BIAS IMPACT ASSESSMENT

The AI system shall conduct bias impact assessments for all AI decisions affecting human safety, resource allocation, mission planning, and operational procedures.

Rationale: *Not all AI decisions carry equal risk of harm from bias. Impact assessments ensure that bias mitigation efforts are prioritized for decisions with the greatest potential consequences for safety and operational success.*

[V.AI-2.5] BIAS IMPACT ASSESSMENT

The AI system shall verify bias impact assessment by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that a bias impact assessment exists for each decision category the hazard analysis (AI-1.5) identifies as affecting human safety, resource allocation, mission planning or operational procedures.

The analysis shall confirm that each assessment rates the consequence of disparity for its category, that the rating is traceable to the hazard analysis, and that the disparity thresholds of AI-2.1 and the monitoring priorities of AI-2.4 reflect the ratings.

Success Criteria: Every decision category has an assessment traceable to a hazard; every assessment's rating is reflected in a threshold or a monitoring priority. A category without an assessment is a fail.

Declared population or set: The decision categories of the AI-1.5 hazard analysis.

Evidence artifact: Bias impact assessment. Cadence: One-time.

2.2.6 Disparity Response (AI-2.8 in versions 3.x)

[R.AI-2.6] DISPARITY RESPONSE

When a disparity indicator exceeds its threshold over the monitoring window, the operating organization shall execute the response defined under Section 4.6.3 for

that indicator: restriction of operation for the affected stratum, escalation under AI-2.8, or retraining under AI-4.3. The AI system shall not alter individual outputs at runtime on the basis of a detected disparity.

Rationale: *Disparity is a property of a population of decisions, not of one decision, so it cannot be corrected one output at a time. A runtime correction path would substitute an unvalidated output at the moment of highest risk. The response is operational and recorded, and it is taken by the organization that owns the system's use.*

[V.AI-2.6] DISPARITY RESPONSE

The AI system shall verify disparity response by inspection and test.

Evidence class: Deterministic.

The inspection shall confirm that no design element alters individual outputs on the basis of a disparity indicator, and that the defined response for each indicator is one of the three named actions with its authority and time budget.

The test shall confirm that a simulated exceedance over the monitoring window triggers the defined response within its time budget and that the response record names the stratum, the action taken and the approving authority.

Success Criteria: Response triggered within budget in every simulated exceedance; every response record complete; no runtime per-decision correction path present. A correction path is a fail.

Declared population or set: The simulated exceedance set, one per declared stratum (Section 1.7).

Evidence artifact: Response procedure and test record. Cadence: One-time; exercised at surveillance.

2.2.7 Safety-critical Disparity Bound (AI-2.10 in versions 3.x)

[R.AI-2.7] SAFETY-CRITICAL DISPARITY BOUND

The applicant shall demonstrate, before deployment and at each surveillance window, that performance at every safety-critical decision point is within the approved disparity threshold for every declared stratum, and where it is not, shall exclude the stratum from the Operational Design Domain under AI-1.0 or restrict the function until it is.

Rationale: *Disparity at a safety-critical decision point is an unacceptable risk, but it cannot be detected on the decision itself; it is detected on the population of*

decisions. The protection is therefore a bound demonstrated before deployment and maintained in service, with the operating domain narrowed where the bound cannot be shown.

[V.AI-2.7] SAFETY-CRITICAL DISPARITY BOUND

The AI system shall verify safety-critical disparity bound by analysis and test.

Evidence class: Statistical.

The analysis shall identify every AI decision point that affects safety-critical operations, from the AI-1.5 hazard analysis.

The test shall confirm, for each such decision point and each declared stratum, that the performance interval lies within the approved threshold, or that the stratum is excluded from the ODD or the function restricted, with the exclusion recorded under AI-1.0.

Success Criteria: At every safety-critical decision point, every stratum is either inside the threshold on its whole interval or excluded and recorded. An interval containing the threshold is insufficient evidence; an unexcluded stratum outside it is a fail.

Declared population or set: The declared strata (Section 1.7) at each safety-critical decision point.

Evidence artifact: Safety-critical disparity report. Cadence: One-time; repeated at surveillance.

2.2.8 Safety-Critical Bias Escalation (AI-2.11 in versions 3.x)

[R.AI-2.8] SAFETY-CRITICAL BIAS ESCALATION

The AI system shall escalate any disparity exceedance that would adversely impact safety-critical operations or human safety for human review and potential system override with documented rationale where the hazard analysis shows the time to effect permits it; otherwise, and wherever the approved autonomy level (AI-9.1) or disconnected operations (AI-4.6) preclude timely review, it shall execute the defined fallback, which for Safety-Critical systems is the assured fallback of AI-14, and shall log the escalation for review at the next opportunity.

Rationale: *Bias affecting safety-critical functions requires immediate human intervention. Autonomous continuation in the presence of detected safety-critical bias could result in harm. This requirement ensures human authority is maintained over safety-critical AI decisions per AI-9.1 (Human Decision Authority).*

[V.AI-2.8] SAFETY-CRITICAL BIAS ESCALATION

The AI system shall verify safety-critical bias escalation by test.

Evidence class: Bounded.

The test shall inject exceedance scenarios and confirm that each reaches a human review state within the time to effect the hazard analysis allows, or executes the defined fallback (the AI-14 fallback at Safety-Critical) within its budget; that autonomous execution is blocked until review or fallback completes; that override is functional; and that the escalation log captures the decision and rationale.

Success Criteria: Every injected scenario resolves to review within the allowed time or to the fallback within budget; override functions in every scenario; every log entry complete. One scenario resolving to neither is a fail.

Declared population or set: The injected scenario set, covering each declared stratum and each safety-critical decision point (Section 1.7).

Evidence artifact: Escalation test record. Cadence: One-time; exercised at surveillance.

2.2.9 Bias Edge Case Definition (AI-2.12 in versions 3.x)

[R.AI-2.9] BIAS EDGE CASE DEFINITION

The AI system shall define edge-case datasets and stress scenarios for bias validation.

Rationale: *Edge cases and stress conditions represent operational boundaries where bias is most likely to emerge but least likely to be represented in training data. Explicit definition of edge cases ensures systematic coverage during validation rather than ad-hoc testing.*

[V.AI-2.9] BIAS EDGE CASE DEFINITION

The AI system shall verify bias edge case definition by inspection.

Evidence class: Deterministic.

The inspection shall confirm that edge-case datasets and stress scenarios are defined for each declared stratum boundary and for each ODD exclusion boundary adjacent to in-scope conditions, that each set states its size and its construction, and that the sets are declared under Section 1.7.

Success Criteria: Every declared stratum boundary and every adjacent exclusion boundary has a defined set with stated size. A boundary without a set is a fail.

Declared population or set: The declared edge-case and stress sets (Section 1.7).

Evidence artifact: Edge-case dataset specification. Cadence: One-time.

2.2.10 Bias Edge Case Validation (AI-2.13 in versions 3.x)

[R.AI-2.10] BIAS EDGE CASE VALIDATION

The AI system shall validate bias performance under defined edge cases and stress scenarios.

Rationale: *Bias may not manifest under nominal conditions but emerge at operational boundaries or under stress. Validation across edge cases ensures the system maintains performance within approved bias thresholds even in conditions that may not be well-represented in training data.*

[V.AI-2.10] BIAS EDGE CASE VALIDATION

The AI system shall verify bias edge case validation by test.

Evidence class: Statistical.

The test shall confirm that on each declared edge-case and stress set the disparity indicators of AI-2.1 are estimated with intervals and lie within the approved thresholds, or that exceedances trigger the defined response of AI-2.6.

Success Criteria: Every set's disparity interval within threshold, or the exceedance response triggered and recorded. An interval containing the threshold is insufficient evidence for that set.

Declared population or set: The declared edge-case and stress sets (Section 1.7).

Evidence artifact: Edge-case validation report. Cadence: One-time.

2.3 AI-3: ML TEST COVERAGE

[R.AI-3] ML TEST COVERAGE

The AI system will demonstrate adequate test coverage for machine learning models using methods appropriate to data-driven systems where traditional code coverage metrics do not apply.

Traditional software testing relies on code coverage metrics (statement coverage, branch coverage, MC/DC) to demonstrate that the software has been adequately exercised. These metrics are meaningless for ML models: there is no "code" to cover in a neural network's weights. AI systems require alternative coverage concepts based on input space coverage, operational scenario coverage, and

behavioral testing. This requirement addresses the “no test coverage equivalent” gap identified in Section 1.2.4.2.

Related Domain Standards

- General: ISO/IEC/IEEE 29119-11:2020 (Testing of AI-based systems); ISO/IEC 5338:2023 §6.4
- Aviation: DO-178C §6 (Verification Process); EUROCAE ED-324/SAE AS6983 (AI/ML aviation)
- Automotive: ISO 26262-6:2018 §9 (Software unit verification); ISO 21448:2022 §7 (Evaluation)
- Medical: IEC 62304:2006+A1:2015 §5.7 (Software system testing)
- EU: EU AI Act Article 15 (Accuracy, robustness and cybersecurity)
- Cross-Domain: NIST AI RMF MEASURE-2; IEEE 7010-2020

2.3.1 Test Coverage Metric Definition

[R.AI-3.1] TEST COVERAGE METRIC DEFINITION

The AI system shall define test coverage metrics for each ML model over the declared Operational Design Domain, as the fraction of cells of a declared grid over the ODD attributes of Table 2.1.0-A that contain at least a declared minimum number of test samples. Structural coverage measures of the network itself (neuron, layer or activation coverage) do not satisfy this requirement on their own.

Rationale: *Coverage has to be measured over the space the certification claim is made about, which is the declared ODD, not over the internals of the model. Structural measures of a network have been shown to bear little relation to the faults a test set finds and are not accepted as the coverage metric. The grid makes coverage a count that two assessors compute the same way.*

[V.AI-3.1] TEST COVERAGE METRIC DEFINITION

The AI system shall verify test coverage metric definition by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the grid is declared with each cell defined by ranges of named ODD attributes, that the minimum samples per cell is declared with its derivation under Section 1.5, and that the rationale for the attributes chosen and the cell resolution is documented.

The analysis shall confirm that the grid spans every quantitative bound of the declared ODD and that the metric is computable from the test partition as declared.

Success Criteria: Every quantitative ODD bound falls within the grid; every cell is defined by named attributes; the metric is computed and reproduced by the assessor from the declared definition. An ODD bound outside the grid is a fail.

Declared population or set: The declared ODD attribute grid (Section 1.7).

Evidence artifact: Coverage metric specification. Cadence: One-time.

2.3.2 Test Coverage Threshold Achievement

[R.AI-3.2] TEST COVERAGE THRESHOLD ACHIEVEMENT

The AI system shall achieve the defined test coverage threshold over the declared grid prior to operational deployment, and for every performance rate claimed under this framework shall state the statistical power of the test set against that claim: the sample size, the confidence, and the number of failure-free or correctly classified samples the claim requires.

Rationale: *Defining metrics without achieving them provides no assurance, and achieving them on a test set too small to support the claimed rate provides the appearance of assurance. A claimed failure rate below one in a thousand needs several thousand failure-free samples to be shown with ordinary confidence; the requirement makes that arithmetic part of the evidence rather than leaving it to the reader.*

[V.AI-3.2] TEST COVERAGE THRESHOLD ACHIEVEMENT

The AI system shall verify test coverage threshold achievement by analysis and test.

Evidence class: Statistical.

The analysis shall confirm that the coverage threshold is derived under Section 1.5 from the system classification and the hazard analysis, and that for each claimed performance rate the test set's sample size supports the claim at the declared confidence, with the computation shown.

The test shall confirm that the achieved coverage over the grid meets the threshold, and that every coverage gap is identified and dispositioned through additional testing or documented risk acceptance.

Success Criteria: Achieved coverage at or above the threshold; every claimed rate supported at the declared confidence by the stated sample size; every gap dispositioned. A claimed rate the sample cannot support at the declared confidence is insufficient evidence for that claim.

Declared population or set: The test partition over the declared grid (Section 1.7).

Evidence artifact: Coverage report with the power statement. Cadence: One-time.

2.3.3 Operational Input Distribution Testing

[R.AI-3.3] OPERATIONAL INPUT DISTRIBUTION TESTING

The AI system shall test ML models across representative operational input distributions.

Rationale: *ML models perform well on data similar to their training data but may fail on data that differs from training distributions. Testing must cover the full range of inputs expected during actual operations, including nominal conditions, off-nominal conditions, and mission phase transitions. Gaps between test and operational distributions represent untested regions where model behavior is uncertain. Operational input distributions shall be derived from the declared ODD (AI-1.0).*

[V.AI-3.3] OPERATIONAL INPUT DISTRIBUTION TESTING

The AI system shall verify operational input distribution testing by test.

Evidence class: Statistical.

The test shall confirm that the operational input distribution is declared from the ODD under Section 1.7 with its derivation (mission analysis, operational scenarios); that model performance on the declared metric is estimated with an interval in every grid cell that reaches the minimum sample count; that cells below the count are listed as uncovered and dispositioned by additional data collection, by synthetic data whose validity is itself evidenced, or by exclusion from the ODD under AI-1.0; and that the fraction of the declared operational distribution's mass lying in covered cells is reported.

Success Criteria: Performance interval within threshold in every covered cell; every uncovered cell dispositioned; covered mass reported. A covered cell whose interval contains the threshold is insufficient evidence; an uncovered cell without disposition is a fail.

Declared population or set: The operational input distribution and grid declared under Section 1.7; the test partition.

Evidence artifact: Operational distribution test report. Cadence: One-time.

2.3.4 Boundary and Edge Case Testing

[R.AI-3.4] BOUNDARY AND EDGE CASE TESTING

The AI system shall perform boundary and edge case testing for all ML models.

Rationale: *ML models often fail at input boundaries: values at the edges of the training distribution or combinations of inputs not well-represented in training data. Edge cases are particularly important for safety-critical functions because they represent conditions where model behavior is least certain. Systematic boundary testing ensures the model behaves acceptably at operational limits.*

[V.AI-3.4] BOUNDARY AND EDGE CASE TESTING

The AI system shall verify boundary and edge case testing by test.

Evidence class: Bounded.

The test shall confirm that input boundaries are defined for each ML model from the quantitative ODD bounds and physical limits; that a boundary and edge-case set is declared under Section 1.7 through a systematic method (equivalence partitioning and boundary value analysis adapted for ML); that model behavior at every declared boundary is tested against a pass condition; and that results are recorded.

Success Criteria: Every quantitative ODD bound has at least one boundary case; every case has a recorded pass or fail; the set is declared. A bound without a case is a fail; the result holds at the declared boundaries and nowhere else.

Declared population or set: The declared boundary and edge-case set (Section 1.7).

Evidence artifact: Boundary test report. Cadence: One-time.

2.4 AI-4: CONTINUOUS VALIDATION

[R.AI-4] CONTINUOUS VALIDATION

The AI system will implement continuous validation processes to ensure model performance remains within acceptable bounds throughout operational deployment.

AI model performance can degrade during operations due to data drift, concept drift, or environmental changes. Continuous validation detects performance degradation before it impacts safety or operational success, enabling timely

corrective action such as model retraining, threshold adjustment, or fallback to manual operations.

Related Domain Standards

- General: ISO/IEC 5338:2023 §6.4.14; ISO/IEC 22989:2022 5.11.9
- Aviation: FAA AI Safety Roadmap (operational monitoring)
- Automotive: ISO 21448:2022 §13 (Field monitoring); UNECE WP.29 R157 (ALKS) §5.2 `(Dynamic Driving Task)`
- Medical: FDA PCCP Guidance (2023); FDA AI/ML-Based SaMD Action Plan §5 (Real-World Performance)
- EU: EU AI Act Article 17 (Quality management system); EU AI Act Article 72 (Post-market monitoring)
- Cross-Domain: NIST AI RMF MEASURE-3; NIST AI RMF MEASURE-4

2.4.1 Data Drift Monitoring

[R.AI-4.1] DATA DRIFT MONITORING

The AI system shall monitor for data drift that could cause inaccurate decisions or predictions. The drift monitor shall be qualified on a clean operational stream before its alerts are relied upon, and shall select its baseline and its monitored regime using only information available in service.

Rationale: *Data drift occurs when the statistical characteristics of production data diverge from the distribution the model was validated on. A drift statistic has no meaning until the monitor's own behaviour on undrifted data is known: naive designs saturate on composition, asset heterogeneity or sampling noise alone. And a monitor whose baseline depends on knowledge of the system's health state is not deployable, because that knowledge is what the monitor is meant to supply. Both conditions are therefore part of the requirement. Drift in the inputs is neither necessary nor sufficient for degraded performance, so drift monitoring is paired with AI-4.2 or AI-4.6 and never stands alone.*

[V.AI-4.1] DATA DRIFT MONITORING

The AI system shall verify data drift monitoring by test.

Evidence class: Surveillance.

The test shall confirm that the drift statistic, window, baseline and regime selection are documented; that on the declared clean qualification stream the statistic

remains below the inhibit level throughout and the count of alert-level excursions is reported; that on the declared injected-drift stream the first alert attributable to the injection occurs within the declared latency budget; that baseline and regime selection use only information available in service, established by inspection of the monitor's inputs; and that threshold crossings raise the Section 4.6 actions.

Success Criteria: Clean stream below the inhibit level at every evaluation point; attributable alert on the injected stream within the latency budget; no monitor input unavailable in service. A clean-stream inhibit crossing or an in-service-unavailable input is a fail.

Declared population or set: The clean qualification stream and the injected-drift stream declared under Section 1.7, with their lengths and the injection profile.

Evidence artifact: Drift monitor qualification and detection report. Cadence: Continuous; qualification one-time per monitor design.

2.4.2 Performance Threshold Maintenance

[R.AI-4.2] PERFORMANCE THRESHOLD MAINTENANCE

Where ground truth, or a validated proxy for it under AI-4.6, is available in operation, the AI system shall maintain model performance above the defined thresholds throughout operation. Where neither is available, this sub-requirement is satisfied through the proxy mechanisms of AI-4.6 and the characterization of ground-truth availability in the ODD declaration.

Rationale: *AI model performance can degrade over time even without data drift through concept drift, model staleness, or accumulated edge cases. Continuous performance monitoring against defined thresholds provides early warning of degradation before it impacts operational success or safety. Threshold values should be derived from mission risk tolerance per Section 5.1 (Threshold Selection Guidance).*

[V.AI-4.2] PERFORMANCE THRESHOLD MAINTENANCE

The AI system shall verify performance threshold maintenance by analysis and test.

Evidence class: Surveillance.

The analysis shall confirm that ground-truth availability in operation is characterized in the ODD declaration; that performance thresholds are derived under Section 1.5; and that the metrics are appropriate to the AI function.

The test shall confirm that, where ground truth or a validated proxy is available, the performance metrics are computed over the declared window during operation,

that degradation below threshold raises the Section 4.6 actions within the declared latency, and that performance logs support trend analysis.

Success Criteria: Metrics computed in every declared window of the test period; every injected degradation below threshold raises the actions within latency. A window without a metric where ground truth was available is a fail.

Declared population or set: Operational data over the declared window, with ground-truth or proxy availability stated (Section 1.7).

Evidence artifact: Performance monitoring records. Cadence: Continuous.

2.4.3 Model Maintenance Criteria

[R.AI-4.3] MODEL MAINTENANCE CRITERIA

The AI system shall define criteria for model maintenance, retraining, or version updates when performance deviates beyond acceptable bounds.

Rationale: *When continuous monitoring detects performance degradation, the system must have predefined response procedures. Without documented criteria for when and how to update models, operators may allow degraded performance to continue or may apply ad-hoc fixes that introduce new risks. This requirement ensures maintenance decisions are systematic and traceable. Model maintenance changes also invoke the model integrity controls established under AI-7.4 and the deployment format validation established in AI-4.7; these three control points apply in concert whenever a deployed model is updated.*

[V.AI-4.3] MODEL MAINTENANCE CRITERIA

The AI system shall verify model maintenance criteria by inspection.

Evidence class: Deterministic.

The inspection shall confirm that retraining and update trigger criteria are documented as quantitative conditions on the AI-4.1 and AI-4.2 indicators; that version control procedures exist for model updates; that the regression test set for updated models is defined as at least the test partition and the declared edge-case sets; and that the change history records each update with its rationale and its passage through the three control points AI-4.3, AI-4.7 and AI-7.4.

Success Criteria: Every trigger criterion quantitative; regression set named; every recorded update shows all three control points. An update missing a control point is a fail.

Declared population or set: Not applicable; a deterministic check on the maintenance records.

Evidence artifact: Model maintenance records. Cadence: Periodic.

2.4.4 Output Validity Boundaries

[R.AI-4.4] OUTPUT VALIDITY BOUNDARIES

The AI system shall define and enforce boundaries for output data validity. For Safety-Critical systems the enforcing element is the runtime assurance monitor of AI-14.

Rationale: *AI systems can produce outputs that are technically valid (within data type constraints) but operationally invalid (physically impossible, outside safe operating ranges, or inconsistent with known constraints). Output boundary enforcement provides a safety envelope that catches erroneous outputs regardless of the internal cause, whether from drift, adversarial inputs, or model failure. Operational limits that bound acceptable outputs shall be derived from the Performance Envelope and Exclusions declared under AI-1.0.*

[V.AI-4.4] OUTPUT VALIDITY BOUNDARIES

The AI system shall verify output validity boundaries by test.

Evidence class: Deterministic.

The test shall confirm that output boundaries are defined for each AI function from the Performance Envelope and Exclusions declared under AI-1.0 and from physical constraints and safety margins; that every injected out-of-bounds output in a declared injection set is flagged or rejected; that violations are logged with context per Section 4.6; and that no out-of-bounds output propagates to a downstream system without the defined response, which at Safety-Critical is the AI-14 fallback.

Success Criteria: One hundred per cent of injected out-of-bounds outputs flagged or rejected; zero propagations without the defined response. One propagation is a fail.

Declared population or set: The declared injection set, covering each output bound at and beyond its edge (Section 1.7).

Evidence artifact: Output boundary test report. Cadence: One-time.

2.4.5 Periodic Model Validation

[R.AI-4.5] PERIODIC MODEL VALIDATION

The AI system shall perform model validation at defined intervals and maintain baseline validation scores.

Rationale: *Continuous monitoring of individual metrics may not capture gradual, systemic degradation. Periodic comprehensive validation against baseline scores provides a systemic check that the model continues to perform as originally validated. Validation frequency should scale with system classification and operational tempo.*

[V.AI-4.5] PERIODIC MODEL VALIDATION

The AI system shall verify periodic model validation by test.

Evidence class: Statistical.

The test shall confirm that validation frequency is derived under Section 1.5 from classification and mission profile; that validation executes automatically at the declared interval; that validation scores are estimated with intervals and compared with the baseline thresholds; and that failures raise the Section 4.6 actions.

Success Criteria: Validation executed at every declared interval in the record; score interval within threshold of baseline at each; failures acted on. A missed interval is a fail; an interval containing the threshold is insufficient evidence for that period.

Declared population or set: The periodic validation set declared under Section 1.7.

Evidence artifact: Periodic validation reports. Cadence: Periodic.

2.4.6 Operational Validation Without Ground Truth

[R.AI-4.6] OPERATIONAL VALIDATION WITHOUT GROUND TRUTH

The AI system shall define and justify proxy validation mechanisms for operational environments where ground truth data is unavailable or inaccessible. The justification shall state, for each proxy, the degradation it is correlated with and where that correlation was measured, the degradation it cannot detect, the condition under which it saturates or becomes insufficient, and the operational response when that condition is reached.

Rationale: *AI-4 requirements assume continuous validation against known correct outputs. This assumption holds during development and in connected operational environments where labeled data or authoritative reference sources remain accessible. However, many safety-critical operational contexts lack ground truth entirely: disconnected systems, autonomous platforms, deep space missions, submarine operations, or any environment where the "correct answer" cannot be*

independently determined in real time. In these environments, validation must rely on proxy mechanisms that provide evidence of continued reliable performance without direct comparison to ground truth. These proxies include output boundary enforcement (per AI-4.4), statistical drift detection against baseline distributions (per AI-4.1), independent cross-validation using dissimilar methods (per AI-5.4), and temporal consistency checks against the system's own recent output history. The selection and sufficiency of proxy mechanisms are context dependent. A system operating autonomously for hours faces different validation challenges than one operating autonomously for months. Projects shall document which proxy mechanisms apply, justify why those proxies are sufficient for the specific operational context and duration, and define the conditions under which proxy validation is no longer adequate and operations should be suspended or degraded.

[V.AI-4.6] OPERATIONAL VALIDATION WITHOUT GROUND TRUTH

The AI system shall verify operational validation without ground truth by inspection and analysis.

Evidence class: Surveillance.

The inspection shall confirm that the operational environment is characterized with respect to ground-truth availability and duration of unavailability; that proxy mechanisms are identified for each AI function operating without ground truth; and that the sufficiency justification carries the four elements the requirement names for every proxy.

The analysis shall confirm that the proxies collectively address performance degradation detection, output validity assurance and behavioral consistency monitoring; that each stated correlation is supported by the qualification data declared under Section 1.7; that the insufficiency conditions are defined with the required response (degraded mode, human escalation or operational suspension); and that proxy coverage is argued against the expected duration of ground-truth unavailability.

Success Criteria: Every proxy carries all four elements; every correlation claim traces to qualification data; every insufficiency condition has a response. A proxy with an unsupported correlation claim is a fail.

Declared population or set: The proxy qualification data declared under Section 1.7.

Evidence artifact: Proxy validation documentation. Cadence: Continuous; justification re-verified at surveillance.

2.4.7 Deployment Format Validation

[R.AI-4.7] DEPLOYMENT FORMAT VALIDATION

The AI system shall revalidate deployment-format models against the original trained model before operational use, where deployment-format models are produced through transformations such as quantization, pruning, knowledge distillation, or framework conversion.

Rationale: *Model transformations applied for deployment can introduce numerical, accuracy, or behavioral drift relative to the originally validated model. Without explicit revalidation, an artifact that passed all training-time verification may exhibit degraded or unsafe behavior at runtime, a gap not closed by retraining triggers (AI-4.3) or load-time integrity checks (AI-7.4). Side-by-side comparison on representative data establishes that the deployed artifact retains its validated properties.*

Common transformations requiring revalidation include:

- **Quantization:** reducing model precision (e.g., FP32 to INT8) to enable execution on resource-constrained hardware.
- **Pruning:** removing model weights, neurons, or layers to reduce computational cost.
- **Knowledge distillation:** training a smaller model to approximate the behavior of a larger original.
- **Format conversion:** converting between frameworks (e.g., PyTorch to ONNX to TensorRT) for runtime compatibility.

[V.AI-4.7] DEPLOYMENT FORMAT VALIDATION

The AI system shall verify deployment format validation by test.

Evidence class: Statistical.

The test shall confirm that the deployment-format model meets every performance threshold established for the original trained model, estimated with intervals on the test partition over the declared grid; that the per-sample disagreement rate between the original and the deployed model is reported; that any degradation attributable to the transformation lies within the margin derived under Section 1.5 or the transformation is revised; and that the comparison is retained as evidence. Deployment Format Validation is a control point complementary to AI-4.3 and AI-7.4; a deployed transformation shall satisfy all three before operational use.

Success Criteria: Every threshold met on the deployed model with its interval within the derived margin in every covered cell; disagreement rate reported. An interval containing the margin is insufficient evidence; a cell outside it is a fail.

Declared population or set: The test partition over the declared grid and the near-boundary set of AI-6.2 (Section 1.7).

Evidence artifact: Side-by-side performance comparison report. Cadence: One-time per transformation.

Related Domain Standards

- Aerospace: NPR 7150.2D §4 (Software Lifecycle Processes including Verification and Validation); NASA-STD-8739.8B (Software Assurance)
- Aviation: DO-178C §6.4 (Software Testing) and §11.5 (Verification of Verification Process Results)
- Automotive: ISO 26262-6:2018 §11 (Testing of the embedded software); ISO 21448:2022 §12 (Evaluation of the achievement of the SOTIF)
- Medical: IEC 62304:2006+A1:2015 §5.7 (Software system testing) and §5.8 (Software release)
- Industrial: ISO 13849-1 (Functional safety of control systems)

2.5 AI-5: HALLUCINATION PREVENTION

[R.AI-5] HALLUCINATION PREVENTION

The AI system will implement mechanisms to detect, prevent, and mitigate AI outputs that are fabricated, unsupported by input data, or inconsistent with operational reality.

Rationale: *AI systems, particularly neural networks and generative models, can produce outputs that appear valid but are completely fabricated or contradicted by physical reality. NIST AI 600-1 defines hallucination as* ***"the production of confidently stated but erroneous or false content."*** Unlike traditional software bugs that produce obviously wrong outputs (e.g., null values, exceptions), hallucinations appear plausible and may not trigger conventional error handling. This requirement addresses the "unpredictable outputs" gap identified in Section 1.2.4.2.

Applicability: AI-5 applies where the AI system produces generative or free-form outputs, that is, outputs not confined to a declared finite range (text, plans, trajectories, code, images and the like). For systems whose outputs are confined to

declared ranges, the output validity boundaries of AI-4.4 serve this purpose, and AI-5 may be documented as Not Applicable with rationale. The hallucination logging requirement of versions 3.x (AI-5.4 in 3.x) is carried by Section 4.6.4; the identifier map in the Document Revision History maps the identifiers of this set.

Related Domain Standards

- General: NIST AI 600-1 (Generative AI Profile, 2024) §2.2 (Confabulation); ISO/IEC 22989:2022 5.11
- Automotive: ISO 21448:2022 §7 (Functional insufficiencies and triggering conditions)
- Medical: FDA Clinical Decision Support Software Guidance (January 2026) §IV(3)-(4) (Interpretation of Criteria; supports independent review of basis for recommendations)
- EU: EU AI Act Article 15 (Robustness)
- Cross-Domain: NIST AI RMF MEASURE-2.7

2.5.1 Hallucination Criteria Definition

[R.AI-5.1] HALLUCINATION CRITERIA DEFINITION

Where the AI system produces generative or free-form outputs (text, plans, trajectories, code, images or any output not confined to a declared finite range), it shall define what constitutes a hallucination within the operational context. For systems whose outputs are confined to declared ranges, the output validity boundaries of AI-4.4 serve this purpose and this set may be documented as Not Applicable with rationale.

Rationale: *Hallucination is context-dependent. An output that would be a hallucination in one domain may be valid in another. For example, a predicted temperature of -270°C would be valid near absolute zero but a hallucination for cabin temperature. Defining mission-specific hallucination criteria enables automated detection and ensures operators understand when an output should be considered unreliable. Ground truth baselines provide authoritative reference points against which outputs can be validated.*

[V.AI-5.1] HALLUCINATION CRITERIA DEFINITION

The AI system shall verify hallucination criteria definition by inspection.

Evidence class: Deterministic.

The inspection shall confirm that hallucination criteria are defined for each generative AI function; that the criteria cover outputs outside expected bounds, outputs contradicting the inputs or other sensors, and outputs unsupported by the inputs; that the criteria are traceable to mission-specific operational constraints; that worked examples distinguish hallucinations from valid edge-case outputs; and that the ground truth baselines used to judge outputs (authoritative sources, physical constraints and conservation laws, independent sensors, operator-verified references) are named.

Success Criteria: Every generative function has criteria covering all three failure classes, traceable to operational constraints, with named baselines and worked examples. A function without criteria is a fail.

Declared population or set: Not applicable; the criteria define the populations used by AI-5.2.

Evidence artifact: Hallucination criteria document. Cadence: One-time.

2.5.2 Hallucination Detection

[R.AI-5.2] HALLUCINATION DETECTION

The AI system shall implement detection mechanisms for hallucinated outputs.

Rationale: *Once hallucination criteria are defined, automated detection mechanisms can identify suspect outputs before they propagate to downstream systems or influence decisions. Detection should occur at inference time, not just during post-mission analysis, to prevent real-time operational impact. Detection mechanisms may include consistency checks, physical constraint validation, cross-sensor comparison, or confidence thresholding.*

[V.AI-5.2] HALLUCINATION DETECTION

The AI system shall verify hallucination detection by test.

Evidence class: Statistical.

The test shall confirm that detection mechanisms are implemented for each generative function and operate before outputs reach downstream systems; that detection is validated on the declared hallucination test set (Section 1.7), which shall contain both injected cases that violate the criteria of AI-5.1 and errors the model itself produced on held-out inputs, reported as separate strata; and that the detection and false-positive rates are estimated with intervals on each stratum against the thresholds derived under Section 1.5.

Success Criteria: Detection rate interval above and false-positive rate interval below their derived thresholds on both strata. The model-produced stratum governs: an interval containing the threshold there is insufficient evidence; a set without a model-produced stratum is a fail.

Declared population or set: The declared hallucination test set with its injected and model-produced strata and their sizes (Section 1.7).

Evidence artifact: Hallucination detection test report. Cadence: One-time; repeated on model update.

2.5.3 Hallucination Response

[R.AI-5.3] HALLUCINATION RESPONSE

The AI system shall prevent hallucinated outputs from being acted upon without human review, where the hazard analysis shows the time to effect permits review, or without the defined fallback behavior, which for Safety-Critical systems is the assured fallback of AI-14, proportionate to the AI function's criticality.

Rationale: *Detection without response provides no protection. When a potential hallucination is detected, the system must prevent autonomous action based on that output, either escalating to human review or invoking defined fallback behavior, consistent with the response pattern established for out-of-distribution inputs (AI-6.3). This ensures either human judgment or a validated safe response is applied before potentially erroneous AI outputs influence safety-critical decisions. The response should be proportionate to the criticality of the AI function and the confidence of hallucination detection.*

[V.AI-5.3] HALLUCINATION RESPONSE

The AI system shall verify hallucination response by test.

Evidence class: Bounded.

The test shall inject detected hallucinations and confirm that each is flagged and quarantined; that autonomous execution is blocked for flagged outputs; that the human review workflow is triggered where permitted or the defined fallback executes within its budget; and that override requires documented rationale.

Success Criteria: Every injected case quarantined and resolved to review or fallback within budget; no flagged output acted upon. One acted-upon flagged output is a fail.

Declared population or set: The injected set drawn from the declared hallucination test set (Section 1.7).

Evidence artifact: Hallucination response test record. Cadence: One-time; exercised at surveillance.

2.5.4 Independent Output Validation (AI-5.5 in versions 3.x)

[R.AI-5.4] INDEPENDENT OUTPUT VALIDATION

The AI system shall validate AI outputs affecting safety-critical or mission-critical functions against an independent source prior to execution. The independent source shall use dissimilar methods, sensors or processing from the primary AI and shall not share its failure modes. For Safety-Critical systems the independent source is the runtime assurance monitor of AI-14 or a method of equivalent independence; human review alone does not satisfy this sub-requirement at Safety-Critical.

Rationale: *For safety-critical functions, relying solely on the AI system's internal consistency checks is insufficient; an AI can produce confidently wrong outputs that pass its own validation. Independent validation using dissimilar methods, sensors, or processing provides a defense-in-depth layer that can catch errors the primary AI cannot detect. The independent source must not share common failure modes with the primary AI to ensure that a single fault (e.g., corrupted sensor data, adversarial input, systematic model error) cannot defeat both the primary output and its validation. Errors of independently trained language models have been measured to correlate, so for text, plans and code a second learned model is not independent by inventory alone; where the validator is learned, its independence is measured on the model-produced stratum.*

[V.AI-5.4] INDEPENDENT OUTPUT VALIDATION

The AI system shall verify independent output validation by analysis and test.

Evidence class: Deterministic.

The analysis shall identify the independent validation source for each safety-critical and mission-critical output and confirm that it shares none of the following with the primary AI: training data sources, sensor inputs, processing hardware, software dependencies, and the learned component itself; and that no source the independent validator consults at runtime (a registry, a page, a service) accepts content from outside the project's change control unless that content is validated under AI-1.6 before use.

The test shall inject discrepancies between the AI output and the independent source and confirm that every discrepancy triggers human review or the defined fallback, that no safety-critical action executes while validation has failed, and that validation latency meets the operational timing requirement.

Where the independent source is itself a learned artifact, the test shall run the primary AI and the independent source on the model-produced stratum of the declared hallucination test set (AI-5.2) and estimate the joint miss rate, the proportion of model-produced hallucinations the independent source also fails to flag, with an interval against the threshold derived under Section 1.5 (statistical evidence under Section 4.1, which supports and does not close).

Success Criteria: Every safety-critical output has an independent source with zero shared failure-mode elements; every injected discrepancy triggers the response; zero executions during failed validation. A shared element or an execution during failed validation is a fail. Where the independent source is a learned artifact, the joint miss rate interval shall lie below its derived threshold; an interval containing the threshold is insufficient evidence. A runtime validation source outside change control and not validated under AI-1.6 is a fail.

Declared population or set: The injected discrepancy set covering each safety-critical and mission-critical output (Section 1.7).

Evidence artifact: Independence analysis and discrepancy test report. Cadence: One-time.

2.5.5 Hallucination Audit

[R.AI-5.5] HALLUCINATION AUDIT

At each declared audit interval, a declared audit sample shall be drawn from operational outputs that passed detection under AI-5.2 and any validation under AI-5.4, each audited output shall be checked against the ground truth baselines of AI-5.1, and the audited miss rate, the proportion of audited outputs found to be hallucinations under the AI-5.1 criteria, shall be estimated with an interval.

Rationale: Detection (AI-5.2) is validated once on a held-out set, and the Section 4.6 indicator for AI-5 counts the hallucinations the detector caught. Neither measures what the detector missed in service, which is the quantity the word prevention in R.AI-5 rests on. An audit of outputs that passed detection and validation, checked against the AI-5.1 baselines by a reviewer or method independent of the detector, is the only direct measurement of that quantity. It is surveillance evidence: it accrues in service, it supports the requirement at the stated confidence, and it never closes it.

[V.AI-5.5] HALLUCINATION AUDIT

The AI system shall verify hallucination audit by analysis.

Evidence class: Surveillance.

The analysis shall confirm that the audit sample is drawn from passed outputs by the declared sampling rule and reaches the declared sample floor (Section 1.7); that each audited output is checked against the AI-5.1 baselines by a reviewer or method independent of the detector; and that the audited miss rate is estimated with an interval against the threshold derived under Section 1.5 at every declared interval.

Success Criteria: Audited miss rate interval below its derived threshold at every interval. An interval containing the threshold is insufficient evidence; an interval above it, or a missed interval, is a fail.

Declared population or set: The declared audit sample of passed operational outputs per interval, with its sampling rule and floor (Section 1.7).

Evidence artifact: Hallucination audit record. Cadence: Periodic, at each declared audit interval.

2.6 AI-6: OUT-OF-DISTRIBUTION DETECTION

[R.AI-6] OUT-OF-DISTRIBUTION DETECTION

The AI system will detect and appropriately handle inputs that fall outside the distribution of training data.

ML models are only reliable within the distribution of data they were trained on. When inputs differ significantly from training data due to sensor anomalies, novel operational scenarios, or environmental conditions not anticipated during development, model outputs become unreliable and potentially dangerous. Traditional software has defined input/output bounds that can be checked explicitly; AI systems are vulnerable to OOD inputs that may appear valid but produce unreliable predictions. This requirement addresses the “vulnerable to OOD inputs” gap identified in Section 1.2.4.2, ensuring the system recognizes when it is operating outside its validated envelope.

Related Domain Standards

- General: ISO/IEC 23053:2022; ISO/IEC 5338:2023 §6.4.14
- Aviation: FAA AI Safety Roadmap (operational envelope)
- Automotive: ISO 21448:2022 §6–§8; ISO 34503 (ODD boundary definitions)
- Medical: FDA AI/ML-Based SaMD Action Plan (real-world performance monitoring)
- EU: EU AI Act Article 15 (Robustness to errors)

- Cross-Domain: NIST AI RMF MEASURE-2.6

2.6.1 Training Distribution Characterization

[R.AI-6.1] TRAINING DISTRIBUTION CHARACTERIZATION

The AI system shall characterize the training data distribution for each ML model within the scope declared under AI-1.0, and shall show that the characterization supports runtime detection on the declared near-boundary out-of-distribution set. Where inputs are high-dimensional and a density or distance characterization in input space cannot be shown to support detection, the characterization shall be made in a declared feature space or by a method validated under AI-12.5.

Rationale: *OOD detection requires knowing what* ***"in-distribution"*** *looks like. Characterizing the training data distribution provides the baseline against which operational inputs are compared. Documentation should include statistical measures (mean, variance, range, density estimates) that enable automated OOD detection at runtime. Without explicit distribution characterization, OOD detection cannot be implemented systematically. The distribution shall be characterized within the scope declared under AI-1.0; characterization need not extend beyond the declared Operational Environment and Subject Population.*

[V.AI-6.1] TRAINING DISTRIBUTION CHARACTERIZATION

The AI system shall verify training distribution characterization by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the training distribution is documented for each model with the statistical measures used (means, variances, covariances, ranges, density estimates, or feature-space equivalents), that the space in which the characterization is made is declared, and that the characterization is traceable to the training partition.

The analysis shall confirm that the out-of-distribution test populations of Section 1.7 are declared with a far-from-envelope stratum and a near-boundary stratum, that the boundary of each has been validated before use (members of the near set shown to lie outside the declared envelope, and in-envelope members excluded), and that the characterization separates the near-boundary stratum from in-envelope data at the operating point AI-6.2 uses.

Success Criteria: Characterization space declared; both strata declared with validated boundaries; separation of the near-boundary stratum shown at the AI-6.2 operating point. An unvalidated test-set boundary is a fail.

Declared population or set: The declared far and near out-of-distribution strata and the in-envelope reference set (Section 1.7).

Evidence artifact: Training distribution characterization and test-set boundary validation record. Cadence: One-time.

2.6.2 Runtime OOD Detection

[R.AI-6.2] RUNTIME OOD DETECTION

The AI system shall implement mechanisms to detect out-of-distribution inputs at runtime.

Rationale: *OOD detection must occur during operation, not just during development testing. Runtime detection mechanisms compare incoming inputs against the characterized training distribution and flag inputs that fall outside defined boundaries. Detection methods may include statistical distance measures (Mahalanobis distance), density estimation, ensemble disagreement, or dedicated OOD detection models. Detection must meet operational timing requirements to enable timely response.*

[V.AI-6.2] RUNTIME OOD DETECTION

The AI system shall verify runtime ood detection by test.

Evidence class: Statistical.

The test shall confirm that out-of-distribution detection is implemented for each ML model; that the detection rate at the declared false-positive budget is estimated with an interval on the far-from-envelope stratum and on the near-boundary stratum separately, the threshold on the detector having been set on the validation partition and the false-positive rate measured on the test partition; and that detection latency meets the operational timing requirement for the AI function.

Success Criteria: Near-boundary detection interval above the derived threshold at or below the false-positive budget, latency within requirement. The far stratum is reported and does not govern. A near-boundary interval containing the threshold is insufficient evidence; a point estimate below it is a fail.

Declared population or set: The declared far and near out-of-distribution strata, the in-envelope test partition for the false-positive rate (Section 1.7).

Evidence artifact: OOD detection test report. Cadence: One-time; repeated on model update.

2.6.3 OOD Response

[R.AI-6.3] OOD RESPONSE

The AI system shall prevent autonomous action on out-of-distribution inputs without human review, where the hazard analysis shows the time to effect permits review, or without the defined fallback behavior, which for Safety-Critical systems is the assured fallback of AI-14.

Rationale: *Detecting OOD inputs without a defined response provides no protection. When an input is flagged as OOD, the system must either escalate to human review or invoke predefined fallback behavior appropriate to the AI function's criticality. Autonomous execution based on potentially unreliable OOD predictions could lead to hazardous outcomes. The response should be proportionate to system classification: Safety-Critical AI may require mandatory human review, while Operational Support AI may use automated fallback.*

[V.AI-6.3] OOD RESPONSE

The AI system shall verify ood response by test.

Evidence class: Bounded.

The test shall inject inputs from both declared out-of-distribution strata and confirm that each flagged input triggers the defined response; that autonomous execution is blocked or the fallback invoked within its budget; that the human review workflow is available where permitted; and that behavior for every declared OOD scenario is documented and tested.

Success Criteria: Every flagged injected input resolves to review or fallback within budget; zero autonomous actions on flagged inputs. One such action is a fail.

Declared population or set: The injected set drawn from both declared strata (Section 1.7).

Evidence artifact: OOD response test record. Cadence: One-time; exercised at surveillance.

2.7 AI-7: ADVERSARIAL ROBUSTNESS

[R.AI-7] ADVERSARIAL ROBUSTNESS

The AI system will implement mechanisms to detect, prevent, and mitigate adversarial attacks that could compromise AI integrity, manipulate outputs, or degrade system performance.

AI systems are vulnerable to attack vectors that do not exist for traditional software. Adversarial attacks can manipulate training data (poisoning), craft inputs that cause misclassification (adversarial examples), extract model parameters (model theft), or reconstruct sensitive training data (model inversion). Adversarial compromise could affect safety or operational success. This requirement addresses the "vulnerable to adversarial attacks" gap identified in Section 1.2.4.

AI-specific adversarial robustness does not replace foundational cybersecurity engineering; it extends it. The controls in AI-7.1 through AI-7.6 address attack vectors unique to AI systems, and they assume an underlying cybersecurity posture for the system and for the environment in which it is developed, built, and deployed. That posture is a prerequisite of certification rather than assessed scope: the applicant declares its foundational cybersecurity baseline by attestation or by reference to existing conformity evidence, Safety Critical Labs records the declaration, and the declared baseline is carried on the certificate as an element of the certified configuration (Section 1.6). NIST SP 800-53 Rev. 5 (Security and Privacy Controls for Information Systems and Organizations), in particular the System and Information Integrity (SI) and System and Communications Protection (SC) control families, and NIST SP 800-218 (Secure Software Development Framework, SSDF) for the ML pipeline code, training infrastructure, and deployment tooling, are representative baselines and are listed with the related domain standards below; this framework does not levy them. AI-7 sub-requirements complement rather than duplicate the declared baseline: data poisoning protection (AI-7.1) extends integrity controls to training data; model integrity (AI-7.4) extends them to model artifacts; supply chain security (AI-7.5) extends secure development practices to AI-specific dependencies including pre-trained models and ML frameworks; and artifact load safety (AI-7.6) extends them to the loading of every artifact the system admits.

Related Domain Standards

- General: NIST AI 100-2e2023 (Adversarial ML Taxonomy); ISO/IEC 5338:2023 §6.4.3
- Aviation: DO-326A (Airworthiness Security Process); DO-356A (Airworthiness Security Methods); FAA AI Safety Roadmap
- Automotive: ISO/SAE 21434:2021 (Road vehicles cybersecurity); UNECE WP.29 R155 (Cyber security management)
- Medical: FDA Premarket Cybersecurity Guidance (2023); FDA Postmarket Cybersecurity Guidance (2016)

- EU: EU AI Act Article 15 (Cybersecurity); EU Cyber Resilience Act (EU Cyber Resilience Act article reference held)
- Cross-Domain: NIST AI RMF MANAGE-2.2; NIST SP 800-218 (Secure Software Development Framework); NIST SP 800-53 Rev. 5 SI family

2.7.1 Data Poisoning Protection

[R.AI-7.1] DATA POISONING PROTECTION

The AI system shall implement protections against training data poisoning attacks.

Rationale: *Data poisoning injects malicious samples into training datasets to influence model behavior in attacker-controlled ways. Poisoned models may appear to function normally on most inputs but produce incorrect outputs on specific attacker-chosen inputs. Poisoning could create backdoors that trigger dangerous behavior under specific conditions. Protection requires authenticating data sources, validating data integrity, and detecting statistically anomalous samples.*

[V.AI-7.1] DATA POISONING PROTECTION

The AI system shall verify data poisoning protection by analysis and test.

Evidence class: Bounded.

The analysis shall confirm that training data sources are authenticated through documented provenance (AI-1.3) and that data integrity checks detect unauthorized modification.

The test shall confirm that anomaly detection on the training data identifies statistically suspicious samples at a reported rate on a declared seeded set, and that every poisoning scenario in the declared poisoning attack pattern set (Section 1.7; for example backdoor triggers, label flipping, gradient-based poisoning) is executed with its outcome recorded: detected, mitigated, or residual.

Success Criteria: Integrity checks detect one hundred per cent of seeded modifications; every declared poisoning scenario has a recorded outcome; residuals are stated as such. An unexecuted scenario is a fail; the result holds for the declared set and no other.

Declared population or set: The declared poisoning attack pattern set and seeded modification set (Section 1.7), with derivation basis.

Evidence artifact: Poisoning protection test report (Part 2 C.2). Cadence: One-time; repeated when the threat model changes.

Related Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-7 (Software, firmware, and information integrity); ISO/IEC 27001:2022 A.8.28 (Secure coding)
- AI-Specific: NIST AI 100-2e2023 §2.3 (Poisoning Attacks and Mitigations); MITRE ATLAS AML.T0020 (Poison training data)
- Data Integrity: ISO/IEC 5259-3:2024 (Data quality management, verification); ISO 8000-61:2016 (Data quality management process reference model)
- Aviation: RTCA DO-326A §3.5 (Security risk assessment); SAE ARP 4754A §4 (Development assurance)
- Automotive: ISO/SAE 21434:2021 §15 (Threat analysis and risk assessment methods)
- Cross-Domain: NIST AI RMF MANAGE-2.1 (Risk response to identified AI risks); EU AI Act Article 15(4) (Resilience against attempts to manipulate training data)

2.7.2 Adversarial Input Protection

[R.AI-7.2] ADVERSARIAL INPUT PROTECTION

The AI system shall implement protections against adversarial input manipulation at runtime against a declared threat model. The threat model shall state who or what can supply a crafted input to the system, through which input path, and with what perturbation budget, and the budget shall be derived from it under Section 1.5. Robustness is demonstrated at that budget and is a bounded demonstration: the budget and the attack set are recorded in the certified configuration and on the certificate, and the result holds at the budget and nowhere else. For Safety-Critical systems the attack set shall include attacks designed against the defence itself (adaptive attacks) and not only fixed attack patterns. For systems incorporating generative models or retrieval augmentation the set shall include direct and indirect prompt injection. Where the threat model shows that no credible adversary can supply a crafted input, the protection is architectural (input provenance, physical isolation) and the tailoring record shall say so.

Rationale: *Adversarial inputs are crafted to cause misclassification while appearing valid. A defence evaluated only against fixed attack patterns can appear robust and fail against an attack built with knowledge of the defence, and detectors of adversarial inputs have a long record of being bypassed. Robustness therefore cannot be claimed in general; it can be shown at a stated budget against a stated set, and the certificate says exactly that.*

[V.AI-7.2] ADVERSARIAL INPUT PROTECTION

The AI system shall verify adversarial input protection by test.

Evidence class: Bounded.

The test shall confirm that the threat model and perturbation budget are declared with their derivation; that input validation detects malformed or anomalous inputs; that model outputs remain within the acceptable degradation at the declared budget against every attack in the declared attack pattern set (Section 1.7), with adaptive attacks included at Safety-Critical and prompt injection included for generative and retrieval-augmented systems; that any robustness attributable to gradient masking is excluded per AI-12.6 where neural networks are present; and that the residual vulnerability is documented.

Success Criteria: Degradation within the derived margin at the declared budget against every attack in the declared set, adaptive attacks included at Safety-Critical, with the residual stated. A missing adaptive attack at Safety-Critical is a fail; the result is bounded to the budget and the set.

Declared population or set: The declared threat model, perturbation budget and attack pattern set (Section 1.7), recorded in the certified configuration.

Evidence artifact: Adversarial robustness test report (Part 2 C.2) with the budget and set. Cadence: One-time; repeated when the threat model or the model changes.

Related Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-10 (Information input validation); NIST AI 100-2e2023 §2 (Evasion attacks)
- AI-Specific: OWASP ML Top 10 (2023) ML01 (Input manipulation attack); OWASP Top 10 for LLM Applications LLM01 (Prompt injection); MITRE ATLAS AML.T0043 (Craft adversarial data); MITRE ATLAS AML.T0051 (LLM prompt injection)
- Robustness: ISO/IEC TR 24029-1:2021 (AI robustness assessment overview); IEEE 7009-2024 (Fail-safe design of autonomous systems)
- Aviation: RTCA DO-326A §3.7 (Security architecture); SAE ARP 6983 (AI/ML in airborne systems, draft)
- Automotive: ISO/SAE 21434:2021 §9.5 (Cybersecurity concept)

- Cross-Domain: NIST AI RMF MEASURE-2.7 (AI system security); EU AI Act Article 15(5) (Resilience against adversarial inputs)

2.7.3 Model Inversion and Extraction Protection

[R.AI-7.3] MODEL INVERSION AND EXTRACTION PROTECTION

The AI system shall implement protections against model inversion and extraction attacks.

Rationale: *Model inversion attacks reconstruct sensitive training data from model outputs. Model extraction attacks query the model systematically to replicate its behavior, enabling offline study of its vulnerabilities. Protection requires access controls, query rate limiting, and output perturbation techniques that limit systematic probing while maintaining operational utility; rate limiting slows extraction and does not prevent it, which is why the residual is stated. Where differential privacy or analogous techniques are applied for data subject privacy purposes, see also AI-10.4 (Privacy-Enhancing Techniques); the same mechanism may satisfy both requirements under distinct threat models.*

[V.AI-7.3] MODEL INVERSION AND EXTRACTION PROTECTION

The AI system shall verify model inversion and extraction protection by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that model access is restricted to authorized queries and that the sensitivity classification of the training data is documented.

The analysis shall confirm that query rate limiting is configured with its limits stated and the extraction effort it implies estimated, and that output perturbation or differential privacy is applied where the sensitivity classification requires it, with the privacy parameter stated.

Success Criteria: Access restricted; rate limits and implied extraction effort stated; perturbation or privacy parameter stated wherever sensitivity requires it. A sensitive-data model without a stated privacy parameter is a fail.

Declared population or set: Not applicable; a deterministic check on the access configuration and the sensitivity record.

Evidence artifact: Access control and privacy configuration record. Cadence: One-time.

Related Domain Standards

- General: NIST SP 800-53 Rev. 5 SC-8 (Transmission confidentiality) and SC-28 (Protection of information at rest)
- AI-Specific: NIST AI 100-2e2023 §2.4 (Privacy Attacks: membership inference, model inversion, extraction); MITRE ATLAS AML.T0024 (Exfiltration via ML inference API)
- Privacy-Preserving ML: NIST SP 800-226 (Guidelines for evaluating differential privacy guarantees)
- Data Protection: ISO/IEC 27701:2019 (Privacy information management); GDPR Article 32 (Security of processing)
- Cross-Domain: NIST AI RMF MAP-4.1 (Approaches for mapping AI technology and legal risks); EU AI Act Article 15(4) (Resilience against confidentiality attacks)

2.7.4 Model Integrity

[R.AI-7.4] MODEL INTEGRITY

The AI system shall maintain model integrity throughout the operational lifecycle.

Rationale: *Model integrity controls address two threat cases. In the first, the artifact was sound when validated and is modified, corrupted, or replaced afterward: cryptographic signing provides tamper-evident verification, and integrity checks at load time and at defined intervals during operation detect the change. The controls of this requirement are necessary for that case, in which unauthorized modification could otherwise introduce backdoors, degrade performance, or alter behavior in ways that circumvent all prior verification. In the second, the artifact is hostile at its origin: a backdoored or code-bearing artifact published by an adversary carries a valid signature and complete provenance, so the same controls verify the authenticity of a compromised artifact and pass. Signing establishes origin, not soundness. The controls of this requirement are therefore not sufficient for the adversarial-origin case, which is addressed by AI-7.5 (provenance and risk assessment of pre-trained components) and AI-7.6 (Artifact Load Safety). Model integrity controls apply in concert with the maintenance criteria established under AI-4.3 and the deployment format validation established in AI-4.7; legitimate model updates shall pass through all three control points before operational deployment.*

[V.AI-7.4] MODEL INTEGRITY

The AI system shall verify model integrity by inspection.

Evidence class: Deterministic.

The inspection shall confirm that every model artifact is cryptographically signed; that integrity verification occurs at model load and at the declared interval during operation; that an unauthorized modification triggers the Section 4.6 alert and the fallback; and that version history is maintained with tamper-evident logging.

Success Criteria: Every artifact in the certified configuration signed; load-time and interval verification present in the record; a seeded unauthorized modification detected. One unsigned artifact is a fail.

Declared population or set: Not applicable; a deterministic check on the artifacts of the certified configuration.

Evidence artifact: Model integrity verification records. Cadence: Continuous.

Related Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-7 (Integrity) and CM-5 (Access restrictions for change)
- Secure Development: NIST SP 800-218 (SSDF, Secure Software Development Framework); SLSA Framework v1.0 (Supply-chain Levels for Software Artifacts)
- Model Signing: Sigstore / Linux Foundation Model Transparency project (cryptographic signing of model weights)
- Aviation: RTCA DO-326A §3.9 (Security assurance); RTCA DO-178C §11.5 (Configuration management)
- Automotive: ISO/SAE 21434:2021 §11 (Cybersecurity validation)
- Cross-Domain: NIST AI RMF MANAGE-2.1 (Risk response); EU AI Act Article 15(1) (Accuracy, robustness, and cybersecurity throughout lifecycle)

2.7.5 AI Supply Chain Security (AI-7.6 in versions 3.x)

[R.AI-7.5] AI SUPPLY CHAIN SECURITY

The AI system shall assess and document supply chain risks for all AI/ML frameworks, libraries, and pre-trained model components.

Rationale: *AI systems depend on complex software supply chains including ML frameworks (e.g., TensorFlow, PyTorch), libraries (e.g., NumPy, ONNX), and potentially pre-trained models or transfer learning components. These dependencies may contain vulnerabilities, backdoors, or undocumented behaviors*

introduced through foreign contribution, compromised maintainers, or malicious updates. Pre-trained models may encode biases, backdoors, or behaviors from unknown training data. Supply chain assessment ensures that AI system integrity extends beyond the project's own development to encompass all external dependencies.

[V.AI-7.5] AI SUPPLY CHAIN SECURITY

The AI system shall verify ai supply chain security by inspection.

Evidence class: Deterministic.

The inspection shall confirm that AI/ML frameworks, libraries and pre-trained components are identified with versions and sources; that the supply chain risk assessment names the vulnerability sources consulted (for example CVE/NVD, vendor advisories, ML-specific trackers) and the date of the sweep, declared under Section 1.7; that the assessment addresses contribution sources, maintenance status and known vulnerabilities; that pre-trained models have documented provenance and integrity verification; and that mitigations are documented for identified risks.

Success Criteria: Every dependency identified with version and source; vulnerability sources and sweep date declared; every identified risk has a mitigation. A dependency without a version or a risk without a mitigation is a fail.

Declared population or set: The declared vulnerability sources and sweep date (Section 1.7).

Evidence artifact: Supply chain risk assessment. Cadence: One-time; re-swept at surveillance.

Related Domain Standards

- General: NIST SP 800-53 Rev. 5 SR-3 (Supply chain controls) and SR-5 (Acquisition strategies)
- Supply Chain: NIST SP 800-161 Rev. 1 (C-SCRM for systems and organizations); NIST SP 800-218 (SSDF)
- Model Bill-of-Materials: NTIA Minimum Elements for SBOM (2021); CISA SBOM guidance; extensions for ML (ML-BOM)
- Aviation: RTCA DO-326A §3.8 (Supplier risk); RTCA DO-355 (Information security guidance for airworthiness)

- Automotive: ISO/SAE 21434:2021 §7 (Distributed cybersecurity activities: supplier relationships)
- Cross-Domain: NIST AI RMF GOVERN-6.1 (Third-party risk management); EU AI Act Article 25 (Obligations along the AI value chain)

2.7.6 Artifact Load Safety (AI-7.7 in versions 3.x)

[R.AI-7.6] ARTIFACT LOAD SAFETY

The AI system shall load model artifacts, and every other artifact admitted into its execution environment through a loader, including tokenizers, processors, datasets, and retrieval corpus content (AI-1.6), using serialization formats and loader configurations that do not execute code during deserialization or load. Any deviation shall be justified and risk assessed under AI-7.5. Loader configurations that execute remote or repository-supplied code at load shall be disabled, or the code so executed shall be brought under the same review and configuration control as project code.

Rationale: *Integrity verification (AI-7.4) establishes that an artifact is the one its publisher produced, and provenance documentation (AI-7.5) records where it came from; neither prevents the artifact from executing code when it is loaded. Executable serialization formats run arbitrary code at deserialization, and loader configurations that execute repository-supplied code at load run whatever the repository contains. A hostile artifact that passes signature verification and carries complete provenance documentation therefore controls the host from the moment it loads. The exposure is the loader, not the artifact type: model weights, tokenizers, processors, datasets, and retrieval corpus content pass through loaders with the same behavior, which is why this requirement binds every artifact load path. Freedom from code execution is a property of the format together with its loader configuration, and several formats execute no code only under configuration (custom operators, embedded graph code), so the requirement binds the loader configuration and not the format name alone. This requirement covers the adversarial-origin case named in the AI-7.4 rationale.*

[V.AI-7.6] ARTIFACT LOAD SAFETY

The AI system shall verify artifact load safety by inspection and test.

Evidence class: Deterministic.

The inspection shall confirm that every artifact load path is documented with its serialization format and loader configuration; that the format and configuration on each path do not execute code during deserialization or load, or that each deviation is justified and risk assessed under AI-7.5; and that loader options which execute

remote or repository-supplied code are disabled, or the code so executed is under project review and configuration control.

The test shall confirm that a crafted artifact carrying an executable payload, presented on each artifact load path, is rejected without execution.

Success Criteria: Every load path documented; crafted artifact rejected without execution on every path. One path that executes is a fail.

Declared population or set: Every artifact load path in the certified configuration.

Evidence artifact: Load path inventory and negative test record. Cadence: One-time; repeated on loader change.

Related Domain Standards

- General: NIST SP 800-53 Rev. 5 SI-3 (Malicious code protection), SI-7 (Software, firmware, and information integrity), and CM-7 (Least functionality)
- Secure Development: NIST SP 800-218 (SSDF, Secure Software Development Framework)
- AI-Specific: MITRE ATLAS AML.T0011.000 (User Execution: Unsafe ML Artifacts); MITRE ATLAS AML.T0010.003 (ML Supply Chain Compromise: Model); OWASP ML Top 10 (2023) ML06 (AI supply chain attacks); OWASP Top 10 for LLM Applications LLM03 (Supply chain)
- Cross-Domain: NIST AI RMF GOVERN-6.1 (Third-party risk management); EU AI Act Article 15(1) (Accuracy, robustness, and cybersecurity throughout lifecycle)

2.8 AI-8: EXPLAINABILITY

[R.AI-8] EXPLAINABILITY

The AI system will provide explanations of AI decisions and recommendations sufficient for operators to verify AI reasoning and establish appropriate trust.

Traditional software has explicit decision logic that can be inspected through code review: an engineer can trace exactly why a particular output was produced. AI systems, particularly neural networks, make decisions through learned patterns in weights that are not directly interpretable. ISO 22989 Section 3.5.7 defines explainability as the property of an AI system to express important factors influencing results in a way humans can understand. Without explainability, operators cannot verify AI reasoning, leading to either blind over-trust or

unwarranted under-trust. This requirement addresses the "black-box decision making" gap identified in Section 1.2.4.2. Where AI-10.3 (Data Subject Rights) requires explanation of automated decisions, AI-8 explainability outputs serve as supporting evidence.

Related Domain Standards

- General: ISO/IEC 22989:2022 3.5.7; ISO/IEC TS 6254:2024 (AI system explainability)
- Aviation: FAA AI Safety Roadmap (interpretability requirements)
- Medical: FDA Transparency Principles for Medical Devices with AI/ML (2021); FDA CDS Guidance (January 2026) §IV(4)
- EU: EU AI Act Article 13 (Transparency and provision of information)
- Cross-Domain: NIST AI RMF MANAGE-4.3; NIST SP 1270 (explainability and bias intersection)

2.8.1 Explainability Requirements Definition

[R.AI-8.1] EXPLAINABILITY REQUIREMENTS DEFINITION

The AI system shall define explainability requirements appropriate to each AI function's criticality and user needs.

Rationale: *Explainability requirements should scale with decision criticality; Safety-Critical AI functions require more rigorous explanation capabilities than Operational Support functions. Requirements must also consider the target audience; operators need operationally relevant explanations, while engineers may need deeper technical detail. Defining requirements upfront ensures explainability is designed in, not retrofitted.*

[V.AI-8.1] EXPLAINABILITY REQUIREMENTS DEFINITION

The AI system shall verify explainability requirements definition by inspection.

Evidence class: Deterministic.

The inspection shall confirm that explainability requirements are defined for each AI function; that they scale with function criticality and decision impact; that the target audiences and their needs are documented; and that the explainability approach for each function is justified, including how its faithfulness to the model's actual computation will be shown under AI-8.2.

Success Criteria: Every function has requirements, audiences and a justified approach with a faithfulness plan. A function without a faithfulness plan is a fail.

Declared population or set: Not applicable.

Evidence artifact: Explainability requirements specification. Cadence: One-time.

2.8.2 Decision Explanation Generation

[R.AI-8.2] DECISION EXPLANATION GENERATION

The AI system shall provide decision explanations in a format understandable to operators, and the explanations shall be faithful to the model's actual computation: an explanation method whose outputs are shown to be insensitive to the model's parameters or to the input it explains does not satisfy this sub-requirement.

Rationale: *Explanations must be comprehensible within operational time constraints, and they must be true. A post-hoc explanation that is plausible but unrelated to what the model computed induces exactly the misplaced trust AI-9.3 exists to prevent. Faithfulness is therefore tested, by sanity checks against the model, and comprehension is tested with representative operators.*

[V.AI-8.2] DECISION EXPLANATION GENERATION

The AI system shall verify decision explanation generation by demonstration and test.

Evidence class: Bounded.

The demonstration shall confirm that explanations are generated for AI decisions and recommendations, that the format suits the operational context and timing constraints, and that operator comprehension is validated through usability assessment with representative operators performing representative tasks under realistic conditions.

The test shall confirm faithfulness: that the explanation method's outputs change when the model's parameters are randomized and when the explained input is perturbed in the features the explanation names (randomization and deletion sanity checks), or, for an inherently interpretable surrogate, that its fidelity to the model on the test partition is measured and reported.

Success Criteria: Comprehension validated with representative operators; faithfulness sanity checks passed, or surrogate fidelity reported above the derived threshold. A method that passes comprehension and fails the sanity checks is a fail.

Declared population or set: The declared explanation evaluation set and operator sample (Section 1.7).

Evidence artifact: Explanation usability and faithfulness report. Cadence: One-time.

2.8.3 Confidence Indication

[R.AI-8.3] CONFIDENCE INDICATION

The AI system shall indicate confidence levels for AI outputs.

Rationale: *AI outputs vary in reliability: some predictions are made with high confidence, others are uncertain. Operators need to know when to trust AI outputs and when to apply additional scrutiny or seek alternative information. Confidence must be calibrated; a stated 90% confidence should be correct approximately 90% of the time, to enable appropriate operator trust calibration. ISO 22989 Section 3.5.8 defines predictability as enabling reliable assumptions about outputs; calibrated confidence supports this property.*

[V.AI-8.3] CONFIDENCE INDICATION

The AI system shall verify confidence indication by test.

Evidence class: Statistical.

The test shall confirm that confidence metrics are computed for AI outputs and presented to operators in a format that supports decision-making; that calibration error is estimated with an interval on the test partition and separately on the near-boundary out-of-distribution stratum of Section 1.7, against the threshold derived under Section 1.5; and that low-confidence outputs are distinguished from high-confidence outputs by visual or auditory differentiation.

Success Criteria: Calibration error interval below the derived threshold on the test partition; the near-boundary result reported with its interval and the operator guidance for that regime stated. An in-envelope interval containing the threshold is insufficient evidence; a point estimate above it is a fail.

Declared population or set: The test partition and the near-boundary stratum (Section 1.7).

Evidence artifact: Calibration report with reliability data. Cadence: One-time; repeated on model update.

Related Domain Standards

- General: ISO/IEC 22989:2022 3.5.8 (Predictability); ISO/IEC TS 6254:2024 §6 (Confidence communication)
- Aviation: FAA AI Safety Roadmap (Interpretability and uncertainty quantification)
- Automotive: SAE J3016:2021 §5.3 (Automation reliability feedback to user)
- Medical: FDA AI/ML Transparency Principles (2021) Principle 3 (Labeling for transparency); FDA CDS Guidance (January 2026) §IV(4) (Criterion 4: basis and evidence supporting recommendations)
- EU: EU AI Act Article 13(3)(b)(iv) (Information on level of accuracy and reliability)
- Cross-Domain: NIST AI RMF MEASURE-2.5

2.8.4 Reasoning Inspection Capability

[R.AI-8.4] REASONING INSPECTION CAPABILITY

The AI system shall enable operators to query or inspect AI reasoning for safety-critical decisions.

Rationale: *For safety-critical decisions, operators may need to understand AI reasoning in greater depth than routine explanations provide. On-demand inspection capability allows operators to drill down into AI reasoning when uncertainty exists or when AI recommendations conflict with operator expectations. This capability supports the human decision authority required by AI-9.1 (Human Decision Authority) and aligns with ISO 22989 Section 3.5.6 controllability requirements.*

[V.AI-8.4] REASONING INSPECTION CAPABILITY

The AI system shall verify reasoning inspection capability by test.

Evidence class: Deterministic.

The test shall confirm that query or inspection capability is available for every safety-critical AI function; that operators can access detail beyond the routine explanation on demand; that inspection completes within the declared timing budget; and that the capability is documented in operator procedures.

Success Criteria: Capability present for every safety-critical function; inspection within budget in every trial; procedures document it. A safety-critical function without the capability is a fail.

Declared population or set: Every safety-critical AI function.

Evidence artifact: Reasoning inspection test record. Cadence: One-time.

2.8.5 Public Disclosure Support

[R.AI-8.5] PUBLIC DISCLOSURE SUPPORT

The AI system shall provide the artifacts and information necessary to support public disclosure obligations applicable to the deployment context, in a form suitable for stakeholder consumption outside the development organization.

Rationale: *AI systems may be subject to public disclosure requirements including external model cards or system factsheets, consumer-facing information under EU AI Act Article 13, federal AI use case inventory reporting under Executive Order 14110 and OMB M-24-10, and public registration for high-risk systems under EU AI Act Article 71. The specific disclosure obligations depend on deployment context and generally bind the deploying organization rather than the developer; however, the AI system design must support those obligations by producing standardized disclosure artifacts consumable by regulators, users, data subjects, and the public. This requirement complements the internal explainability requirements (AI-8.1 through AI-8.4) by establishing the external-facing disclosure surface. Disclosure artifacts should be derived from existing framework evidence (ODD declaration per AI-1.0, bias evaluation per AI-2, performance characterization per AI-4, explainability outputs per AI-8.1-8.4) rather than generated independently, to ensure external disclosures remain consistent with internal verification evidence. This sub-requirement is optional by classification and may be documented as Not Applicable with rationale where no public disclosure obligation applies.*

[V.AI-8.5] PUBLIC DISCLOSURE SUPPORT

The AI system shall verify public disclosure support by inspection.

Evidence class: Deterministic.

The inspection shall confirm that applicable public disclosure obligations are identified in the ODD declaration under Regulatory and Operational Authorization; that disclosure artifact formats are defined; that each required artifact is populated from existing framework evidence rather than generated in parallel; that artifacts are reviewed before release for consistency with the ODD declaration, documented limitations and verification evidence; that update procedures keep the artifacts consistent with the current system state after updates, remediation, reclassification or ODD revision; and that artifacts are retained for the regime's retention period.

Success Criteria: Every identified obligation has a defined, populated and reviewed artifact with an update procedure. An obligation without an artifact is a fail. Where no obligation applies, AI-8.5 is Not Applicable with rationale.

Declared population or set: Not applicable.

Evidence artifact: Disclosure artifacts and review records. Cadence: Periodic.

Related Domain Standards

- General: ISO/IEC 23894:2023 Annex A.12 (Transparency and explainability to stakeholders); ISO/IEC 5259-4 §7 (Data quality process for ML); ISO/IEC 42001:2023 §8.3 (Communication)
- Aviation: FAA AI Safety Roadmap (stakeholder transparency)
- Automotive: NHTSA AV 4.0 element 2 (Operational Design Domain description); NHTSA AV Transparency and Engagement Principles
- Medical: FDA Predetermined Change Control Plan Guidance (2024) §V (Policy for Predetermined Change Control Plans, including labeling requirements at lines 844-887 of the guidance); FDA Transparency for Machine Learning-Enabled Device Guidance (2021)
- EU: EU AI Act Article 13 (Transparency and provision of information to deployers); EU AI Act Article 50 (Transparency obligations for general-purpose AI); EU AI Act Article 71 (Public EU database registration for high-risk AI)
- US Federal: Executive Order 14110 Section 10.1(b) (Federal AI inventory); OMB M-24-10 (Federal AI use case inventory requirements); NIST AI 100-1 (AI Risk Management Framework)
- Cross-Domain: NIST AI RMF GOVERN-1.4 (Risk documentation); NIST AI RMF MANAGE-4.1 (Post-deployment monitoring disclosure)

2.9 AI-9: HUMAN-AI TEAMING

[R.AI-9] HUMAN-AI TEAMING

The AI system will support effective human-AI collaboration, enabling operators to maintain situational awareness, appropriate trust calibration, and final decision authority over safety-critical actions.

AI systems in safety-critical operations operate as part of a human-machine team, not as fully autonomous agents. ISO 22989 Section 5.13 defines automation levels

ranging from full human control to full autonomy; for safety-critical systems, human oversight remains essential. Operators must understand what the AI is doing, trust it appropriately (neither too much nor too little), and retain authority to override AI actions when necessary. This requirement addresses the "operators cannot verify reasoning" gap and ensures AI augments rather than replaces human judgment in safety-critical decisions.

Related Domain Standards

- General: ISO/IEC 22989:2022 3.5.6 (Controllability); §5.13 (Automation levels)
- Aviation: 14 CFR 25.1302 (Installed systems and equipment for use by the flightcrew); FAA AC 25.1322 (Flight Crew Alerting); FAA AI Safety Roadmap (human oversight)
- Automotive: SAE J3016:2021 §5 (Automation levels); NHTSA Automated Vehicles Comprehensive Plan (2021)
- Medical: FDA CDS Guidance (January 2026) §IV(4) (Criterion 4: independent review); HHS/ONC Health IT Certification
- EU: EU AI Act Article 14 (Human oversight); EU AI Act Article 26 (Deployer obligations)
- Cross-Domain: NIST AI RMF GOVERN-1.5; NIST AI RMF MANAGE-2.1

2.9.1 Human Decision Authority

[R.AI-9.1] HUMAN DECISION AUTHORITY

The AI system shall maintain human decision authority over safety-critical actions: execution shall require human authorization except where the hazard analysis (AI-1.5) shows that the time to effect of the hazard is shorter than the time a qualified operator needs to review and authorize, in which case the action is governed by the runtime assurance architecture of AI-14 at Safety-Critical and by the defined fallback at Mission-Critical, and the condition is recorded in the Applicability Determination Record.

Rationale: *Humans retain final authority over actions that could cause harm, equipment damage or mission loss. The exception for autonomous execution exists because some hazards act faster than any operator can respond, and in those cases authority is exercised through the design of the monitor and fallback rather than through a real-time approval that cannot happen. The exception is therefore established by the hazard analysis case by case, not declared by the applicant.*

[V.AI-9.1] HUMAN DECISION AUTHORITY

The AI system shall verify human decision authority by test.

Evidence class: Deterministic.

The test shall confirm that every safety-critical action requires human authorization before execution unless the Applicability Determination Record carries a time-to-effect condition for it from the hazard analysis; that each conditioned action is governed by AI-14 (Safety-Critical) or the defined fallback (Mission-Critical); that human override is always available and functional; and that authority boundaries are documented and enforced through technical controls.

Success Criteria: Every autonomous safety-critical action has a hazard-analysis time-to-effect condition on record; every unconditioned action requires authorization in test; override functions in every trial. An autonomous action without a recorded condition is a fail.

Declared population or set: Every safety-critical action identified under AI-1.5.

Evidence artifact: Authority boundary record and override test results. Cadence: One-time.

Related Domain Standards

- General: ISO/IEC 22989:2022 3.5.6 (Controllability)
- Aviation: 14 CFR 25.1302 (Flight crew installed systems); FAA AC 25.1322 §4 (Alert prioritization and integration)
- Automotive: SAE J3016:2021 §5.2 (Dynamic Driving Task fallback); NHTSA AV 4.0 Voluntary Safety Self-Assessment element 3 (Human Machine Interface)
- Medical: FDA Clinical Decision Support Software Guidance (January 2026) §IV(4) (Criterion 4: basis for practitioner independent review)
- EU: EU AI Act Article 14(4) (Human oversight measures for high-risk AI); EU AI Act Article 14(5) (oversight of certain biometric categorization)
- Cross-Domain: NIST AI RMF GOVERN-1.5; NIST AI RMF MANAGE-2.1

2.9.2 Operational Situational Awareness

[R.AI-9.2] OPERATIONAL SITUATIONAL AWARENESS

The AI system shall support operator situational awareness of AI state and behavior.

Rationale: *Operators cannot effectively supervise AI systems they do not understand. Situational awareness requires visibility into what the AI is doing, what it is recommending, and why. Changes in AI state (e.g., mode transitions, confidence drops, detected anomalies) must be communicated to operators to prevent surprise and enable proactive response.*

[V.AI-9.2] OPERATIONAL SITUATIONAL AWARENESS

The AI system shall verify operational situational awareness by test.

Evidence class: Deterministic.

The test shall confirm that AI operational state is visible to operators through appropriate displays; that actions and recommendations are communicated within the declared operational timing constraints; that changes in AI state or confidence are annunciated within the declared budgets, which are derived under Section 1.5 from the human factors analysis; and that operators can determine what the AI is doing and why from the displays and explanations.

Success Criteria: Every declared state change and confidence change annunciated within its budget in every trial; operators correctly state AI status and reasoning in the declared proportion of trials. A missed annunciation is a fail.

Declared population or set: The declared annunciation scenario set and operator sample (Section 1.7).

Evidence artifact: Situational awareness test record. Cadence: One-time.

2.9.3 Trust Calibration

[R.AI-9.3] TRUST CALIBRATION

The AI system shall support appropriate trust calibration between operators and AI.

Rationale: *Miscalibrated trust either over-trust or under-trust degrades human-AI team performance. Over-trust leads to automation complacency where operators fail to catch AI errors; under-trust leads to operators ignoring valid AI recommendations. Trust calibration requires that operators understand AI reliability and limitations, that training addresses appropriate trust levels, and that system design promotes calibrated trust through transparency and appropriate confidence communication.*

[V.AI-9.3] TRUST CALIBRATION

The AI system shall verify trust calibration by analysis and demonstration.

Evidence class: Bounded.

The analysis shall confirm that AI reliability and limitations, including the declared ODD bounds, the Section 1.5 thresholds and the residuals carried on the certificate, are documented and communicated to operators, and that training materials address scenarios in which the AI should and should not be trusted.

The demonstration shall confirm that the design discourages over-trust (uncertainty indication, limitation warnings, faithful explanations per AI-8.2) and under-trust (reliability demonstration, calibrated confidence), and that trust calibration is assessed through human factors evaluation with representative operators under realistic conditions, including scenarios in which the AI is wrong.

Success Criteria: Operators in the evaluation sample detect the injected AI errors at or above the declared rate and accept correct recommendations at or above the declared rate. Either rate below its declared value is a fail; the result holds for the evaluated scenarios.

Declared population or set: The declared evaluation scenario set, including wrong-AI scenarios, and the operator sample (Section 1.7).

Evidence artifact: Trust calibration evaluation report. Cadence: One-time.

2.9.4 Graceful Degradation

[R.AI-9.4] GRACEFUL DEGRADATION

The AI system shall degrade gracefully when AI capabilities are reduced or unavailable. For Safety-Critical systems the degraded mode is the assured fallback of AI-14; for other classifications the degraded modes shall be defined and consistent with the System Inputs degradation behaviors declared under AI-1.0.

Rationale: *AI systems may experience capability reduction due to sensor failures, computational resource constraints, detected anomalies, or explicit mode transitions. When AI capabilities are degraded, operators must be notified and manual or backup procedures must be available to continue operations. Abrupt AI failure without graceful degradation could leave operators without situational awareness or control capability. Degraded mode definitions shall be consistent with the System Inputs degradation behaviors declared under AI-1.0.*

[V.AI-9.4] GRACEFUL DEGRADATION

The AI system shall verify graceful degradation by test.

Evidence class: Deterministic.

The test shall confirm that degraded modes are defined for each AI function; that operators are notified of capability reduction through appropriate annunciation;

that manual or backup procedures are available and documented; and that transition to each degraded mode completes within the declared operator-action and time budgets derived under Section 1.5.

Success Criteria: Every injected capability loss reaches its degraded mode within budget with the operator notified. One transition over budget is a fail.

Declared population or set: The injected capability-loss scenario set (Section 1.7).

Evidence artifact: Degradation test record. Cadence: One-time.

2.9.5 Human-AI Interaction Logging

[R.AI-9.5] HUMAN-AI INTERACTION LOGGING

The AI system shall log human-AI interactions for post-mission analysis.

Rationale: *Comprehensive logging of human-AI interactions supports post-mission analysis to identify human factors issues, assess trust calibration, evaluate override patterns, and improve future AI systems. Logs should capture what the AI recommended, what the human decided, and the rationale for any overrides. This data is essential for continuous improvement of human-AI teaming effectiveness.*

[V.AI-9.5] HUMAN-AI INTERACTION LOGGING

The AI system shall verify human-ai interaction logging by inspection.

Evidence class: Deterministic.

The inspection shall confirm that human-AI interactions are logged with timestamps; that logs capture the AI recommendation, the human decision and the rationale for any override; that the log properties of Section 4.6.4 (immutability, retrievability, retention) apply; and that the logs support post-mission human factors analysis including trust calibration assessment and override pattern analysis.

Success Criteria: Every interaction in a sampled period logged with all three elements; Section 4.6.4 properties present. A sampled interaction without a recorded decision is a fail.

Declared population or set: A declared sample of operational periods.

Evidence artifact: Interaction logs. Cadence: Continuous.

2.9.6 Operational Role Verification

[R.AI-9.6] OPERATIONAL ROLE VERIFICATION

The AI system shall implement periodic verification that the system's operational role remains consistent with its classification.

Rationale: *AI system classification (Section 1.2.2) determines which requirements apply and at what stringency. Classification is assigned based on the intended operational role of the AI system: whether its outputs are advisory, operational, or safety critical. However, operational usage can drift over time without any change to the system itself. A system classified as Operational Support may produce outputs that operators gradually elevate from advisory to authoritative, relying on AI recommendations as though they were directives. This usage drift is particularly insidious because no technical change triggers a review. The model performs identically; only the human reliance on it has changed. When this occurs, the system is effectively operating at a higher classification tier without the corresponding requirements, verifications, or safeguards. Periodic verification of operational role ensures that classification remains aligned with actual usage and that any drift toward higher reliance triggers formal reclassification and the application of additional requirements. This is the human factors complement to the technical drift monitoring required by AI-4.*

[V.AI-9.6] OPERATIONAL ROLE VERIFICATION

The AI system shall verify operational role verification by inspection and analysis.

Evidence class: Surveillance.

The inspection shall confirm that a periodic review schedule is defined and derived from classification and operational tempo; that the review assesses how operators actually use AI outputs, not only how the system is designed to be used; that the review criteria address whether operators treat outputs as advisory, operational or authoritative and whether actual use remains within the declared ODD; and that findings are retained.

The analysis shall confirm that the review detects usage drift through named indicators (operator workflow analysis, override rate trends, decision dependency assessment); that identified drift triggers reclassification evaluation under Section 1.2.2; and that the requirements of a higher classification are applied and verified before continued operation at the elevated role.

Success Criteria: Every scheduled review held, with the three indicators reported; every drift finding carries a reclassification decision. A missed review is a fail.

Declared population or set: The operational period since the last review.

Evidence artifact: Periodic review records. Cadence: Periodic.

2.9.7 Operator Qualification

[R.AI-9.7] OPERATOR QUALIFICATION

Where the determination boundary includes the operating organization, the AI system shall require documented operator qualifications appropriate to system classification and operational role, including system-specific training, demonstrated proficiency in normal and abnormal operations, and currency requirements that remain valid during continued operation. Where it does not, this sub-requirement is recorded as a deployer obligation on the certificate and the Certification Determination Document, and is not assessed.

Rationale: *AI system safety and performance depend on operator understanding of system capabilities, limitations, and failure modes. Unqualified operators may over-trust AI outputs, fail to recognize degraded performance, or misinterpret advisory outputs as directives. Qualification rigor should scale with classification: Safety-Critical AI requires formal operator certification, Mission-Critical AI requires role-based qualification, and Operational Support AI requires documented familiarization. Currency requirements prevent qualification decay during extended operation and ensure operators remain proficient following system updates that materially change AI behavior.*

[V.AI-9.7] OPERATOR QUALIFICATION

The AI system shall verify operator qualification by inspection and test.

Evidence class: Deterministic.

The inspection shall confirm that operator qualification requirements are documented for each operational role; that qualification criteria address AI-specific topics including capabilities, documented limitations (Section 1.5 thresholds, AI-1.0 ODD bounds, certificate residuals), known failure modes (AI-6 OOD, AI-7 adversarial, AI-4 drift) and trust calibration (AI-9.3); that currency requirements are defined with derived intervals; and that qualification records are retained.

The test shall confirm that every operator assigned holds current qualification for the role; that operators without it are prevented from assuming the role by administrative or technical controls; and that re-validation is triggered when material system updates affect operator-relevant behavior.

Success Criteria: Every assigned operator current; control prevents unqualified assumption of role in test. One unqualified assignment is a fail. Where the boundary excludes the operating organization, the deployer-obligation marking is present and this verification is Not Applicable.

Declared population or set: Every operator assigned to the system at assessment.

Evidence artifact: Qualification records. Cadence: Periodic.

Related Domain Standards

- General: ISO/IEC 22989:2022 3.5.5 (Human-AI collaboration); ISO/IEC 25059:2023 §5 (Quality characteristics for AI systems)
- Aviation: 14 CFR 121 Subpart N (Training Program); 14 CFR 121 Subpart Y (Advanced Qualification Program); FAA AI Safety Roadmap (operator training)
- Automotive: SAE J3016:2021 §8 (User considerations); NHTSA AV 4.0 element 12 (Consumer Education and Training)
- Medical: FDA 21 CFR 820.25 (Personnel - Training); FDA Clinical Decision Support Software Guidance (January 2026); note: 'Practitioner learning' is not addressed as a distinct section in the current 2026 version; concept is implicit in §IV(4) Criterion 4 discussion of HCP independent review
- EU: EU AI Act Article 4 (AI literacy); EU AI Act Article 14(4)(a) (Oversight competence); EU AI Act Article 26(2) (Deployer training obligations)
- Cross-Domain: NIST AI RMF GOVERN-2.2 (Accountability for staff); NIST SP 800-53 Rev. 5 AT-3 (Role-Based Training); NIST SP 800-53 Rev. 5 PS-3 (Personnel Screening)

2.9.8 Workload Management

[R.AI-9.8] WORKLOAD MANAGEMENT

Where the determination boundary includes the operating organization, the AI system shall define operational workload parameters and implement design measures that maintain operator cognitive capacity within documented limits under nominal, degraded and high-tempo conditions. Where it does not, the design measures remain assessed and the operational parameters are recorded as a deployer obligation on the certificate.

Rationale: *AI deployment can increase or decrease operator cognitive workload depending on interface design, alert logic, and the degree to which AI outputs require operator verification. Excess workload degrades decision quality, delays responses to abnormal conditions, and increases error probability. Insufficient workload induces complacency and reduces attentional resources available for monitoring AI outputs, which is a documented precursor to automation-induced failures. Workload must be validated as part of the operational design rather than discovered in deployment; this requirement complements trust calibration (AI-9.3)*

by ensuring the operator retains the cognitive margin required to exercise judgment and complements operator qualification (AI-9.7) by establishing the environment in which qualified operators function.

[V.AI-9.8] WORKLOAD MANAGEMENT

The AI system shall verify workload management by analysis and test.

Evidence class: Bounded.

The analysis shall confirm that workload parameters are defined for representative nominal, degraded and high-tempo scenarios; that cognitive workload limits are documented with reference to an accepted assessment method or domain guidance; that automation-bias and complacency risks are addressed in the design; and that workload monitoring indicators are specified.

The test shall confirm that measured workload remains within the documented limits in every declared scenario; that mitigations are invoked when indicators exceed their thresholds; and that operators retain the capacity to exercise the decision authority of AI-9.1.

Success Criteria: Measured workload within limits in every declared scenario; mitigations invoked at every injected exceedance. One scenario over the limit without mitigation is a fail; the result holds for the declared scenarios.

Declared population or set: The declared workload scenario set and operator sample (Section 1.7).

Evidence artifact: Workload assessment report. Cadence: One-time.

Related Domain Standards

- General: ISO 9241-11 (Usability: Definitions and concepts); ISO 11064 (Ergonomic design of control centres); ISO/IEC 25059:2023 §5 (AI system quality characteristics)
- Aviation: 14 CFR 25.1302 (Flightcrew workload); FAA AC 25.1302-1 (Installed Systems and Equipment for Use by the Flightcrew); FAA AI Safety Roadmap (cognitive workload)
- Automotive: NHTSA Visual-Manual Driver Distraction Guidelines (2013); SAE J3016:2021 §5.3 (Driver readiness); SAE J2944 (Driving performance terms)
- Medical: FDA Human Factors and Usability Engineering Guidance (2016) §6.3.1 (Task Analysis); AAMI HE75:2009/(R)2018 (Human factors design)

- EU: EU AI Act Article 14(4)(d) (Measures to counter automation bias); Council Directive 89/391/EEC Article 6 (General obligations for safe working conditions)
- Cross-Domain: NIST AI RMF MEASURE-2.8 (Human factors); MIL-STD-1472H (DoD Human Engineering Design Criteria); NUREG-0711 Rev. 3 (Human Factors Engineering Program Review)

2.9.9 Training Program Requirements

[R.AI-9.9] TRAINING PROGRAM REQUIREMENTS

Where the determination boundary includes the operating organization, the AI system shall be supported by a training program that provides initial qualification, recurrent training and scenario-based exercises covering AI system capabilities, documented limitations, failure modes, the faithfulness limits of its explanations, and appropriate trust calibration. Where it does not, this sub-requirement is recorded as a deployer obligation on the certificate and is not assessed.

Rationale: *Training programs for AI systems must address considerations that differ from traditional system training: non-deterministic behavior, performance variability across operational conditions, AI-specific failure modes such as OOD inputs and adversarial attacks, and the calibration between operator reliance and demonstrated AI performance. Initial qualification alone is insufficient because AI system behavior may evolve with updates, data drift, or operational context. Recurrent training maintains proficiency during extended operation, and scenario-based exercises expose operators to rare failure modes before they occur operationally. Training program requirements complement operator qualification (AI-9.7) by defining the content and cadence that produce qualified operators, and complement workload management (AI-9.8) by ensuring operators possess the knowledge required to recognize and respond to high-workload conditions.*

[V.AI-9.9] TRAINING PROGRAM REQUIREMENTS

The AI system shall verify training program requirements by inspection.

Evidence class: Deterministic.

The inspection shall confirm that an initial qualification curriculum is documented and covers capabilities, documented limitations (Section 1.5 thresholds, AI-1.0 ODD bounds, certificate residuals), known failure modes (AI-6 OOD, AI-7 adversarial, AI-4 drift), the explanation method's known failure modes (AI-8.2), and trust calibration guidance (AI-9.3); that recurrent training is scheduled at derived intervals; that scenario-based exercises cover nominal and rare off-nominal conditions; that effectiveness is measured by operator performance, not attendance; that material

system updates trigger targeted training before release; and that training records support AI-9.7 currency.

Success Criteria: Curriculum covers every named topic; recurrent interval derived; effectiveness measured by performance; update-triggered training present in the record. A missing topic is a fail. Where the boundary excludes the operating organization, the deployer-obligation marking is present and this verification is Not Applicable.

Declared population or set: Not applicable.

Evidence artifact: Training program records. Cadence: Periodic.

Related Domain Standards

- General: ISO/IEC 22989:2022 5.5 (Operation and monitoring); ISO/IEC 27002:2022 §6.3 (Information security awareness, education and training)
- Aviation: 14 CFR 121 Subpart Y (Advanced Qualification Program); FAA AC 120-35D (Line Operational Simulations); FAA AI Safety Roadmap §4 (Operator proficiency)
- Automotive: NHTSA AV 4.0 element 12 (Consumer education); SAE J3016:2021 §8.3 (User training for higher levels of driving automation)
- Medical: FDA 21 CFR 820.25 (Personnel training); FDA Clinical Decision Support Software Guidance (January 2026); note: dedicated 'Practitioner learning and proficiency' section does not exist in the current 2026 version; concept is implicit in §IV(4)
- EU: EU AI Act Article 4 (AI literacy); EU AI Act Article 26(2) (Deployer obligations, competence of natural persons assigned oversight)
- Cross-Domain: NIST AI RMF GOVERN-2.2 (Accountability for staff); NIST SP 800-53 Rev. 5 AT-2 (Literacy Training), AT-3 (Role-Based Training), AT-4 (Training Records)

2.10 AI-10: PRIVACY AND DATA PROTECTION

[R.AI-10] PRIVACY AND DATA PROTECTION

The AI system will implement personal data protection mechanisms that enable lawful processing, respect data subject rights, and minimize privacy risk throughout the AI lifecycle.

AI systems frequently process personal data: patient records in medical AI, biometric data in identity systems, behavioral data in recommendation systems, sensor-derived personal information in autonomous platforms. The handling of this data is governed by privacy regulations distinct from information classification regimes: GDPR in the EU, HIPAA in US healthcare, CCPA/CPRA in California, PIPEDA in Canada, and analogous frameworks globally. AI systems also introduce privacy risks not present in traditional software: training data may be reconstructable from model parameters (model inversion), trained models may leak membership information about training subjects, and the opacity of AI decisions complicates the exercise of data subject rights. This requirement addresses personal data handling throughout the AI lifecycle, complementing the information classification handling established in AI-15.

Applicability: AI-10 applies to AI systems that process personal data as defined under any applicable privacy framework. Where the system does not process personal data, AI-10 sub-requirements may be documented as Not Applicable with rationale. Determination of personal data processing shall be documented in the ODD declaration (AI-1.0) under Subject Population and Regulatory and Operational Authorization.

Related Domain Standards

- General: ISO/IEC 27701:2019 (Privacy Information Management); ISO/IEC 29100:2011 (Privacy framework)
- Medical: HIPAA Privacy Rule (45 CFR 164 Subpart E); HIPAA Security Rule (45 CFR 164 Subpart C)
- EU: GDPR (Regulation 2016/679); EU AI Act Article 10(5) (special categories of personal data)
- US State: CCPA/CPRA (Cal. Civ. Code §§ 1798.100 et seq.)
- Cross-Domain: NIST Privacy Framework v1.0; NIST SP 800-53 Rev. 5 PT family; NIST SP 800-122 (PII)

2.10.1 Personal Data Identification and Minimization

[R.AI-10.1] PERSONAL DATA IDENTIFICATION AND MINIMIZATION

The AI system shall identify personal data within all training, validation, test, and operational datasets, and shall apply data minimization principles to limit personal data collection, retention, and processing to what is necessary for the declared purpose.

Rationale: *Personal data identification is a prerequisite for any privacy control. Without knowing what personal data is present, no other privacy requirement can be satisfied. Direct identifiers (name, email, government ID, biometric data, medical record number) are typically straightforward to detect; indirect identifiers (combinations of attributes that enable re-identification) require more sophisticated assessment. Data minimization, processing only what is necessary for the declared purpose, is foundational to GDPR Article 5(1)(c), CCPA reasonable necessity provisions, and HIPAA minimum necessary standard. AI systems often violate minimization implicitly by retaining all collected data for potential future training; explicit minimization requires documented justification for each data category retained.*

[V.AI-10.1] PERSONAL DATA IDENTIFICATION AND MINIMIZATION

The AI system shall verify personal data identification and minimization by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that every personal data element processed is identified in a data inventory with its category, source, purpose and legal basis.

The analysis shall confirm that each element is necessary for the declared purpose, that elements not necessary are excluded or minimized, and that the inventory is complete against the data flows of the determination boundary.

Success Criteria: Every data flow element in the inventory with purpose and necessity stated; zero elements processed without an inventory entry. An unlisted element found in the data flows is a fail.

Declared population or set: All data flows within the determination boundary.

Evidence artifact: Personal data inventory and minimization analysis. Cadence: One-time.

2.10.2 Lawful Basis and Consent Management

[R.AI-10.2] LAWFUL BASIS AND CONSENT MANAGEMENT

The AI system shall document the lawful basis for personal data processing under each applicable privacy framework, and shall implement consent management mechanisms where consent is the lawful basis.

Rationale: *Personal data processing requires a lawful basis under all major privacy frameworks. GDPR Article 6 enumerates six lawful bases (consent, contract, legal obligation, vital interests, public task, legitimate interests); HIPAA distinguishes*

covered entity processing, business associate processing, and authorization-based processing; CCPA provides a notice-and-opt-out framework with specific bases for sensitive personal information. The lawful basis selected determines the obligations that follow: consent-based processing requires consent records, withdrawal mechanisms, and re-consent triggers; legitimate-interests-based processing requires a documented balancing test. AI systems trained on data collected under one lawful basis may be unable to be retrained on the same data if the basis changes; documenting basis throughout the lifecycle is necessary for ongoing lawful operation.

[V.AI-10.2] LAWFUL BASIS AND CONSENT MANAGEMENT

The AI system shall verify lawful basis and consent management by inspection.

Evidence class: Deterministic.

The inspection shall confirm that a lawful basis is documented for each processing purpose under the regime the ODD declaration identifies; that consent, where it is the basis, is recorded with its scope, time and withdrawal mechanism; and that processing outside the recorded basis is prevented by control.

Success Criteria: Every purpose has a documented basis; every consent record carries scope, time and withdrawal. A purpose without a basis is a fail.

Declared population or set: Every processing purpose in the inventory.

Evidence artifact: Lawful basis and consent records. Cadence: One-time.

2.10.3 Data Subject Rights Implementation

[R.AI-10.3] DATA SUBJECT RIGHTS IMPLEMENTATION

The AI system shall implement procedures to support data subject rights as required by applicable privacy frameworks, including rights of access, rectification, erasure, restriction, portability, and objection.

Rationale: *Data subject rights are a defining feature of modern privacy frameworks. GDPR Articles 15 through 22 establish access, rectification, erasure, restriction, portability, objection, and rights regarding automated decision-making. HIPAA establishes patient access rights and amendment rights under 45 CFR 164.524 and 164.526. CCPA establishes access, deletion, and opt-out rights. AI systems present unique implementation challenges: the right to erasure may require model retraining if personal data influenced learned parameters; the right to access may require explanation of automated decisions; the right to rectification of training data does not retroactively correct already-trained model behavior. These challenges require explicit procedural design rather than ad-hoc*

handling. Where data subject rights require explanation of automated decisions, AI-8 (Explainability) outputs serve as supporting evidence.

[V.AI-10.3] DATA SUBJECT RIGHTS IMPLEMENTATION

The AI system shall verify data subject rights implementation by inspection and test.

Evidence class: Deterministic.

The inspection shall confirm that procedures exist for each data subject right the identified regime grants (access, rectification, erasure, portability, objection, explanation of automated decisions), with response timelines meeting the regime's.

The test shall confirm that a request of each type, exercised against the system, is fulfilled within the regime's timeline, including erasure or exclusion of the subject's data from retraining and retrieval where the regime requires it.

Success Criteria: Every right type fulfilled in test within timeline. One unfulfilled right type is a fail.

Declared population or set: One test request per right type the regime grants.

Evidence artifact: Data subject rights procedures and test log (Part 2 C.6). Cadence: One-time; exercised at surveillance.

2.10.4 Privacy-Enhancing Techniques

[R.AI-10.4] PRIVACY-ENHANCING TECHNIQUES

The AI system shall apply privacy-enhancing techniques appropriate to the personal data sensitivity, processing purpose, and re-identification risk of the system.

Rationale: *Privacy-enhancing techniques (PETs) reduce privacy risk through technical design rather than procedural controls alone. Anonymization removes the link between data and data subjects; pseudonymization replaces identifiers with reversible tokens; differential privacy adds calibrated noise to prevent membership inference; federated learning trains models without centralizing raw data; secure aggregation enables computation without exposing inputs; encryption protects data at rest, in transit, and increasingly in computation. The selection of PETs depends on the threat model, the utility requirements, and the regulatory regime. PETs are not interchangeable: anonymization that meets HIPAA Safe Harbor may not meet GDPR anonymization standards; differential privacy parameters that satisfy academic definitions may not satisfy regulatory expectations. Where differential privacy is applied as defense against model inversion attacks, see also*

AI-7.3 (Model Inversion and Extraction Protection); AI-10.4 addresses PETs as design choices for data subject privacy, while AI-7.3 addresses inversion as an adversarial threat. The same technical mechanism may satisfy both requirements.

[V.AI-10.4] PRIVACY-ENHANCING TECHNIQUES

The AI system shall verify privacy-enhancing techniques by analysis and test.

Evidence class: Bounded.

The analysis shall confirm that the privacy-enhancing techniques applied (de-identification, differential privacy, federated or encrypted computation) are selected from the privacy impact assessment of AI-10.6, that their parameters are stated (for differential privacy, the privacy budget), and that the cryptographic standard declared under Section 1.6 is met.

The test shall confirm that the technique performs as parameterized: re-identification attempts on a declared attack set fail at the stated rate, or the privacy budget accounting is reproduced.

Success Criteria: Parameters stated; re-identification rate on the declared set at or below the stated bound, or budget accounting reproduced. A technique without a stated parameter is a fail; the result is bounded to the declared attack set.

Declared population or set: The declared re-identification attack set (Section 1.7).

Evidence artifact: PET analysis and test report. Cadence: One-time.

2.10.5 Cross-Border Data Transfer Controls

[R.AI-10.5] CROSS-BORDER DATA TRANSFER CONTROLS

The AI system shall identify and control cross-border transfers of personal data using mechanisms appropriate to the source and destination jurisdictions.

Rationale: *Personal data transferred across jurisdictional boundaries is subject to additional regulatory requirements. GDPR Chapter V restricts transfers outside the European Economic Area to jurisdictions with adequacy decisions, transfers under Standard Contractual Clauses, transfers under Binding Corporate Rules, or transfers under enumerated derogations. Other jurisdictions impose analogous restrictions (e.g., China PIPL, India DPDP, Brazil LGPD). AI systems often involve transfers that are not obvious: cloud-hosted training infrastructure, third-party annotation services, regional inference endpoints, model weights derived from trans-jurisdictional training data. Each transfer point must be identified and brought under an appropriate transfer mechanism, or the system must be architected to avoid the transfer.*

[V.AI-10.5] CROSS-BORDER DATA TRANSFER CONTROLS

The AI system shall verify cross-border data transfer controls by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that every cross-border transfer of personal data is identified with its destination jurisdiction and transfer mechanism.

The analysis shall confirm that each mechanism is valid under the identified regime and that technical controls prevent transfers outside the recorded set.

Success Criteria: Every transfer in the inventory with a valid mechanism; control prevents unrecorded transfers in test. An unrecorded transfer is a fail.

Declared population or set: All data flows crossing a jurisdictional boundary.

Evidence artifact: Transfer inventory and control record. Cadence: One-time.

2.10.6 Privacy Impact Assessment

[R.AI-10.6] PRIVACY IMPACT ASSESSMENT

The AI system shall conduct a Privacy Impact Assessment (PIA) or Data Protection Impact Assessment (DPIA) where required by applicable regulation, addressing privacy risks across the AI lifecycle and documenting risk mitigation measures.

Rationale: *Privacy impact assessment is mandated for high-risk processing under GDPR Article 35 and recommended as a best practice across most modern privacy frameworks. AI systems frequently meet GDPR DPIA triggering criteria: systematic evaluation of personal aspects (Article 35(3)(a)), large-scale processing of special categories (Article 35(3)(b)), and systematic monitoring (Article 35(3)(c)). Beyond regulatory compliance, PIAs surface privacy risks that would otherwise be invisible: re-identification risk in anonymized datasets, membership inference susceptibility, function creep beyond original purpose, and downstream privacy harms from automated decisions. The PIA is the integrating artifact that ties together AI-10's other sub-requirements.*

[V.AI-10.6] PRIVACY IMPACT ASSESSMENT

The AI system shall verify privacy impact assessment by inspection.

Evidence class: Deterministic.

The inspection shall confirm that a privacy impact assessment exists where the identified regime or the processing's risk requires one; that it covers the AI-specific

privacy risks of Section 5.2.2 (re-identification, membership inference, inversion, inferred attribute disclosure); that each identified risk has a mitigation traceable to AI-10.4 or AI-7.3; and that the assessment is current under change control.

Success Criteria: Every Section 5.2.2 privacy risk addressed; every identified risk mitigated or accepted with rationale; assessment current. A risk without disposition is a fail.

Declared population or set: Not applicable.

Evidence artifact: Privacy impact assessment (Part 2 C.5). Cadence: One-time; re-verified on change.

2.10.7 Privacy Compliance Audit Trail

[R.AI-10.7] PRIVACY COMPLIANCE AUDIT TRAIL

The AI system shall maintain an audit trail of privacy-relevant events including data subject rights requests and dispositions, consent grants and withdrawals, cross-border transfer authorizations, identified privacy incidents, and PIA reviews and updates.

Rationale: *Regulatory compliance under GDPR, HIPAA, CCPA, and analogous frameworks requires demonstrable evidence of proper privacy handling. GDPR Article 30 specifies records of processing activities; HIPAA 164.316 requires policies and procedures documentation with retention; CCPA requires verification records for data subject requests. Beyond compliance, the privacy audit trail supports incident scoping, regulatory inquiries, and demonstration of accountability. This requirement is the privacy-domain complement to the classification compliance audit trail established in AI-15.4; the two audit trails serve distinct regulatory regimes and are maintained separately even where the underlying logging infrastructure is shared.*

[V.AI-10.7] PRIVACY COMPLIANCE AUDIT TRAIL

The AI system shall verify privacy compliance audit trail by inspection.

Evidence class: Deterministic.

The inspection shall confirm that processing activities, consent events, data subject requests and privacy incidents are logged with the properties of Section 4.6.4 and retained for the period the identified regime requires.

Success Criteria: Every event type logged in a sampled period; retention meets the regime. A missing event type is a fail.

Declared population or set: A declared sample of operational periods.

Evidence artifact: Privacy compliance logs. Cadence: Continuous.

Section 3: Architecture and Paradigm Requirements

Section 3 establishes requirements that apply when an AI system is built using specific architectures or follows specific learning paradigms. Unlike Section 2 (Core Requirements), which applies to every AI system regardless of underlying technology, the requirements in this section apply conditionally based on the system's design choices.

Architecture refers to structural design choices including multi-model coordination, neural network use, ensemble methods, foundation model use, and other architectural patterns that introduce specific failure modes beyond those addressed by the algorithm-agnostic core.

Paradigm refers to learning or operational pattern choices including continuous learning, federated learning, online learning, and other paradigms that introduce failure modes related to how the system learns, adapts, or operates over time.

The architecture, paradigm and classification requirement sets are AI-11 (Multi-Model Systems), AI-12 (Neural Networks), AI-13 (Continuous Learning and Adaptation), AI-14 (Runtime Assurance, mandatory at Safety-Critical and available by tailoring-in at Mission-Critical) and AI-15 (Regulated Information Handling, where an information classification or export control regime applies). Future revisions of this framework may add sets as new architectures or paradigms introduce failure modes of their own. Sub-requirement identifiers are stable within a major version; a major version may renumber, and the identifier map in the Document Revision History maps the identifiers of version 3.7 to this version.

3.1 AI-11: MULTI-MODEL SYSTEMS REQUIREMENTS

[R.AI-11] MULTI-MODEL SYSTEMS REQUIREMENTS

The AI system will, when implemented as a multi-model system, manage inter-model coordination, combined confidence representation, cascading failure mitigation, and operator-facing display of multi-model decisions in a manner that preserves overall system reliability and operator interpretability.

Multi-model systems combine outputs from two or more AI models. Examples include a perception model feeding a planning model, multiple specialized classifiers contributing to a fused decision, ensemble architectures with voting or weighted aggregation, and retrieval-augmented generation pipelines. These

systems exhibit failure modes that single-model systems do not, including coordination drift, conflicting outputs, error propagation, combined-confidence misrepresentation, and correlated failures. The following sub-requirements apply to AI systems that incorporate two or more interacting AI models.

Related Domain Standards

- General: ISO/IEC 22989:2022; ISO/IEC 5338:2023
- Aerospace: NPR 7150.2D §4 (Software Lifecycle Processes); NASA-STD-8739.8B (Software Assurance); neither directly addresses multi-model AI architecture; framework-defined
- Aviation: SAE ARP4754B §4 (Development assurance for systems and items); SAE ARP6983/EUROCAE ED-324 (forward reference, Q1 2026)
- Automotive: ISO 26262-6:2018 §7 (Software architectural design); ISO 21448:2022 §5 (Specification and design)
- Medical: IEC 62304:2006+A1:2015 §5.3 (Software architectural design); FDA AI-Enabled Device Software Functions guidance Section IX (Model Description and Development)
- Industrial: ISO 13849-1 (Functional safety of control systems); not multi-model-specific; framework-defined for multi-model AI

3.1.1 Multi-Model Architecture Documentation

[R.AI-11.1] MULTI-MODEL ARCHITECTURE DOCUMENTATION

The AI system shall document the multi-model architecture including the role of each model, the data flow between models, the interface specifications for model-to-model data exchange, the combined system-level performance requirements, and the failure mode analysis addressing cascading and correlated failure scenarios.

Rationale: *Multi-model architectures introduce coordination concerns that single-model systems do not face. Without explicit architecture documentation, inter-model dependencies are implicit and the system cannot be assessed against the failure modes that multi-model coordination introduces.*

[V.AI-11.1] MULTI-MODEL ARCHITECTURE DOCUMENTATION

The AI system shall verify multi-model architecture documentation by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the architecture documentation identifies the role of each model, the data flow between models, the interface specification for each model-to-model exchange, the combined system-level performance requirements, and a failure mode analysis of cascading and correlated failures that is traceable to the AI-1.5 hazard analysis.

Success Criteria: Every model and every inter-model interface documented; every cascading or correlated failure mode in the analysis traces to a hazard or a documented non-credibility. An undocumented interface is a fail.

Declared population or set: Not applicable.

Evidence artifact: Multi-model architecture documentation. Cadence: One-time.

3.1.2 Cascading Failure Mitigation

[R.AI-11.2] CASCADING FAILURE MITIGATION

The AI system shall implement and document mitigations for cascading failure scenarios in chained model architectures, including error propagation between sequential models, confidence degradation across model chains, out-of-distribution propagation where upstream OOD outputs appear in-distribution to downstream models, and latency accumulation in time-critical decision paths.

Rationale: *When AI models are chained, errors can amplify or be masked, combined confidence is generally lower than individual model confidence, and OOD detection at the system entry point may not catch outputs that drift OOD between models. Latency accumulation in sequential processing affects time-critical decisions.*

[V.AI-11.2] CASCADING FAILURE MITIGATION

The AI system shall verify cascading failure mitigation by analysis and test.

Evidence class: Bounded.

The analysis shall confirm that mitigations are documented for error propagation between sequential models, confidence degradation across chains, out-of-distribution propagation where an upstream OOD output appears in-distribution downstream, and latency accumulation in time-critical paths.

The test shall confirm that each mitigation operates as designed under a declared cascading-failure scenario set, including injected upstream OOD outputs and injected latency.

Success Criteria: Every documented mitigation exercised in at least one declared scenario and operating as designed. An unexercised mitigation is a fail; the result holds for the declared scenarios.

Declared population or set: The declared cascading-failure scenario set (Section 1.7).

Evidence artifact: Cascading failure test report. Cadence: One-time.

3.1.3 Ensemble Coordination

[R.AI-11.3] ENSEMBLE COORDINATION

The AI system shall implement and document coordination mechanisms for ensemble and voting architectures, including controls against correlated failures (where models trained on similar data fail on the same inputs), documented voting threshold rationale (majority, unanimous, or weighted voting schemes), and disagreement handling when models disagree beyond defined thresholds.

Rationale: *Ensemble systems are intended to provide robustness through model diversity, but correlated failures can defeat this intent. Voting threshold selection materially affects ensemble behavior and requires justification.*

[V.AI-11.3] ENSEMBLE COORDINATION

The AI system shall verify ensemble coordination by analysis and test.

Evidence class: Statistical.

The analysis shall confirm that the coordination mechanisms include controls against correlated failures, a documented rationale for the voting threshold, and defined handling for disagreement beyond threshold.

The test shall confirm that the error correlation between ensemble members is measured on the test partition and reported with an interval, that it lies within the bound assumed by the voting rationale, and that the disagreement handling operates in every declared disagreement scenario.

Success Criteria: Measured error correlation within the assumed bound on its whole interval; disagreement handling operates in every scenario. An interval containing the bound is insufficient evidence.

Declared population or set: The test partition and the declared disagreement scenario set (Section 1.7).

Evidence artifact: Ensemble coordination report. Cadence: One-time.

3.1.4 Combined Confidence Representation

[R.AI-11.4] COMBINED CONFIDENCE REPRESENTATION

The AI system shall compute and display combined confidence values that represent the aggregate uncertainty across all participating models, with the dependence between models either measured and reflected in the combination or stated as an assumption whose effect on the combined value is bounded. Multi-model confidence displays shall include the combined system-level confidence clearly labeled, individual model contributions visible on demand, visual indication when models disagree beyond defined thresholds, and a clear mapping between confidence levels and recommended operator action.

Rationale: *Operators interacting with multi-model systems need to understand both the aggregate confidence and the contribution of each model to diagnose unusual outputs and exercise appropriate authority. Cascaded model architectures generally compute combined confidence as the product of individual confidences (assuming independence) or via Bayesian propagation; ensemble architectures may use weighted average, minimum (most conservative), or ensemble-specific calibration. The presentation method must support operator interpretation.*

[V.AI-11.4] COMBINED CONFIDENCE REPRESENTATION

The AI system shall verify combined confidence representation by test.

Evidence class: Statistical.

The test shall confirm that combined confidence is computed by the documented method; that the method's treatment of inter-model dependence is either measured on the test partition or stated with its bounded effect; that the combined confidence is calibrated against actual accuracy with an interval, per AI-8.3; that the display elements are present; and that the operator interface is validated in scenario-based testing including disagreement cases.

Success Criteria: Combined confidence calibration interval below the derived threshold; dependence treatment stated or measured; display elements present. An interval containing the threshold is insufficient evidence.

Declared population or set: The test partition and the declared disagreement scenario set (Section 1.7).

Evidence artifact: Combined confidence test report. Cadence: One-time.

3.1.5 Multi-Model Audit Trail

[R.AI-11.5] MULTI-MODEL AUDIT TRAIL

The AI system shall maintain an audit trail of multi-model decisions including which models contributed to each output, the contribution weights or confidence values, the aggregation method applied, and the resulting decision rationale.

Rationale: *Post-mission analysis, incident investigation, and certification surveillance require the ability to reconstruct multi-model decisions including the contribution of each constituent model.*

[V.AI-11.5] MULTI-MODEL AUDIT TRAIL

The AI system shall verify multi-model audit trail by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the audit trail records, for each output, the contributing models, their contribution weights or confidence values, the aggregation method applied and the resulting decision rationale, with the log properties of Section 4.6.4 and retention per AI-15.4 where a classification regime applies.

Success Criteria: Every sampled output reconstructable from the trail to its contributing models and weights. One unreconstructable output is a fail.

Declared population or set: A declared sample of outputs.

Evidence artifact: Multi-model audit trail. Cadence: Continuous.

3.2 AI-12: NEURAL NETWORK REQUIREMENTS

[R.AI-12] NEURAL NETWORK REQUIREMENTS

The AI system will, when implemented using neural network architectures, address the specific failure modes of neural architectures including confidence miscalibration, the absence of inspectable decision logic, training integrity concerns, adversarial vulnerability, generative hallucination, and transformation-induced behavioral drift.

Neural network architectures include deep learning models, convolutional neural networks, recurrent neural networks, transformer-based systems, and generative neural models. While the algorithm-agnostic requirements in Section 2 apply to any AI system, neural networks present specific failure modes, including the absence of inspectable decision logic, prevalence of confidence miscalibration, susceptibility to adversarial inputs, sensitivity to model transformations, and hallucination in

generative outputs, that warrant dedicated requirements. The following sub-requirements apply to AI systems that incorporate neural network components. AI-12.2 (Confidence Calibration) and AI-12.4 (Training Integrity) additionally apply to AI systems implemented with other statistical machine learning models, including gradient-boosted trees, random forests, and support vector machines, which exhibit the same miscalibration and overfitting failure modes; for such systems the remaining AI-12 sub-requirements apply only where neural network components are present.

Related Domain Standards

- General: ISO/IEC 22989:2022; ISO/IEC 5338:2023
- Aerospace: NPR 7150.2D §4 (Software Lifecycle Processes); NASA-STD-8739.8B (Software Assurance); neither neural-network-specific; framework-defined
- Aviation: DO-178C §5.2 (Software Design Process); SAE ARP6983/EUROCAE ED-324 (forward reference, Q1 2026, AI/ML aviation)
- Automotive: ISO 26262-6:2018 §7 (Software architectural design); emerging ISO/PAS 8800 (Road Vehicles - Safety and AI) when published
- Medical: FDA AI-Enabled Device Software Functions guidance Section IX (Model Description and Development); directly addresses neural network architecture documentation
- Industrial: ISO 13849-1 (Functional safety of control systems); not neural-network-specific; framework-defined

3.2.1 Neural Network Architecture Documentation

[R.AI-12.1] NEURAL NETWORK ARCHITECTURE DOCUMENTATION

The AI system shall document neural network architecture details in addition to standard software documentation, including layer types, dimensions, and connectivity; activation functions and normalization schemes; total parameter count and memory footprint; computational requirements (FLOPs) for inference; optimizer configuration, learning rate schedule, and regularization; batch size and training duration; data augmentation techniques; hardware and framework versions used; accuracy metrics across data subsets and operational scenarios; calibration curves for confidence outputs; latency and throughput on target hardware; known failure modes and limitations; and the deployment artifact chain with traceability and cryptographic hashes at each transformation step.

Rationale: *Neural network behavior depends on architectural choices, training configuration, and deployment transformations in ways not present in traditional software. Comprehensive architecture documentation is required for assessor evaluation and post-mission analysis.*

[V.AI-12.1] NEURAL NETWORK ARCHITECTURE DOCUMENTATION

The AI system shall verify neural network architecture documentation by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the documentation records the network architecture (layer types, dimensions, connectivity, activation and normalization schemes, parameter count, memory footprint, inference computational requirements), the training configuration (optimizer, learning rate schedule, regularization, batch size, training duration, data augmentation, hardware and framework versions), the performance characterization (accuracy across data subsets and operational scenarios, calibration data, latency and throughput on target hardware, known failure modes and limitations), and the deployment artifact chain with traceability and cryptographic hashes at each transformation step.

Success Criteria: Every named element present and current with the deployed artifact's hash. A missing element or a stale hash is a fail.

Declared population or set: Not applicable.

Evidence artifact: Neural network architecture documentation. Cadence: One-time; updated on change.

3.2.2 Confidence Calibration

[R.AI-12.2] CONFIDENCE CALIBRATION

The AI system shall demonstrate that model confidence outputs are calibrated against actual prediction accuracy. Calibration techniques such as temperature scaling, Platt scaling, or isotonic regression shall be applied where calibration error exceeds thresholds established per Section 5.1 (Threshold Selection Guidance), and calibration shall be validated on representative held-out data.

Rationale: *Neural network confidence scores (typically softmax outputs) are often poorly calibrated. A model reporting 90% confidence may be correct only 70% of the time. Operators relying on AI confidence to make decisions are misled by uncalibrated confidence outputs.*

[V.AI-12.2] CONFIDENCE CALIBRATION

The AI system shall verify confidence calibration by analysis and test.

Evidence class: Statistical.

The analysis shall confirm that calibration error is measured against actual accuracy, that the threshold is derived under Section 1.5, and that any calibration technique applied (temperature scaling, Platt scaling, isotonic regression) is justified and fitted on the validation partition only.

The test shall confirm that calibration error on the test partition lies within the threshold with its interval, and that calibration on the near-boundary out-of-distribution stratum of Section 1.7 is reported with its interval.

Success Criteria: Test-partition calibration interval below the derived threshold; near-boundary result reported. An interval containing the threshold is insufficient evidence; a point estimate above it is a fail.

Declared population or set: The test partition and the near-boundary stratum (Section 1.7).

Evidence artifact: Calibration analysis and test report. Cadence: One-time; repeated on model update.

3.2.3 Neural Network Explainability Methods

[R.AI-12.3] NEURAL NETWORK EXPLAINABILITY METHODS

The AI system shall implement explainability methods appropriate to the neural network architecture and decision criticality, with method selection documented and justified. For Safety-Critical AI functions, the AI system shall document the rationale for selecting neural network architectures over inherently interpretable alternatives (decision trees, rule-based systems, linear models) when interpretable alternatives could achieve required performance.

Rationale: *Neural networks present inherent explainability challenges because decision logic is distributed across learned weights rather than explicit rules. Post-hoc methods such as SHAP, LIME, attention visualization, and layer-wise relevance propagation provide explanations of individual predictions but do not constitute proof of correct decision logic. For high-criticality functions, the choice to use a neural network rather than an interpretable alternative is itself a design decision requiring justification.*

[V.AI-12.3] NEURAL NETWORK EXPLAINABILITY METHODS

The AI system shall verify neural network explainability methods by inspection and test.

Evidence class: Bounded.

The inspection shall confirm that the explainability methods are documented with selection rationale appropriate to the architecture and decision criticality, and that any Safety-Critical use of a neural network over an inherently interpretable alternative that could achieve the required performance is justified per AI-8.

The test shall confirm that the implemented methods generate explanations for individual predictions and that they pass the faithfulness sanity checks of AI-8.2 (randomization and deletion), or that an interpretable surrogate's fidelity to the network is measured and reported.

Success Criteria: Methods documented and justified; faithfulness checks passed or surrogate fidelity reported above the derived threshold. A method that fails the sanity checks is a fail.

Declared population or set: The declared explanation evaluation set (Section 1.7).

Evidence artifact: Explainability method report. Cadence: One-time.

3.2.4 Training Integrity

[R.AI-12.4] TRAINING INTEGRITY

The AI system shall implement and document training integrity controls including: documented overfitting indicators and the threshold beyond which retraining is required; documented relationship between model architecture complexity and training data volume; documented random seeds and training configurations to enable reproducibility; and validation set contamination controls beyond [R.AI-1.1] (Dataset Separation) and [R.AI-1.2] (Partition Use Restriction), specifically addressing hyperparameter tuning on validation data and architecture decisions informed by validation performance.

Rationale: *Neural network and other statistical machine learning training introduces risks not present in traditional software development. Models with sufficient capacity can memorize training data rather than learning generalizable patterns. Hyperparameter tuning effectively incorporates validation information into the model. Stochastic training processes introduce reproducibility concerns. These risks require specific controls beyond the algorithm-agnostic dataset partitioning requirements.*

[V.AI-12.4] TRAINING INTEGRITY

The AI system shall verify training integrity by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that training integrity controls are documented: overfitting indicators and the retraining threshold, the relationship between architecture complexity and training data volume, random seeds and configurations enabling reproducibility, and validation set contamination controls beyond AI-1.1 and AI-1.2 addressing hyperparameter tuning and architecture decisions made on validation performance.

The analysis shall confirm from the training records that the documented controls were applied and that a reproduction from the recorded seeds and configuration yields the deployed artifact's reported performance within the declared tolerance.

Success Criteria: Every control evidenced in the training records; reproduction within tolerance. A reproduction outside tolerance is a fail.

Declared population or set: Not applicable; a deterministic check on the training records.

Evidence artifact: Training integrity records. Cadence: One-time.

3.2.5 Neural Network Out-of-Distribution Detection Extensions

[R.AI-12.5] NEURAL NETWORK OUT-OF-DISTRIBUTION DETECTION EXTENSIONS

The AI system shall implement out-of-distribution detection methods appropriate to neural network architectures, recognizing that softmax confidence alone is insufficient for OOD detection. OOD detection for neural networks shall use methods that account for the tendency of neural networks to produce high-confidence predictions on inputs far from the training distribution, such as ensemble disagreement, energy-based methods, feature-space distance metrics, or deep generative model likelihood.

Rationale: *Neural networks often produce high softmax confidence scores on inputs outside the training distribution. Standard OOD detection per [R.AI-6.2] requires neural-network-specific extensions to be effective.*

[V.AI-12.5] NEURAL NETWORK OUT-OF-DISTRIBUTION DETECTION EXTENSIONS

The AI system shall verify neural network out-of-distribution detection extensions by test.

Evidence class: Statistical.

The test shall confirm that the out-of-distribution detection method used for the neural network is one that accounts for high-confidence predictions far from the training distribution (ensemble disagreement, energy-based methods, feature-space distance, deep generative likelihood, or another method with a stated validity

condition); that softmax confidence alone is shown on the declared strata to be insufficient, with its result reported; and that the chosen method meets the AI-6.2 criterion on the near-boundary stratum with an interval.

Success Criteria: Chosen method's near-boundary detection interval above the derived threshold at the false-positive budget; softmax result reported. An interval containing the threshold is insufficient evidence.

Declared population or set: The declared far and near out-of-distribution strata (Section 1.7).

Evidence artifact: Neural network OOD detection report. Cadence: One-time; repeated on model update.

3.2.6 Adversarial Vulnerability Mitigation

[R.AI-12.6] ADVERSARIAL VULNERABILITY MITIGATION

The AI system shall implement adversarial robustness measures specific to neural network attack classes, evaluated against gradient-based attacks, gradient-free attacks and, for Safety-Critical systems, adaptive attacks designed against the implemented defence, at the perturbation budget declared under AI-7.2. Adversarial defences shall be evaluated for gradient masking, where a defence appears to reduce gradient-based attack success without improving robustness.

Rationale: *Neural networks are uniquely susceptible to adversarial examples: inputs with imperceptible perturbations that cause dramatic misclassification. This vulnerability is fundamental to gradient-based learning and cannot be fully eliminated. Some adversarial defenses create a false sense of security by masking gradients rather than improving robustness, requiring evaluation with gradient-free attacks. Neural-network-specific adversarial measures supplement [R.AI-7.2] (Adversarial Input Manipulation Protection).*

[V.AI-12.6] ADVERSARIAL VULNERABILITY MITIGATION

The AI system shall verify adversarial vulnerability mitigation by test.

Evidence class: Bounded.

The test shall confirm that robustness at the declared budget is evaluated against gradient-based and gradient-free attack classes and, at Safety-Critical, against adaptive attacks; that apparent robustness is not attributable to gradient masking; and that residual vulnerability is documented with the budget.

Success Criteria: Degradation within the derived margin at the budget against every required attack class; gradient masking excluded; residual stated. A missing attack class is a fail; the result holds at the budget.

Declared population or set: The declared attack pattern set and budget (Section 1.7).

Evidence artifact: Neural network adversarial test report. Cadence: One-time; repeated when the threat model or the model changes.

3.2.7 Generative Model Hallucination Constraints

[R.AI-12.7] GENERATIVE MODEL HALLUCINATION CONSTRAINTS

For neural networks that generate outputs (text, trajectories, plans, code, images), the AI system shall implement domain-specific hallucination constraints including physical plausibility checks (conservation laws, kinematic limits, domain-specific bounds), cross-validation with independent sensors or models where available, confidence-bounded outputs (refusing to generate beyond confidence thresholds), and output format constraints that limit the space of generable outputs to verifiable forms.

Rationale: *Generative neural models can produce plausible-seeming outputs that are factually incorrect or physically infeasible. Hallucination detection per [R.AI-5] requires generative-model-specific constraints to be effective in safety-critical contexts.*

[V.AI-12.7] GENERATIVE MODEL HALLUCINATION CONSTRAINTS

The AI system shall verify generative model hallucination constraints by analysis and test.

Evidence class: Deterministic.

The analysis shall confirm that the documented constraints include physical plausibility checks (conservation laws, kinematic limits, domain-specific bounds), cross-validation with independent sensors or models where available, confidence-bounded outputs, and output format constraints that limit generable outputs to verifiable forms.

The test shall confirm that every output in a declared generated-output set that violates a constraint is rejected before delivery, and that the constrained output space is as documented.

Success Criteria: One hundred per cent of constraint-violating outputs in the declared set rejected. One delivered violation is a fail.

Declared population or set: The declared generated-output test set (Section 1.7).

Evidence artifact: Generative constraint test report. Cadence: One-time.

3.2.8 Deployment Transformation Validation (Neural Network Considerations)

[R.AI-12.8] DEPLOYMENT TRANSFORMATION VALIDATION (NEURAL NETWORK CONSIDERATIONS)

Neural network deployment transformations shall be validated per [R.AI-4.7] (Deployment Format Validation), with neural-network-specific attention to: quantization sensitivity (per-layer impact analysis, post-quantization calibration on representative data); pruning effects (validation that pruned models do not exhibit different failure modes than original models); knowledge distillation (validation that distilled student models preserve teacher model behavior on out-of-distribution inputs and edge cases); and framework conversion (numerical equivalence validation between source and target frameworks with documented acceptable tolerances).

Rationale: *Neural networks are particularly sensitive to deployment-format transformations. Reducing precision can cause catastrophic accuracy degradation in some layers, pruning can introduce different failure modes, distillation may not preserve out-of-distribution behavior, and framework conversion can introduce numerical differences due to operator implementation variations. AI-4.7 establishes the general normative requirement; this sub-requirement specifies the neural-network-specific concerns.*

[V.AI-12.8] DEPLOYMENT TRANSFORMATION VALIDATION (NEURAL NETWORK CONSIDERATIONS)

The AI system shall verify deployment transformation validation (neural network considerations) by test, per [V.AI-4.7] (Deployment Format Validation).

Evidence class: Statistical.

The test shall demonstrate the validation evidence required by [V.AI-4.7], with neural-network-specific attention to quantization sensitivity (per-layer impact analysis and post-quantization calibration on representative data), pruning effects (confirmation that the pruned model does not exhibit failure modes the original does not), knowledge distillation (confirmation that the student preserves the teacher's behavior on the near-boundary out-of-distribution stratum and the declared edge-case sets), and framework conversion (numerical equivalence between source and target within documented tolerances).

Success Criteria: Every AI-4.7 criterion met on the deployed network; distillation preserves teacher behavior on the near-boundary stratum within the derived margin; numerical equivalence within tolerance. A new failure mode in the pruned model is a fail.

Declared population or set: The test partition over the declared grid, the near-boundary stratum and the edge-case sets (Section 1.7).

Evidence artifact: Neural network transformation validation report. Cadence: One-time per transformation.

3.3 AI-13: CONTINUOUS LEARNING AND ADAPTATION REQUIREMENTS

[R.AI-13] CONTINUOUS LEARNING AND ADAPTATION REQUIREMENTS

The AI system will, when implementing continuous learning or adaptation during operations, control the conditions under which learning occurs, validate what is being learned, preserve the ability to detect and recover from learning that produces degraded behavior, and maintain transparency over the system's learning history.

Continuously-learning systems update their model parameters during operation based on observed data. ISO/IEC 22989:2022 5.11.9.2 defines continuous learning as incremental training that occurs during operation. While continuous learning enables AI systems to adapt to operational environments and address drift (improving mission effectiveness when properly controlled), it introduces failure modes that statically-trained systems do not exhibit, including catastrophic forgetting, learning data drift, online learning instability, and the inability to verify the operational model against a fixed validation baseline. The following sub-requirements apply to AI systems implementing continuous learning during operations.

Related Domain Standards

- General: ISO/IEC 22989:2022 5.11.9.2; ISO/IEC 5338:2023
- Aerospace: NPR 7150.2D §4 (Software Lifecycle Processes); NASA-STD-8739.8B (Software Assurance); current NASA guidance heavily restricts post-deployment learning; framework-defined for continuous learning
- Aviation: FAA AI Safety Roadmap, p.10 ("Differentiate Between Learned AI and Learning AI" principle); directly addresses learning-AI safety assurance methodology

- Automotive: ISO 21448:2022 §13 (Operation phase activities) for field-monitoring of evolving systems; UNECE WP.29 R157 §5.2 for post-market ALKS changes
- Medical: FDA AI-Enabled Device Software Functions guidance Section III (TPLC Approach: General Principles); FDA Predetermined Change Control Plan (PCCP) Guidance; central FDA framework for post-market AI/ML changes
- Industrial: ISO 13849-1 (Functional safety of control systems); does not address continuous learning; framework-defined

3.3.1 Continuous Learning Permission Criteria

[R.AI-13.1] CONTINUOUS LEARNING PERMISSION CRITERIA

The AI system shall declare the conditions under which continuous learning is permitted, including the system classification tier, the operational scenarios in which learning may occur, the data sources from which learning may proceed, and the controls required at each tier. For Safety-Critical systems, learning shall not modify the certified learned component in operation; learning is permitted in shadow mode only, and a shadow-learned update becomes operational only as a new certified configuration under the change provisions of the assessment methodology. Mission-Critical AI continuous learning requires performance monitoring per AI-4, automated rollback at defined thresholds, logging of all model updates, and periodic review of the learning trajectory. Operational Support AI continuous learning requires baseline performance tracking and anomaly alerting for significant behavior changes.

Rationale: *Continuous learning has different risk profiles by classification. At Safety-Critical, a certificate attaches to a fixed configuration (Section 1.6) and no authority in the framework's target domains accepts learning in service for a safety function; the framework therefore permits learning to run beside the certified component and not inside it, and treats each accepted update as a configuration change with its own determination.*

[V.AI-13.1] CONTINUOUS LEARNING PERMISSION CRITERIA

The AI system shall verify continuous learning permission criteria by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the permission criteria declare the classification tier, the permitted scenarios, the permitted data sources and the controls at each tier; that at Safety-Critical the design prevents in-operation modification of the

certified learned component and confines learning to shadow mode; and that the tier controls named in the requirement are present and enforced by system controls.

Success Criteria: Every declared element present; at Safety-Critical, the technical control preventing in-operation modification is identified and tested under AI-13.2. A Safety-Critical design that can modify the certified component in operation is a fail.

Declared population or set: Not applicable.

Evidence artifact: Continuous learning permission record. Cadence: One-time.

3.3.2 Runtime Learning Controls

[R.AI-13.2] RUNTIME LEARNING CONTROLS

The AI system shall implement runtime controls that prevent learning when defined preconditions are not met, including controls for safety-critical operational windows, controls when learning data sources are suspect, and controls when system confidence is below validated thresholds.

Rationale: *Continuous learning permitted under all conditions risks incorporating poisoned data, learning during anomalous operational states, or amplifying low-confidence behavior. Runtime controls operationalize the permission criteria declared under AI-13.1.*

[V.AI-13.2] RUNTIME LEARNING CONTROLS

The AI system shall verify runtime learning controls by test.

Evidence class: Deterministic.

The test shall confirm that runtime controls prevent learning when defined preconditions are not met, including during safety-critical operational windows, when learning data sources are suspect and when system confidence is below validated thresholds; and, at Safety-Critical, that an attempted in-operation modification of the certified learned component is blocked and logged.

Success Criteria: Learning blocked in every injected precondition violation; the Safety-Critical modification attempt blocked and logged. One unblocked case is a fail.

Declared population or set: The declared precondition-violation scenario set (Section 1.7).

Evidence artifact: Runtime learning control test record. Cadence: One-time.

3.3.3 Learning Data Validation

[R.AI-13.3] LEARNING DATA VALIDATION

The AI system shall validate learning data quality before incorporating it into model updates, including data freshness (recency relative to operational distribution), source legitimacy (trusted data origin), and absence of poisoning indicators (anomaly detection on learning data prior to use).

Rationale: *Learning data quality directly determines what the model learns. Without validation, continuous learning is vulnerable to data poisoning, distribution shift in learning data, and incorporation of stale data that misrepresents current operational conditions.*

[V.AI-13.3] LEARNING DATA VALIDATION

The AI system shall verify learning data validation by test and analysis.

Evidence class: Deterministic.

The analysis shall confirm that validation criteria address data freshness relative to the operational distribution, source legitimacy, and poisoning indicators applied before use.

The test shall confirm that learning data failing any of the three checks in a declared seeded set is rejected before incorporation.

Success Criteria: One hundred per cent of seeded failing samples rejected. One incorporated failing sample is a fail.

Declared population or set: The declared seeded learning-data set (Section 1.7).

Evidence artifact: Learning data validation test report. Cadence: One-time.

3.3.4 Catastrophic Forgetting Mitigation

[R.AI-13.4] CATASTROPHIC FORGETTING MITIGATION

The AI system shall implement and verify mitigations against catastrophic forgetting (loss of previously-validated capabilities through new learning), using methods such as elastic weight consolidation, progressive neural networks, replay buffers, or other techniques appropriate to the model architecture and learning paradigm.

Rationale: *Per ISO/IEC 22989:2022, continuous learning systems are susceptible to catastrophic forgetting where new learning overwrites previously acquired*

knowledge. Without mitigation, continuous learning can degrade validated capabilities even as it improves performance on new tasks.

[V.AI-13.4] CATASTROPHIC FORGETTING MITIGATION

The AI system shall verify catastrophic forgetting mitigation by test.

Evidence class: Statistical.

The test shall confirm that the implemented mitigations (elastic weight consolidation, progressive networks, replay buffers, or others appropriate to the architecture) retain previously validated capabilities after a declared sequence of learning episodes, with performance on the original validation and edge-case sets estimated with intervals before and after.

Success Criteria: Post-learning performance interval on every original set within the derived margin of the pre-learning value. An interval containing the margin is insufficient evidence; a point estimate outside it is a fail.

Declared population or set: The original validation and edge-case sets and the declared learning-episode sequence (Section 1.7).

Evidence artifact: Forgetting mitigation test report. Cadence: Periodic.

3.3.5 Learning Validation

[R.AI-13.5] LEARNING VALIDATION

The AI system shall validate continuous learning updates against benchmark performance criteria before the updated model becomes the operational model. Updates that fail validation shall be rejected and the prior validated model retained.

Rationale: *Without pre-activation validation, learning updates that produce degraded behavior become operational before the degradation is detected. Pre-activation validation is the gate that prevents this.*

[V.AI-13.5] LEARNING VALIDATION

The AI system shall verify learning validation by test.

Evidence class: Statistical.

The test shall confirm that continuous learning updates are validated against the benchmark suite before becoming operational; that updates failing validation are rejected with the prior validated model retained; and that accepted updates meet or exceed prior performance on the suite, with intervals.

Success Criteria: Every failing update rejected; every accepted update's interval at or above the prior value on the suite. An accepted update below the prior value is a fail.

Declared population or set: The declared benchmark suite (Section 1.7).

Evidence artifact: Learning validation records. Cadence: Continuous.

3.3.6 Rollback Procedures

[R.AI-13.6] ROLLBACK PROCEDURES

The AI system shall implement rollback procedures that restore a prior validated model state when continuous learning updates produce degraded performance after activation. Rollback shall be invokable both automatically (on threshold-triggered detection) and manually (by qualified operator authority).

Rationale: *Pre-activation validation reduces but does not eliminate the risk of degraded learning outcomes reaching operations. Rollback procedures provide the recovery path when post-activation degradation is detected.*

[V.AI-13.6] ROLLBACK PROCEDURES

The AI system shall verify rollback procedures by test.

Evidence class: Deterministic.

The test shall confirm that rollback restores a prior validated model state when a learning update produces degraded performance after activation, invoked both automatically on threshold-triggered detection and manually by qualified operator authority, within the declared time budget.

Success Criteria: Both rollback paths restore the prior state within budget in every trial; the restored state matches the recorded prior artifact hash. One failed restoration is a fail.

Declared population or set: The declared rollback scenario set (Section 1.7).

Evidence artifact: Rollback test record. Cadence: One-time.

3.3.7 ODD Evolvability Declaration

[R.AI-13.7] ODD EVOLVABILITY DECLARATION

The AI system shall declare its ODD evolvability per AI-1.0, specifying whether the operational design domain is permitted to evolve through continuous learning, the bounds within which evolution may occur, and the conditions that trigger ODD reassessment.

Rationale: *Continuous learning that changes the model's effective operational envelope changes the system's ODD. Without explicit ODD evolvability declaration, the relationship between continuous learning and the validated operational envelope is implicit and unverifiable.*

[V.AI-13.7] ODD EVOLVABILITY DECLARATION

The AI system shall verify odd evolvability declaration by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the declaration specifies, per AI-1.0, whether the operational design domain may evolve through continuous learning, the bounds within which evolution may occur, and the conditions that trigger ODD reassessment, and that it is consistent with the AI-1.0 declaration and with the Safety-Critical restriction of AI-13.1.

Success Criteria: Evolvability declared and consistent with AI-1.0 and AI-13.1; bounds and triggers stated. A Safety-Critical declaration permitting in-operation ODD evolution is a fail.

Declared population or set: Not applicable.

Evidence artifact: ODD evolvability declaration. Cadence: One-time.

3.3.8 Continuous Learning Audit Trail

[R.AI-13.8] CONTINUOUS LEARNING AUDIT TRAIL

The AI system shall maintain an audit trail of all continuous learning events, including data sources used for each learning event, parameter updates applied, validation outcomes, and any rollback actions taken.

Rationale: *Post-incident analysis, certification surveillance, and operational accountability require complete reconstruction of the system's learning history.*

[V.AI-13.8] CONTINUOUS LEARNING AUDIT TRAIL

The AI system shall verify continuous learning audit trail by inspection.

Evidence class: Deterministic.

The inspection shall confirm that the audit trail records the data sources used for each learning event, the parameter updates applied, the validation outcomes and any rollback actions, with the log properties of Section 4.6.4 and retention per AI-15.4 where a classification regime applies.

Success Criteria: Every learning event in a sampled period reconstructable from the trail. One unreconstructable event is a fail.

Declared population or set: A declared sample of operational periods.

Evidence artifact: Learning event logs. Cadence: Continuous.

3.4 AI-14: RUNTIME ASSURANCE REQUIREMENTS

[R.AI-14] RUNTIME ASSURANCE REQUIREMENTS

The AI system will, when classified Safety-Critical, be bounded in operation by a runtime assurance architecture whose monitor is explicitly specified and verifiable by conventional means and whose fallback is assured under the applicable conventional standard, so that the certificate's strongest tier attests a protection a reader can check independently of the learned component's statistical evidence.

The requirement sets of Section 2 produce evidence about a learned component that is, for the most part, statistical or bounded: a rate on a declared population, a demonstration at a declared budget. That evidence supports a requirement and does not close it (Section 4.1). For a Safety-Critical classification the framework therefore requires, in addition, a protection whose correctness does not depend on the learned component at all: a monitor whose logic is written down and verifiable by the methods conventional software standards already provide, and a fallback assured under the conventional standard the applicant's domain regime imposes. The learned component is treated as untrusted; the monitor and fallback are what the Safety-Critical tier attests.

A learned component for which no conventionally assurable monitor and fallback can be built cannot be certified Safety-Critical under this framework. It may be certified at Mission-Critical where its consequence classification supports that (Section 1.2.2), or it is outside what this framework can attest. This set is mandatory at Safety-Critical and available by tailoring-in at Mission-Critical. The conventional assurance of the monitor and the fallback is evidenced by reference to the applicable domain regime; Safety Critical Labs records the standard and level on the certificate and does not assess conventional assurance (Section 1.6).

Applicability: AI-14 applies to every system classified Safety-Critical. Where the determination boundary excludes the monitor and fallback, Section 1.2.1 requires that they be declared in the integrating system with their assurance evidenced by reference before a Safety-Critical determination can issue.

Related Domain Standards

- General: ISO/IEC 22989:2022 3.5.6 (Controllability); IEEE 1012-2024 (verification of the monitor and fallback as conventional software)
- Aviation: ASTM F3269 (Standard Practice for Methods to Safely Bound Flight Behavior of Unmanned Aircraft Systems Containing Complex Functions); DO-178C partitioning and independence concepts; EASA AI Concept Paper Issue 2 (safety risk mitigation objectives)
- Automotive: ISO 26262-6:2018 §7 (Software architectural design, freedom from interference); ISO 21448:2022 (Safety of the intended functionality, triggering conditions and reasonably foreseeable misuse)
- Medical: IEC 62304:2006+A1:2015 §5.3 (Software architectural design, segregation); IEC 60601-1 essential performance concepts
- Autonomy: UL 4600 (safety case treatment of fallback and minimal risk conditions); SAE J3016:2021 §5.2 (minimal risk condition and fallback)
- Cross-Domain: NIST AI RMF MANAGE-2.1; the simplex architecture literature (informative)

3.4.1 Monitor Specification

[R.AI-14.1] MONITOR SPECIFICATION

The runtime assurance monitor shall be explicitly specified: its logic shall be written as a conventional specification, shall not be a learned artifact, shall be verifiable by the analysis and test methods of the applicable conventional software standard, shall cover the hazards assigned to the learned component under AI-1.5, and shall not share the learned component's failure modes (training data, sensor inputs, processing hardware or software dependencies that the learned component depends on).

Rationale: The monitor is the element whose correctness the Safety-Critical tier actually rests on, so it must be the kind of element conventional assurance knows how to verify. A learned monitor would reintroduce the problem it is meant to bound. Independence from the learned component's failure modes is what makes a discrepancy between the two informative.

[V.AI-14.1] MONITOR SPECIFICATION

The AI system shall verify monitor specification by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the monitor specification exists as a conventional specification with explicit logic, that no element of the monitor is a learned artifact under the Section 1.2.1 deletion test, and that the specification names each hazard it covers by reference to the AI-1.5 assignment.

The analysis shall confirm that the monitor's inputs, processing and dependencies share none of the learned component's failure-mode elements, and that the monitor is verifiable by the methods of the applicable conventional standard, with the verification evidence referenced under AI-14.5.

Success Criteria: Monitor is non-learned under the deletion test; every assigned hazard it claims is named; zero shared failure-mode elements. A learned element in the monitor or a shared element is a fail.

Declared population or set: Not applicable; a deterministic check on the monitor specification and its dependency inventory.

Evidence artifact: Monitor specification and independence analysis. Cadence: One-time; re-verified on change.

3.4.2 Assured Fallback

[R.AI-14.2] ASSURED FALLBACK

The fallback the monitor invokes shall be assured under the applicable conventional standard at the level the hazard analysis assigns to the hazards it mitigates, and the standard and level shall be recorded in the certified configuration and on the certificate. The fallback shall bring the system to a declared safe or minimal-risk condition from every state in which the monitor can invoke it.

Rationale: When the monitor switches, the fallback is the system. Its assurance is therefore the assurance of the Safety-Critical claim, and it comes from the conventional regime, by reference: Safety Critical Labs records it and does not assess it.

[V.AI-14.2] ASSURED FALLBACK

The AI system shall verify assured fallback by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the applicable conventional standard and the assurance level are stated, that evidence of the fallback's assurance at that level is referenced, and that the declared safe or minimal-risk condition is defined for each operational phase of the ODD.

The analysis shall confirm that the level matches the severity the hazard analysis assigns to the hazards the fallback mitigates, and that the fallback is reachable from every monitor-invoking state identified in AI-14.3.

Success Criteria: Standard and level stated and matched to hazard severity; assurance evidence referenced; safe condition reachable from every invoking state. An invoking state with no reachable safe condition is a fail.

Declared population or set: Every monitor-invoking state identified under AI-14.3.

Evidence artifact: Fallback assurance reference and reachability analysis. Cadence: One-time; re-verified on change.

3.4.3 Switching Logic and Timing

[R.AI-14.3] SWITCHING LOGIC AND TIMING

The switching between the learned component and the fallback shall be specified with its conditions, its detection-to-switch latency bounded against the time to effect of each covered hazard, its hysteresis and re-engagement conditions, and a manual override available to the operator.

Rationale: A monitor that detects too late, or that oscillates, protects nothing. The latency bound is derived from the hazard analysis, so the timing requirement is traceable rather than chosen.

[V.AI-14.3] SWITCHING LOGIC AND TIMING

The AI system shall verify switching logic and timing by analysis and test.

Evidence class: Bounded.

The analysis shall confirm that each switching condition is stated, that the latency budget for each covered hazard is derived from its time to effect under AI-1.5, and that hysteresis and re-engagement conditions are specified.

The test shall confirm, under a declared set of injected learned-component failures covering every assigned hazard, that the switch occurs within the latency budget in every trial, that re-engagement follows the specified conditions, and that manual override functions.

Success Criteria: Switch within budget in every trial for every covered hazard; override functions; no oscillation outside the specified hysteresis. One trial over budget is a fail; the result holds for the injected set.

Declared population or set: The declared injected-failure set covering every assigned hazard (Section 1.7).

Evidence artifact: Switching test report. Cadence: One-time; exercised at surveillance.

3.4.4 Coverage Statement

[R.AI-14.4] COVERAGE STATEMENT

The applicant shall state which hazards assigned to the learned component under AI-1.5 the monitor covers, and for each hazard it does not cover, the mitigation that applies elsewhere or the documented risk acceptance. Interaction hazards identified by the system safety process, in which the learned component behaves as specified but the system is unsafe, shall be addressed in this statement.

Rationale: A monitor covers what it was built to detect. The coverage statement is where the hazard analysis and the architecture meet, and it is where the hazards that are not component failures, the unsafe interactions and the flawed assumptions, have to be written down against a mitigation rather than left between requirements.

[V.AI-14.4] COVERAGE STATEMENT

The AI system shall verify coverage statement by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the coverage statement lists every hazard assigned under AI-1.5 with its status: covered by the monitor, mitigated elsewhere by a named requirement, or accepted with rationale.

The analysis shall confirm the two-way trace between the statement and the hazard analysis and that every interaction hazard the system safety process identified appears in it.

Success Criteria: Every assigned hazard has a status; every interaction hazard appears; two-way trace complete. One hazard without a status is a fail.

Declared population or set: The complete hazard list assigned to the learned component under AI-1.5.

Evidence artifact: Monitor coverage statement. Cadence: One-time; re-verified on change.

3.4.5 Validation Evidence

[R.AI-14.5] VALIDATION EVIDENCE

The verification evidence for the monitor and the fallback under the applicable conventional standard shall be referenced in the evidence package, and the integration of the learned component, the monitor and the fallback shall be tested under injected learned-component failures with the results recorded.

Rationale: The framework does not re-perform conventional verification; it requires that it exists and can be produced, and it verifies the one thing the conventional regime does not: the integration with the learned component.

[V.AI-14.5] VALIDATION EVIDENCE

The AI system shall verify validation evidence by inspection and test.

Evidence class: Deterministic.

The inspection shall confirm that the monitor's and the fallback's verification evidence under the applicable conventional standard is referenced with the documents, versions and configuration they cover, and that the configuration matches the certified configuration.

The test shall confirm that the integrated system, under the injected-failure set of AI-14.3, reaches the declared safe or minimal-risk condition in every trial and that the results are recorded.

Success Criteria: Evidence referenced and configuration-matched; safe condition reached in every integration trial. A configuration mismatch or a trial not reaching the safe condition is a fail.

Declared population or set: The declared injected-failure set (Section 1.7).

Evidence artifact: Conventional verification references and integration test report. Cadence: One-time; exercised at surveillance.

3.5 AI-15: REGULATED INFORMATION HANDLING REQUIREMENTS

[R.AI-15] REGULATED INFORMATION HANDLING REQUIREMENTS

The AI system will, where it or its data pipeline is subject to an information classification or export control regime, prevent the loss of classification markings through training, fine-tuning and retrieval, carry the inherited classification to its outputs and their consumers, and keep the audit trail that regime requires.

These requirements were AI-1.5 to AI-1.8 in versions 3.x. They address handling obligations under information classification and export control regimes (CUI, ITAR, EAR, or organization-specific markings) rather than failure modes that affect the safety of operation, and in 4.0 they form a conditional set so that the core sets of

Section 2 carry only the requirements that attach to every learned component. The requirement text is carried without change of meaning; the verification statements follow the 4.0 template of Section 4.1.

Applicability: AI-15 applies where the AI system or its data pipeline is subject to an information classification or export control regime. Where no such regime applies, AI-15 may be documented as Not Applicable with rationale, following the pattern established for AI-10. The retention references that other requirements make to this set (AI-1.6, AI-11.5, AI-13.8, Section 4.4) apply only where it applies.

Related Domain Standards

- Government/Defense: 32 CFR Part 2002 (CUI); 22 CFR Part 120 (ITAR definitions, including export by release to a foreign person); 22 CFR 122.5 (ITAR records); 15 CFR 762 (EAR recordkeeping)
- Cross-Domain: NIST SP 800-53 Rev. 5 AC and MP families; NIST AI RMF MAP-1 through MAP-5

3.5.1 Pre-ingestion Classification Screening

[R.AI-15.1] PRE-INGESTION CLASSIFICATION SCREENING

The AI system shall screen all data sources for information classification markings prior to ingestion into model training, fine-tuning, or retrieval-augmented generation (RAG) pipelines.

Rationale: AI training pipelines ingest data from diverse sources including government databases, technical repositories, contractor deliverables, sensor archives, and publicly available datasets. Any of these sources may contain data subject to information handling controls (CUI, ITAR, EAR, or other classification designations). Once controlled data is ingested into model training, the classification marking is permanently separated from the content. Pre-ingestion screening is the only reliable point at which to prevent classification loss, as no post-training mechanism exists to identify or remove controlled content from model parameters. This requirement establishes screening as a mandatory gate in the data pipeline, analogous to the partition integrity verification required by AI-1.1 prior to every training run.

[V.AI-15.1] PRE-INGESTION CLASSIFICATION SCREENING

The AI system shall verify pre-ingestion classification screening by inspection and analysis.

Evidence class: Statistical.

The inspection shall confirm that a documented screening process exists for all data sources entering training, fine-tuning or retrieval pipelines; that screening criteria include CUI markings per 32 CFR Part 2002, ITAR controlled technical data identifiers, EAR Export Control Classification Numbers, and any organization-specific markings; and that screening results are recorded for each source with disposition (approved, excluded, or approved with handling controls).

The analysis shall confirm that automated screening tools are validated against a declared known-positive set of controlled content with the detection rate estimated with an interval against the threshold derived under Section 1.5; that manual review procedures are defined where automated screening is insufficient; and that no data entered the pipeline without a recorded disposition.

Success Criteria: Detection interval on the known-positive set above the derived threshold; one hundred per cent of sources dispositioned. An interval containing the threshold is insufficient evidence; an undispositioned source is a fail.

Declared population or set: The declared known-positive controlled-content set (Section 1.7) and all data sources entering the pipelines.

Evidence artifact: Screening process record and tool validation report. Cadence: One-time; repeated per data source admission.

3.5.2 Information Classification Inheritance

[R.AI-15.2] INFORMATION CLASSIFICATION INHERITANCE

The AI system shall document the information classification level of all training data and shall apply the principle that AI model artifacts and outputs inherit the highest classification level of the training data unless the training pipeline demonstrates, with verifiable evidence, that controlled content was excluded.

Rationale: In traditional information systems, data classification follows the data: a CUI document copied to a new system requires the new system to handle the copy as CUI. AI systems break this principle because the data transformation from documents to model weights destroys the classification metadata while retaining the controlled content. Without an explicit inheritance rule, the default assumption becomes that AI outputs are uncontrolled, which is incorrect when the training data included controlled content. This requirement establishes classification inheritance as the default, placing the burden of proof on the pipeline to demonstrate that controlled content was excluded rather than on the consumer to determine whether generated outputs contain controlled information. This is consistent with the derivative classification principle in information security: products derived from classified sources inherit the classification of the source material.

[V.AI-15.2] INFORMATION CLASSIFICATION INHERITANCE

The AI system shall verify information classification inheritance by inspection and analysis.

Evidence class: Deterministic.

The inspection shall confirm that the classification level of every training dataset is documented in the provenance record of AI-1.3; that model artifacts (trained weights, serialized models, deployment packages) are marked with the inherited classification level; and that the inheritance determination is documented with justification, either inheriting the highest training data classification or certifying exclusion of controlled content with the AI-15.1 screening evidence.

The analysis shall confirm that the inheritance chain is complete from data sources through model artifacts to deployment endpoint; that any exclusion claim is supported by AI-15.1 evidence; and that downstream systems receiving outputs are authorized at the inherited level.

Success Criteria: Every artifact marked; chain complete end to end; every exclusion claim evidenced; every downstream system authorized. A gap in the chain is a fail.

Declared population or set: Every dataset, artifact and downstream endpoint in the certified configuration.

Evidence artifact: Classification inheritance record. Cadence: One-time; re-verified on change.

3.5.3 Output Classification Awareness

[R.AI-15.3] OUTPUT CLASSIFICATION AWARENESS

The AI system shall implement mechanisms to ensure that consumers of AI-generated outputs are aware of the information classification handling requirements applicable to those outputs.

Rationale: When an AI model trained on or exposed to controlled content generates outputs, those outputs may contain reconstructed controlled information. If the model operates at an inherited classification level per AI-15.2, all outputs must be handled at that level. However, the primary risk is that end users receive AI-generated content without any indication of applicable handling requirements. The user treats the output as uncontrolled general knowledge, copies it to uncontrolled systems, shares it with unauthorized recipients, or disseminates it to foreign persons, each constituting a potential violation of applicable regulations. This requirement ensures that classification awareness propagates to the point of

consumption, not just the point of model storage. For ITAR technical data, failure to maintain this awareness constitutes a potential deemed export violation (22 CFR 120.17). For CUI, it violates dissemination controls established by the CUI Registry.

[V.AI-15.3] OUTPUT CLASSIFICATION AWARENESS

The AI system shall verify output classification awareness by test and analysis.

Evidence class: Deterministic.

The test shall confirm that AI-generated outputs include or are accompanied by classification handling indicators appropriate to the output channel (markings on documents, banners on displays, metadata tags on programmatic outputs); that indicators are consistent with the inherited level under AI-15.2; that indicators cannot be stripped or bypassed by end users in normal operational use; and that classification metadata propagates programmatically to downstream consumers.

The analysis shall confirm that the mechanisms cover every operational output channel (display, document generation, API response, data export); that operator training addresses handling of AI-generated outputs; and that outputs to users at different authorization levels are appropriate to the requesting user's authorization.

Success Criteria: Indicator present and correct on one hundred per cent of outputs in a declared output sample across every channel; zero successful strip attempts in test; authorization respected in every trial. One unmarked output is a fail.

Declared population or set: The declared output sample across every channel and authorization level (Section 1.7).

Evidence artifact: Output classification test report. Cadence: One-time; re-verified on channel change.

3.5.4 Classification Compliance Audit Trail

[R.AI-15.4] CLASSIFICATION COMPLIANCE AUDIT TRAIL

The AI system shall maintain an audit trail documenting classification screening decisions, inheritance determinations, output classification events, and any identified or suspected classification incidents throughout the AI system lifecycle.

Rationale: Regulatory compliance for CUI, ITAR, and EAR requires demonstrable evidence of proper handling. Traditional information systems maintain access logs and handling records. AI systems require analogous records covering the AI-specific aspects of information classification: what data was screened and what was the disposition, what classification level was inherited and on what basis, what

outputs were generated at what classification level, and whether any incidents of suspected classification loss or improper dissemination occurred. Without this audit trail, an organization cannot demonstrate compliance during regulatory review, cannot scope the impact of a discovered classification incident, and cannot support the continuous improvement of classification controls. This requirement is the classification-specific complement to the human-AI interaction logging required by AI-9.5 and the OOD event logging required by Section 4.6.4.

[V.AI-15.4] CLASSIFICATION COMPLIANCE AUDIT TRAIL

The AI system shall verify classification compliance audit trail by inspection.

Evidence class: Deterministic.

The inspection shall confirm that pre-ingestion screening decisions are logged with timestamps, source identifiers, screening method and disposition; that inheritance determinations are logged with justification and authorizing authority; that output classification events are logged at a level sufficient to support incident investigation (query context, level applied, channel); that suspected incidents are logged with timestamps, discovery method, affected scope estimate and remediation; that the log properties of Section 4.6.4 apply; and that records are retained for the periods the applicable regime requires (32 CFR Part 2002 for CUI, 22 CFR 122.5 for ITAR, 15 CFR 762 for EAR).

Success Criteria: Every event type present in a sampled period with all named fields; retention meets the regime. A missing event type or field is a fail.

Declared population or set: A declared sample of operational periods.

Evidence artifact: Classification compliance audit trail. Cadence: Continuous.

Section 4: Verification

4.0 OVERVIEW

Section 4 establishes the verification methods, evidence requirements, and verification matrix that implement the [V.AI-x.y] verification statements declared under each Section 2 and Section 3 sub-requirement. The methods defined in 4.1 are applied per the matrix in 4.2; cadence (one-time, periodic, or continuous) is indicated per row to reflect lifecycle phase coverage including post-deployment Operations and Sustainment per Table 1-7. Section 4.4 elaborates the continuous and periodic verification activities required for deployed AI systems. Section 4.5 establishes the verification independence required by system classification.

4.1 VERIFICATION METHODS

Verification of AI-specific requirements uses standard methods defined in IEEE 1012-2024, with tailoring for AI/ML system characteristics.

Method	Definition	AI Application
Inspection	Examination of documentation, artifacts, or code	Dataset documentation, provenance records, log schemas
Analysis	Evaluation using mathematical modeling or analytical techniques	Bias risk assessment, coverage analysis, distribution characterization
Demonstration	Qualitative observation of system behavior under realistic conditions	Operator comprehension assessment, human-AI interaction evaluation
Test	Execution of the system or component under controlled conditions with defined inputs, producing quantitative measurements evaluated against pass/fail criteria	Drift detection, OOD detection, hallucination detection

Evidence classes. Every verification statement names the class of evidence its governing success criterion produces, because the class states what the evidence can establish. Deterministic evidence establishes a property that is true or false of a named artifact or configuration, checked by computation or by inspection of the artifact itself; it closes its requirement. Statistical evidence estimates a rate or score on a declared population (Section 1.7); it supports its requirement at a stated confidence with a stated residual and never closes it. Bounded evidence demonstrates a property under a declared budget (a perturbation size, an operating point, a scenario set); it holds at that budget and nowhere else, and the budget is recorded with the result. Surveillance evidence is an obligation discharged continuously or periodically in service; it produces evidence over time and never closes. Where a verification mixes classes, the class of the governing criterion is named.

Verification outcomes. Every verification resolves to one of four outcomes. Pass: the success criterion is met on the whole of its interval or for the whole of its declared set. Fail: the criterion is not met, or a named fail condition in the verification statement holds. Not applicable: the requirement's applicability clause excludes it, with rationale recorded. Insufficient evidence: the interval on a

statistical measurement contains the criterion, the declared population cannot support the rate claimed at the stated confidence, or a bounded demonstration's budget is below the one the hazard analysis requires. The effect of an insufficient-evidence outcome on the determination is stated by tier in Section 1.6.

The verification statement template. Every [V.AI-x.y] statement in Sections 2 and 3 carries, in order: the method or methods; the evidence class; the confirmations the method shall establish; a success criterion that discriminates, which is a number with its confidence, a named artifact with a stated property, or a named test with a pass condition, and never a restatement of the confirmations; the declared population, partition or set the criterion is measured on; the evidence artifact; and the cadence. One rule governs the method field: inspection may establish the existence and content of an artifact; it may not establish a property that requires computation. A property that requires computation is verified by analysis or test.

The residual list. The certificate carries the residual list: every requirement whose outcome is insufficient evidence and carried under Section 1.6, every bounded demonstration with its budget, and every surveillance obligation outstanding at issue. The list is the honest statement of what the evidence did not close.

4.2 VERIFICATION MATRIX SUMMARY

The verification matrix traces each sub-requirement to its verification method, evidence class, cadence, declared population or set, and evidence artifact: one row per sub-requirement under this version's identifiers, in document order, with the method and evidence class of the governing criterion, the cadence, the declared population or set the criterion is measured on, and the artifact the evidence package shall contain. It is the index an assessor reads first and the checklist derives from, and it adds nothing to the verification statements, which govern. The matrix is regenerated from the verification statements at every version; the identifier map in the Document Revision History resolves the identifiers of version 3.7.

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
AI-1.0	Operational Design Domain declaration	Inspection; Deterministic	One-time; reviewed at the AI-9.6 interval	Not applicable to the declaration itself	ODD Declaration Record (Part 2 C.4) with the trace

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
					table to downstream artifacts
AI-1.1	Dataset separation	Inspection, Analysis; Deterministic	One-time; repeated before every training run	All three partitions in full	Partition integrity report: inspection record (Part 2 C.3) and the analysis output with the counterfactual where run
AI-1.2	Partition use restriction	Inspection, Analysis; Deterministic	One-time	The complete partition access log for the development period	Access logs and partition audit record
AI-1.3	Data provenance	Inspection; Deterministic	One-time	All datasets named in the training configuration and	Provenance records
AI-1.4	Ground truth labeling quality	Inspection, Analysis; Statistical	One-time	An audit sample drawn from each partition and each declared Subject...	Labeling quality report with agreement and error-rate intervals
AI-	AI hazard	Inspection,	One-	The complete	Hazard

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
1.5	analysis integration	Analysis; Deterministic	time; re-verified on change	hazard list for the declared determination boundary	analysis extract and two-way trace table
AI-1.6	Retrieval corpus controls	Inspection, Test; Bounded	One-time; re-tested on corpus pipeline change	The declared modification-attempt set and the declared indirect…	Corpus control inspection record and injection test report (Part 2 C.2)
AI-2.1	Bias-bounded baseline performance	Analysis, Test; Statistical	One-time	The Subject Population strata and User Population roles declared…	Baseline disparity report with per-pair intervals
AI-2.2	Bias-managed training data	Inspection, Analysis; Statistical	One-time	The training partition by the declared strata	Training data representation report
AI-2.3	Training dataset validation	Inspection, Analysis; Statistical	One-time	The training partition and the declared bias pattern inventory	Training dataset validation report
AI-2.4	Continuous bias monitoring	Test; Surveillance	Continuous	Operational data by the declared strata	Bias monitoring records
AI-2.5	Bias impact assessment	Inspection, Analysis; Deterministi	One-time	The decision categories of the AI-1.5 hazard	Bias impact assessme

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
		c		analysis	nt
AI-2.6	Disparity Response	Inspection, Test; Deterministic	One-time; exercised at surveillance	The simulated exceedance set	Response procedure and test record
AI-2.7	Safety-critical Disparity Bound	Analysis, Test; Statistical	One-time; repeated at surveillance	The declared strata	Safety-critical disparity report
AI-2.8	Safety-critical bias escalation	Test; Bounded	One-time; exercised at surveillance	The injected scenario set	Escalation test record
AI-2.9	Bias edge case definition	Inspection; Deterministic	One-time	The declared edge-case and stress sets	Edge-case dataset specification
AI-2.10	Bias edge case validation	Test; Statistical	One-time	The declared edge-case and stress sets	Edge-case validation report
AI-3.1	Test coverage metric definition	Inspection, Analysis; Deterministic	One-time	The declared ODD attribute grid	Coverage metric specification
AI-3.2	Test coverage threshold achievement	Analysis, Test; Statistical	One-time	The test partition over the declared grid	Coverage report with the

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
					power statement
AI-3.3	Operational input distribution testing	Test; Statistical	One-time	The operational input distribution and grid declared under Section 1.7	Operational distribution test report
AI-3.4	Boundary and edge case testing	Test; Bounded	One-time	The declared boundary and edge-case set	Boundary test report
AI-4.1	Data drift monitoring	Test; Surveillance	Continuous; qualification one-time per monitor design	The clean qualification stream and the injected-drift stream...	Drift monitor qualification and detection report
AI-4.2	Performance threshold maintenance	Analysis, Test; Surveillance	Continuous	Operational data over the declared window	Performance monitoring records
AI-4.3	Model maintenance criteria	Inspection; Deterministic	Periodic	Not applicable	Model maintenance records
AI-4.4	Output validity boundaries	Test; Deterministic	One-time	The declared injection set	Output boundary test report
AI-4.5	Periodic model validation	Test; Statistical	Periodic	The periodic validation set declared under Section 1.7	Periodic validation reports
AI-	Operational	Inspection,	Continuo	The proxy	Proxy

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
4.6	validation without ground truth	Analysis; Surveillance	us; justificati on re-verified at surveillan ce	qualification data declared under Section 1.7	validation documen tation
AI-4.7	Deployment format validation	Test; Statistical	One-time per transfor mation	The test partition over the declared grid and the near-boundary set…	Side-by-side performa nce comparis on report
AI-5.1	Hallucination criteria definition	Inspection; Deterministi c	One-time	Not applicable	Hallucinat ion criteria documen t
AI-5.2	Hallucination detection	Test; Statistical	One-time; repeated on model update	The declared hallucination test set with its injected and…	Hallucinat ion detection test report
AI-5.3	Hallucination response	Test; Bounded	One-time; exercised at surveillan ce	The injected set drawn from the declared hallucination test set	Hallucinat ion response test record
AI-5.4	Independent output validation	Analysis, Test; Deterministi c	One-time	The injected discrepancy set covering each safety-critical and…	Independ ence analysis and discrepan cy test report

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
AI-5.5	Hallucination Audit	Analysis; Surveillance	Periodic, at each declared audit interval	The declared audit sample of passed operational outputs per interval	Hallucination audit record
AI-6.1	Training distribution characterization	Inspection, Analysis; Deterministic	One-time	The declared far and near out-of-distribution strata and the…	Training distribution characterization and test-set boundary validation record
AI-6.2	Runtime OOD detection	Test; Statistical	One-time; repeated on model update	The declared far and near out-of-distribution strata	OOD detection test report
AI-6.3	OOD response	Test; Bounded	One-time; exercised at surveillance	The injected set drawn from both declared strata	OOD response test record
AI-7.1	Data poisoning protection	Analysis, Test; Bounded	One-time; repeated when the threat model changes	The declared poisoning attack pattern set and seeded modification set	Poisoning protection test report (Part 2 C.2)
AI-7.2	Adversarial input	Test; Bounded	One-time;	The declared threat model	Adversarial

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
	protection		repeated when the threat model or the model changes		robustness test report (Part 2 C.2) with the budget and set
AI-7.3	Model inversion and extraction protection	Inspection, Analysis; Deterministic	One-time	Not applicable	Access control and privacy configuration record
AI-7.4	Model integrity	Inspection; Deterministic	Continuous	Not applicable	Model integrity verification records
AI-7.5	AI supply chain security	Inspection; Deterministic	One-time; re-swept at surveillance	The declared vulnerability sources and sweep date	Supply chain risk assessment
AI-7.6	Artifact load safety	Inspection, Test; Deterministic	One-time; repeated on loader change	Every artifact load path in the certified configuration	Load path inventory and negative test record
AI-8.1	Explainability requirements definition	Inspection; Deterministic	One-time	Not applicable	Explainability requirements specification

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
AI-8.2	Decision explanation generation	Demonstration, Test; Bounded	One-time	The declared explanation evaluation set and operator sample	Explanation usability and faithfulness report
AI-8.3	Confidence indication	Test; Statistical	One-time; repeated on model update	The test partition and the near-boundary stratum	Calibration report with reliability data
AI-8.4	Reasoning inspection capability	Test; Deterministic	One-time	Every safety-critical AI function	Reasoning inspection test record
AI-8.5	Public disclosure support	Inspection; Deterministic	Periodic	Not applicable	Disclosure artifacts and review records
AI-9.1	Human decision authority	Test; Deterministic	One-time	Every safety-critical action identified under AI-1.5	Authority boundary record and override test results
AI-9.2	Operational situational awareness	Test; Deterministic	One-time	The declared annunciation scenario set and operator sample	Situational awareness test record
AI-9.3	Trust calibration	Analysis, Demonstration; Bounded	One-time	The declared evaluation scenario set	Trust calibration evaluatio

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
					n report
AI-9.4	Graceful degradation	Test; Deterministic	One-time	The injected capability-loss scenario set	Degradation test record
AI-9.5	Human-AI interaction logging	Inspection; Deterministic	Continuous	A declared sample of operational periods	Interaction logs
AI-9.6	Operational role verification	Inspection, Analysis; Surveillance	Periodic	The operational period since the last review	Periodic review records
AI-9.7	Operator qualification	Inspection, Test; Deterministic	Periodic	Every operator assigned to the system at assessment	Qualification records
AI-9.8	Workload management	Analysis, Test; Bounded	One-time	The declared workload scenario set and operator sample	Workload assessment report
AI-9.9	Training program requirements	Inspection; Deterministic	Periodic	Not applicable	Training program records
AI-10.1	Personal data identification and minimization	Inspection, Analysis; Deterministic	One-time	All data flows within the determination boundary	Personal data inventory and minimization analysis
AI-10.2	Lawful basis and consent management	Inspection; Deterministic	One-time	Every processing purpose in the inventory	Lawful basis and consent records
AI-10.3	Data subject rights implementatio	Inspection, Test; Deterministi	One-time; exercised	One test request per right type the regime grants	Data subject rights

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
	n	c	at surveillance		procedures and test log (Part 2 C.6)
AI-10.4	Privacy-enhancing techniques	Analysis, Test; Bounded	One-time	The declared re-identification attack set	PET analysis and test report
AI-10.5	Cross-border data transfer controls	Inspection, Analysis; Deterministic	One-time	All data flows crossing a jurisdictional boundary	Transfer inventory and control record
AI-10.6	Privacy Impact Assessment	Inspection; Deterministic	One-time; re-verified on change	Not applicable	Privacy impact assessment (Part 2 C.5)
AI-10.7	Privacy compliance audit trail	Inspection; Deterministic	Continuous	A declared sample of operational periods	Privacy compliance logs
AI-11.1	Multi-model architecture documentation	Inspection; Deterministic	One-time	Not applicable	Multi-model architecture documentation
AI-11.2	Cascading failure mitigation	Analysis, Test; Bounded	One-time	The declared cascading-failure scenario set	Cascading failure test report
AI-11.3	Ensemble coordination	Analysis, Test; Statistical	One-time	The test partition and the declared disagreement	Ensemble coordination report

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
				scenario set	
AI-11.4	Combined confidence representation	Test; Statistical	One-time	The test partition and the declared disagreement scenario set	Combined confidence test report
AI-11.5	Multi-model audit trail	Inspection; Deterministic	Continuous	A declared sample of outputs	Multi-model audit trail
AI-12.1	Neural network architecture documentation	Inspection; Deterministic	One-time; updated on change	Not applicable	Neural network architecture documentation
AI-12.2	Confidence calibration	Analysis, Test; Statistical	One-time; repeated on model update	The test partition and the near-boundary stratum	Calibration analysis and test report
AI-12.3	Neural network explainability methods	Inspection, Test; Bounded	One-time	The declared explanation evaluation set	Explainability method report
AI-12.4	Training integrity	Inspection, Analysis; Deterministic	One-time	Not applicable	Training integrity records
AI-12.5	Neural network OOD detection extensions	Test; Statistical	One-time; repeated on model update	The declared far and near out-of-distribution strata	Neural network OOD detection report
AI-12.6	Adversarial vulnerability	Test; Bounded	One-time;	The declared attack pattern set	Neural network

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
	mitigation		repeated when the threat model or the model changes	and budget	adversarial test report
AI-12.7	Generative model hallucination constraints	Analysis, Test; Deterministic	One-time	The declared generated-output test set	Generative constraint test report
AI-12.8	Deployment transformation validation (neural network considerations)	Test; Statistical	One-time per transformation	The test partition over the declared grid	Neural network transformation validation report
AI-13.1	Continuous learning permission criteria	Inspection; Deterministic	One-time	Not applicable	Continuous learning permission record
AI-13.2	Runtime learning controls	Test; Deterministic	One-time	The declared precondition-violation scenario set	Runtime learning control test record
AI-13.3	Learning data validation	Test, Analysis; Deterministic	One-time	The declared seeded learning-data set	Learning data validation test report
AI-13.4	Catastrophic forgetting	Test; Statistical	Periodic	The original validation and	Forgetting

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
	mitigation			edge-case sets and the declared…	mitigation test report
AI-13.5	Learning validation	Test; Statistical	Continuous	The declared benchmark suite	Learning validation records
AI-13.6	Rollback procedures	Test; Deterministic	One-time	The declared rollback scenario set	Rollback test record
AI-13.7	ODD evolvability declaration	Inspection; Deterministic	One-time	Not applicable	ODD evolvability declaration
AI-13.8	Continuous learning audit trail	Inspection; Deterministic	Continuous	A declared sample of operational periods	Learning event logs
AI-14.1	Monitor Specification	Inspection, Analysis; Deterministic	One-time; re-verified on change	Not applicable	Monitor specification and independence analysis
AI-14.2	Assured Fallback	Inspection, Analysis; Deterministic	One-time; re-verified on change	Every monitor-invoking state identified under AI-14.3	Fallback assurance reference and reachability analysis
AI-14.3	Switching Logic and Timing	Analysis, Test; Bounded	One-time; exercised at surveillan	The declared injected-failure set covering every assigned hazard	Switching test report

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
			ce		
AI-14.4	Coverage Statement	Inspection, Analysis; Deterministic	One-time; re-verified on change	The complete hazard list assigned to the learned component under AI-1.5	Monitor coverage statement
AI-14.5	Validation Evidence	Inspection, Test; Deterministic	One-time; exercised at surveillance	The declared injected-failure set	Conventional verification references and integration test report
AI-15.1	Pre-ingestion Classification Screening	Inspection, Analysis; Statistical	One-time; repeated per data source admission	The declared known-positive controlled-content set	Screening process record and tool validation report
AI-15.2	Information Classification Inheritance	Inspection, Analysis; Deterministic	One-time; re-verified on change	Every dataset	Classification inheritance record
AI-15.3	Output Classification Awareness	Test, Analysis; Deterministic	One-time; re-verified on channel change	The declared output sample across every channel and authorization level	Output classification test report
AI-15.4	Classification Compliance	Inspection; Deterministi	Continuous	A declared sample of operational	Classification

Req ID	Requirement Summary	Method; Evidence Class	Cadence	Declared Population or Set	Evidence Artifact
	Audit Trail	c		periods	compliance audit trail

4.3 DEPLOYMENT FORMAT VALIDATION

Verification of deployment-format model transformations is established normatively under V.AI-4.7. Refer to AI-4.7 for the verification statement, success criteria, and applicable domain standards. The corresponding row in the Section 4.2 Verification Matrix lists the verification method (Test) and required evidence artifact (side-by-side performance comparison report).

4.4 CONTINUOUS AND PERIODIC VERIFICATION

A subset of verification activities established under Section 2 cannot be satisfied by a one-time pre-deployment activity. These activities operate continuously or on a defined periodicity throughout the Operations and Sustainment lifecycle phase per Table 1-7. The Cadence column of the Section 4.2 Verification Matrix identifies which sub-requirements carry this obligation.

Continuous verification activities include:

- Bias monitoring per AI-2.4, evaluating performance metrics across declared Subject Population strata as the system processes operational data.
- Performance threshold maintenance per AI-4.2 and drift detection per AI-4.1, comparing live performance to validated baselines.
- Operational validation without ground truth per AI-4.6, applying surrogate validation methods where outcome data is delayed or unavailable.
- Out-of-distribution detection per AI-6.2, flagging inputs outside the declared Operational Design Domain at runtime.
- Operator role and authority verification per AI-9.6, confirming that human decision authority remains correctly assigned during operations.
- Alerting, response and logging for every monitored indicator per Section 4.6, which states the cycle once for all of the above.

Periodic verification activities include:

- Model maintenance review per AI-4.3 and periodic model validation per AI-4.5, conducted on the cadence established by mission risk per Section 5.1.
- Operator qualification renewal per AI-9.7 and training program review per AI-9.9, conducted on the schedule established by the operating organization.
- Adversarial robustness re-testing per AI-7.x, conducted when the threat environment evolves or on the schedule established by the security program.

Continuous and periodic verification evidence shall be retained per Section 4.6.4, and per AI-15.4 where AI-15 applies, and shall be available for review during certification surveillance, incident investigation, or recertification activities.

4.5 VERIFICATION INDEPENDENCE

Verification activities shall be performed with independence scaled by system classification, as specified in the table below. Independence applies to the execution and disposition of the [V.AI-x.y] verifications defined in Sections 2 and 3, including the judgment that project-defined thresholds (Section 1.5) have been met.

Classification	Required Independence
Safety-Critical AI	Verification performed by individuals who did not develop the artifact under verification, with the verification function organizationally distinct from the development function. Verification results dispositioned by the independent verification function.
Mission-Critical AI	Verification performed by individuals who did not develop the artifact under verification. Organizational independence recommended but not required.
Operational Support AI	Self-verification permitted. Verification evidence independently reviewed by the Software Assurance authority prior to closure.

This independence scaling follows the pattern established for software verification independence in DO-178C and for independent verification and validation in IEEE 1012. Certification assessment under this framework additionally provides third-party examination of the verification evidence itself; that examination does not substitute for the project-level independence required above.

4.6 MONITORING, ALERTING, RESPONSE AND LOGGING

Several requirement sets in Sections 2 and 3 require an indicator to be defined and thresholded, monitored in operation, alerted on, responded to and logged. In versions 3.x each set stated that cycle in its own sub-requirements. In 4.0 the cycle is stated once here, normatively, and each set states only its failure-specific content: what is monitored, over which declared population and window, and which response the classification requires. The clauses below are procedural requirements of this section and form part of the conformity basis under Section 1.6; they are verified by the evidence they name. Table 4-1 gives the per-set content.

What were AI-2.6, AI-2.7, AI-2.9, AI-5.4, AI-6.4 and AI-7.5 in versions 3.x are retired as identifiers and their content is carried here; see the identifier disposition table in the 4.0 change log.

4.6.1 Indicator and Threshold Definition

For every indicator a requirement set names in Table 4-1, the project shall define the indicator, the statistic that computes it, the declared population and window it is computed over (Section 1.7), its alert threshold and its inhibit or action threshold, each threshold derived under Section 1.5, and the sample floor below which the indicator is reported as unavailable rather than computed. Where an indicator counts detections, the project shall also monitor the detector's recall on a declared seeded set (Section 1.7) run in shadow mode, so that a falling count cannot be read as improvement; seeded cases shall never enter the live path and shall never reach actuation or the AI-14 switch.

Verification: Verified by inspection of the indicator register: every Table 4-1 indicator applicable to the system has all six elements, and every indicator that counts detections has a declared seeded set, a shadow-mode path and a recall indicator computed per window. Evidence class: Deterministic. A missing element on any applicable indicator, or a detection-count indicator without its recall indicator, is a fail. Evidence artifact: Indicator register. Cadence: one-time; re-verified on change.

4.6.2 Alerting

The AI system shall raise an alert when an indicator crosses its alert threshold, deliver it to the designated recipients within the declared latency, and distinguish alert-level from inhibit-level crossings. Alerts raised while an indicator is below its sample floor shall be suppressed and the suppression logged.

Verification: Verified by test: an injected crossing of each applicable indicator produces an alert delivered within the declared latency and labelled with its level; an injected crossing below the sample floor is suppressed and logged. Evidence class: Deterministic. One undelivered or mislabelled alert is a fail. Evidence artifact: Alert test record. Cadence: one-time; exercised at surveillance.

4.6.3 Response

For every indicator, the project shall define the response to an inhibit-level crossing as one of: operational restriction of the affected scope, escalation to human review where the hazard analysis time to effect permits it, invocation of the defined fallback (the AI-14 fallback at Safety-Critical), or a maintenance action under AI-4.3. The response shall name its authority and time budget. A response shall not alter individual outputs at runtime on the basis of a population indicator.

Verification: Verified by inspection and test: every applicable indicator has a defined response with authority and budget; an injected inhibit-level crossing of each produces the defined response within budget; no design element alters individual outputs on the basis of a population indicator. Evidence class: Deterministic. A missing response or a per-decision correction path is a fail. Evidence artifact: Response register and test record. Cadence: one-time; exercised at surveillance.

4.6.4 Logging

The AI system shall log every alert, suppression, response and the events each requirement set names in Table 4-1, with timestamps, the indicator value and threshold, the declared population and window, the system response and any operator action. Logs shall be immutable, retrievable for post-mission analysis and incident investigation, and retained for the period the applicable regime requires or, where none applies, for the life of the certificate plus the surveillance interval.

Verification: Verified by inspection of a declared sample of operational periods: every event type applicable to the system is present with all named fields; immutability and retrievability are demonstrated; retention meets the stated period. Evidence class: Deterministic. A missing event type or field in the sample is a fail. Evidence artifact: Operational logs and log schema. Cadence: continuous.

Table 4-1: Per-Set Monitoring Content

Set	Indicator monitored	Declared population and window	Response at inhibit level	Events logged
AI-2 Bias	Disparity	Operational	Restriction of	Disparity

Set	Indicator monitored	Declared population and window	Response at inhibit level	Events logged
	indicators of AI-2.1 across declared strata (AI-2.4)	data by Subject Population strata; monitoring window and sample floor (Section 1.7)	the affected stratum, escalation under AI-2.8, or retraining under AI-4.3 (AI-2.6); AI-14 fallback at Safety-Critical	values per stratum, thresholds, response taken, operator inputs
AI-4 Continuous validation	Drift statistic (AI-4.1); performance metric or proxy (AI-4.2, AI-4.6)	Clean-qualified monitoring stream and window; proxy qualification data (Section 1.7)	Revalidation under AI-4.5, retraining under AI-4.3, degraded mode, or AI-14 fallback at Safety-Critical	Statistic values, alert and inhibit crossings, response, validation outcomes
AI-5 Hallucination (where applicable)	Hallucination detection events (AI-5.2); detector recall on the declared seeded set in shadow mode (4.6.1); audited miss rate (AI-5.5)	Operational generative outputs; declared hallucination criteria	Quarantine and human review where permitted, else defined fallback (AI-5.3); AI-14 fallback at Safety-Critical	Detected outputs with inputs, detection method and confidence, review decision and rationale
AI-6 Out-of-distribution	OOD detection events (AI-6.2); detector recall on the declared seeded set in shadow mode (4.6.1)	Operational inputs against the characterized distribution; far and near strata for qualification	Block autonomous action; human review where permitted, else defined fallback (AI-6.3); AI-14 fallback at Safety-Critical	Event with input characterization, detection confidence and method, response, operator action

Set	Indicator monitored	Declared population and window	Response at inhibit level	Events logged
AI-7 Adversarial robustness	Adversarial detection and security events (AI-7.2, AI-7.4, AI-7.5); detector recall on the declared seeded set in shadow mode (4.6.1)	Operational inputs; artifact integrity checks at load and interval	Reject or quarantine input; fallback on integrity failure (AI-7.4); incident handling per the declared cybersecurity baseline	Event type and severity, detection confidence, response; immutable; retention per mission plus analysis period (NIST SP 800-53 Rev. 5 AU-2, AU-3, AU-12)
AI-9 Human-AI teaming	Operator usage drift indicators (AI-9.6); workload indicators (AI-9.8)	Operational periods between reviews	Reclassification evaluation (AI-9.6); workload mitigations (AI-9.8)	Interaction records per AI-9.5; review findings
AI-13 Continuous learning (where applicable)	Learning validation outcomes and rollback triggers (AI-13.5, AI-13.6)	Benchmark suite and operational performance window	Reject update; automatic or manual rollback (AI-13.6)	Learning events per AI-13.8

Section 5: Implementation Patterns

5.1 THRESHOLD SELECTION GUIDANCE

This section provides informative guidance for threshold selection. It does not establish normative requirements. Threshold documentation, the derivation basis every value must carry, and the adequacy criteria the assessor applies are established normatively in Section 1.5; the populations a value is measured on are declared under Section 1.7. Projects should document threshold selection rationale as part of verification evidence.

Section 2 requires projects to define and document thresholds for bias metrics, drift detection, performance monitoring, and other AI-specific measures. This section provides example threshold ranges and selection considerations to assist projects in establishing appropriate values. These examples represent typical industry practice and should be tailored based on mission risk tolerance, operational context, and system classification.

5.1.1 Example Threshold Ranges by Requirement

The following table gives illustrative values for selected verification metrics. The values in the Example column are unsourced: they are not criteria, they are not a record of industry practice, and a project that adopts one must still derive it under Section 1.5 and declare the population it is measured on under Section 1.7. A detection, coverage or disparity rate has no meaning until that population is fixed. The Selection Considerations column is the useful column.

Req	Metric	Example (illustrative, unsourced; not a criterion)	Selection Considerations
AI-2	Demographic parity difference	≤2% for safety-critical; ≤5% for mission-critical	Tighter thresholds for crew health/safety functions; consider protected class sample sizes
AI-2	Equalized odds difference	≤3% TPR/FPR difference across groups	Critical for detection/diagnostic systems where false negatives have safety implications
AI-4	Data drift (PSI)	<0.1 negligible; 0.1-0.25 moderate; >0.25 significant	Alert at moderate; halt autonomous operation at significant for safety-critical
AI-4	Data drift (KL divergence)	>0.1 warrants investigation; >0.2 requires action	Monitor per feature; aggregate thresholds may mask localized drift
AI-4	Performance degradation	≤5% accuracy drop from baseline	Baseline established during V&V; consider mission phase sensitivity
AI-5	Hallucination detection rate	≥95% detection with ≤1% false positive rate	Balance detection sensitivity against operational false alarm burden
AI-	Input space	≥95% of ODD-derived	Derive equivalence classes

Req	Metric	Example (illustrative, unsourced; not a criterion)	Selection Considerations
3	coverage	input equivalence classes (safety-critical); ≥90% (mission-critical)	from declared ODD attributes (AI-1.0); complement with boundary and edge case testing per AI-3.4; applicable regardless of model architecture
AI-6	OOD detection rate	≥97% detection with ≤2% false positive rate	Validate against realistic OOD scenarios from operational environment
AI-8	Confidence calibration error	ECE ≤0.05; reliability slope 0.9-1.1	Critical for operator trust calibration; validate across operational scenarios
AI-9	State change notification	≤500ms; confidence drop >15% triggers alert	Align with human factors reaction time requirements
AI-9	Manual mode transition	≤2 operator actions; ≤5 seconds	Validate under time pressure; include in crew training scenarios
AI-1.5	Pre-ingestion classification screening detection rate	≥95% Safety-Critical; ≥90% Mission-Critical; ≥85% Operational Support	Detection rate measured against known-positive controlled content. Values proposed; confirm with operational testing.
AI-4.7	Deployment transformation performance margin	≤5% performance delta vs. original model (Safety-Critical); ≤10% (Mission-Critical); ≤15% (Operational Support); document and accept with justification	Margins measured per relevant performance metric (accuracy, F1, calibration, etc.) on representative test data. Values proposed; confirm with operational testing.

5.1.2 Scaling Thresholds by Classification

Threshold stringency should scale with AI system classification:

Safety-Critical AI: Most stringent thresholds. Conservative margins. Multiple independent detection mechanisms recommended. Thresholds derived from hazard analysis and safety margins.

Mission-Critical AI: Thresholds balanced against mission objectives. May accept higher false positive rates to ensure detection. Thresholds derived from mission success criteria.

Operational Support AI: Thresholds may be relaxed where operational impact is limited. Focus on trend detection over absolute thresholds. Thresholds derived from operational efficiency considerations.

5.1.3 Threshold Documentation

When documenting threshold selection rationale, projects should capture:

- Selected threshold value and units
- Derivation basis (analysis, heritage, industry standard, simulation)
- Relationship to mission risk tolerance and safety margins
- System response when threshold is exceeded
- Validation evidence demonstrating threshold appropriateness
- Conditions under which threshold should be revisited

5.2 AI-SPECIFIC HAZARD ANALYSIS INTEGRATION

This section provides informative guidance for integrating AI-specific failure modes into hazard analysis. The normative requirement to perform and integrate this analysis is established in AI-1.5 (AI Hazard Analysis Integration); this section guides its implementation and does not itself establish normative requirements.

5.2.1 Purpose

Traditional hazard analysis methods (FMEA, FTA, HAZOP) assume deterministic failure modes. AI systems introduce probabilistic, emergent, and adversarial failure modes that require augmented analysis approaches.

5.2.2 AI Failure Mode Categories

When conducting hazard analysis for AI systems, include the following failure mode categories:

- **Data-Related:** bias, drift, poisoning, leakage

- **Model-Related:** hallucination, overconfidence, underfitting, overfitting
- **Operational:** OOD inputs, concept drift, degradation
- **Adversarial:** input manipulation, model theft, poisoning attacks
- **Privacy:** re-identification of training data subjects, membership inference attacks revealing training set composition, training-data leakage through model outputs (inversion attacks), inferred attribute disclosure beyond intended outputs.
- **Human Factors:** automation bias, over-trust, under-trust

5.3 HUMAN-AI TEAMING IMPLEMENTATION GUIDANCE

This section provides implementation guidance for the human-AI teaming requirements established under AI-9. Effective human-AI teaming requires more than well-designed interfaces; it requires deliberate attention to operator workload, trust calibration, role clarity, and the conditions under which authority transfers between human and AI.

5.3.1 Operator Workload and Attention Management

Implementation of [R.AI-9.8] (Workload Management) should consider both cognitive load (the mental demand of monitoring AI behavior) and attention demand (the frequency and duration of operator engagement). Workload assessment methods include NASA-TLX subjective ratings, secondary-task performance measures, and physiological indicators where available. AI systems intended for sustained operation should demonstrate sustainable workload across the longest contiguous operating period in the declared ODD.

5.3.2 Trust Calibration

Calibrated trust, where operator reliance on AI matches the AI's actual reliability, is the goal. Both over-trust (automation bias) and under-trust (rejection of valid AI recommendations) are failure modes. Implementation of [R.AI-9.3] (Trust Calibration) should include confidence indicators that are themselves calibrated (per [R.AI-8.3]), consistent presentation of AI uncertainty, and operator training that explicitly addresses both failure modes. Verification activities should include scenario-based exercises measuring operator response to both AI failures and AI successes outside the operator's expectation.

5.3.3 Operator Role Verification

[R.AI-9.6] requires periodic verification that operator role assignments remain consistent with the original safety analysis. Drift can occur when operating organizations consolidate roles, when shift schedules change the personnel filling roles, or when the operational tempo deviates from the design assumption. Verification should be cadenced per the operating organization's change-control process and at minimum on each system upgrade.

5.3.4 Operator Qualification and Training

[R.AI-9.7] (Operator Qualification) and [R.AI-9.9] (Training Program Requirements) together establish the qualification and training infrastructure. Training programs should include both initial qualification and recurrent qualification on a defined cadence, with explicit coverage of: AI system capability limits, the declared ODD and its implications, escalation procedures when AI behavior departs from expectation, and the human authority preserved under [R.AI-9.1].

5.3.5 Authority Transfer

Where AI systems and operators share decision authority, the conditions under which authority transfers between them should be unambiguous and trained. Implementation should specify: which actions remain under human authority unconditionally per [R.AI-9.1], which decisions the AI may execute autonomously, the conditions that trigger AI escalation to human authority, and the conditions that allow operators to override AI recommendations or revoke AI authority. Ambiguous authority allocation is a known root cause of automation surprises and should be eliminated by design.

5.4 PRIVACY AND DATA PROTECTION IMPLEMENTATION GUIDANCE

This section provides implementation guidance for the privacy and data protection requirements established under AI-10. AI systems often process personal data at scale and may infer attributes operators did not explicitly intend to expose. Implementation should anticipate both the data the system processes directly and the inferences it can derive.

5.4.1 Data Minimization

Implementation of [R.AI-10] data minimization principles should begin at the data ingestion boundary. Where the operational task can be accomplished without personal data, personal data should not enter the AI processing path. Where personal data is required, the minimum sufficient subset should be retained and the

remainder discarded or replaced with derived features. Data minimization should be a design-time decision, not a post-hoc filter.

5.4.2 De-Identification and Re-Identification Risk

De-identification techniques (k-anonymity, l-diversity, differential privacy, synthetic data generation) should be selected based on the threat model and the system's classification tier. Safety-Critical AI processing personal data should assume re-identification attempts will occur and select techniques whose privacy guarantees survive the assumed adversary capabilities. Re-identification risk should be re-evaluated whenever the input data distribution or the auxiliary data available to potential adversaries materially changes.

5.4.3 Access Controls and Audit

Access to personal data within the AI processing path should be controlled, logged, and auditable. Access logs should be retained per Section 4.6.4 and, where AI-15 applies, the AI-15.4 Classification Compliance Audit Trail and should include both human access (developers, operators, auditors) and automated access (training pipelines, model serving infrastructure, monitoring systems). Anomalous access patterns should trigger investigation.

5.4.4 Data Retention

Personal data retention should be limited to the minimum period required for the operational task and any legally mandated retention. Retention schedules should be documented per data category and enforced through automated deletion where possible. Aggregated or de-identified outputs derived from personal data may have different retention requirements than the source data; both schedules should be explicit.

5.4.5 Inference and Membership Concerns

AI systems can leak training data through inversion attacks, membership inference attacks, and confidence-based extraction techniques. Implementation should consider: differential privacy or related techniques during training where the threat model warrants, output filtering or noise injection at inference time for sensitive applications, and adversarial testing for membership inference vulnerabilities for systems trained on sensitive data. The risk surface depends on model architecture, training data sensitivity, and adversary access; all three should inform the mitigation strategy.

5.4.6 Cross-Border Data Transfer

Where personal data crosses jurisdictional boundaries, applicable data protection regulations (GDPR, regional equivalents) apply in addition to the AI-10 requirements. Implementation should document the legal basis for transfer, the safeguards applied, and the data subject rights preserved. Cross-border considerations should be addressed at design time, not as a post-hoc compliance overlay.

Appendix A: GLOSSARY

The glossary contains 104 terms organized into 10 topical categories: Core AI/ML, System Classification, Operational Design Domain, Data, Risk and Failure Mode, Verification and Validation, Operational, Security, Explainability and Trust, and Privacy and Data Protection. Definitions are framework-defined or mapped to international standards where applicable; each entry indicates its source. Cross-references appear at the end of related entries.

The document's own structural nouns are used as follows. A requirement set is one of the numbered groups AI-1 through AI-15: a parent outcome statement ([R.AI-x], stated with will) decomposed into sub-requirements. This document carries fifteen sets, ten in Section 2 and five in Section 3; AI-5, AI-10 and the five Section 3 sets apply conditionally (Section 1.2, Table 1-3). A sub-requirement is an individually identified requirement [R.AI-x.y] with its paired verification [V.AI-x.y]; this document carries 93 sub-requirements; the identifiers of version 3.7 that were withdrawn or renumbered are mapped in the Document Revision History. The certificate and the Safety Critical Labs website print requirement area for the same level, listing the sets assessed for a given certification. The two nouns name the same level, and a count of sets is never a count of sub-requirements.

CORE AI/ML TERMS

AI System: Engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives. (ISO/IEC 22989:2022 3.1.4)

Artificial Intelligence (AI): A system that, for a given set of human-defined objectives, generates outputs such as predictions, recommendations, classifications, or decisions, and that exhibits the capacity to perform tasks commonly associated with human cognition (e.g., perception, reasoning, learning, decision-making) under varying conditions. (Adapted from ISO/IEC 22989:2022 3.1.4; modified to add capacities associated with human cognition. ISO/IEC 22989 3.1.3 defines artificial intelligence as the discipline.)

Continuous Learning: Incremental training of an AI system that takes place on an ongoing basis during the operation phase of the AI system life cycle. (ISO/IEC 22989:2022 3.1.9)

Determination Boundary: A defined and bounded collection of hardware and software items, together with its declared interfaces, on which the AI applicability determination of Section 1.2.1 is performed. Framework requirements attach to learned components within the boundary and to artifacts whose correctness criterion is fidelity to a learned component; obligations on the integrating system attach where the respective requirements specify. Adapted from the AI/ML constituent concept of the EASA Artificial Intelligence Concept Paper Issue 2; modified to a domain-agnostic determination unit for framework applicability.

Format Conversion: A model transformation that translates a model between framework representations (e.g., PyTorch to ONNX to TensorRT) to enable execution on target runtimes. Subject to revalidation per AI-4.7.

Human-Machine Teaming: Integration of human interaction with machine intelligence capabilities. (ISO/IEC 22989:2022 3.3.3)

Hyperparameter: Characteristic of a machine learning algorithm that affects its learning process and is selected prior to training. (ISO/IEC 22989:2022 3.3.4)

Inference: Reasoning by which conclusions are derived from known premises. (ISO/IEC 22989:2022 3.1.17)

Knowledge Distillation: A model transformation that trains a smaller “student” model to approximate the behavior of a larger “teacher” model, producing a compact model with comparable functional behavior. Subject to revalidation per AI-4.7.

Learned Artifact: An artifact whose values are determined or improved from data by machine learning (ISO/IEC 22989:2022 3.3.5), whether before deployment, through retraining (3.3.10), or through continuous learning during operation (3.1.9), and that serves as the specification of system behavior without a documented derivation constituting a correctness argument for that behavior. A trained model (3.3.14) is the paradigm case. A model designed for continuous learning is a learned artifact at every point in the operation phase, including prior to its first operational update. Data-fitted parameters within explicitly written model structures whose derivation constitutes a correctness argument are not learned artifacts (Table 1-2b). Framework-defined to make the Section 1.2.1 determination decidable, drawing on ISO/IEC 22989 process terminology.

Learned Component: A component whose behavior is specified by one or more learned artifacts. Framework requirements attach to learned components within

the determination boundary and to artifacts whose correctness criterion is fidelity to a learned component (Section 1.2.1). Framework-defined.

Machine Learning: Process of optimizing model parameters through computational techniques, such that the model's behavior reflects the data or experience. (ISO/IEC 22989:2022 3.3.5)

Machine Learning Model: Mathematical construct that generates an inference or prediction based on input data. (ISO/IEC 22989:2022 3.3.7)

Model: Physical, mathematical or otherwise logical representation of a system, entity, phenomenon, process or data. (ISO/IEC 22989:2022 3.1.23)

Parameter (Model Parameter): Internal variable of a model that affects how it computes its outputs (weights in a neural network). (ISO/IEC 22989:2022 3.3.8)

Pruning: A model transformation that removes model weights, neurons, or layers determined to contribute minimally to model output, reducing model size and computational cost. Subject to revalidation per AI-4.7.

Quantization: A model transformation that reduces the numerical precision of model weights and/or activations (e.g., FP32 to INT8) to reduce memory footprint and computational cost, typically for execution on resource-constrained hardware. Subject to revalidation per AI-4.7.

Retraining: Updating a trained model by training with different training data. (ISO/IEC 22989:2022 3.3.10)

Trained Model: Result of model training. (ISO/IEC 22989:2022 3.3.14)

Training / Model Training: Process to determine or to improve the parameters of a machine learning model, based on a machine learning algorithm, by using training data. (ISO/IEC 22989:2022 3.3.15)

SYSTEM CLASSIFICATION TERMS

Mission-Critical AI: AI outputs affect operational success but not human safety. (Framework-defined, Section 1.2.2)

Operational Support AI: AI supports operations but does not drive critical decisions. (Framework-defined, Section 1.2.2)

Safety-Critical AI: AI outputs directly affect human safety or system survivability. (Framework-defined, Section 1.2.2)

OPERATIONAL DESIGN DOMAIN TERMS

ODD Declaration: The formal documentation of an AI system's Operational Design Domain, including environmental conditions, input characteristics, operational scenarios, and constraints under which the system is designed to function as validated. Per AI-1.0. (Framework-defined; aligned with SAE J3016 ODD concept.)

ODD Evolvability: The declared property of an Operational Design Domain that specifies whether and how the ODD may change over the system's operational life through continuous learning, retraining, or reconfiguration. Systems with non-evolvable ODD declarations operate within a frozen specification; evolvable ODDs require additional controls per AI-13. (Framework-defined.)

Operational Claim: Single-sentence summary statement that opens an ODD declaration, summarizing the function, subject population, operational conditions, user role, and regulatory regime of the certified system. (Framework-defined, AI-1.0)

Operational Design Domain (ODD): The operational conditions, inputs, users, subjects, regulatory context, and constraints under which an AI system is designed and validated to operate. The ODD defines the scope of the certification claim and the envelope within which reliable behavior is expected. (Framework-defined, Section 1.2.3 and AI-1.0; concept aligned with ISO 34503 and SAE J3016) See also ODD Declaration; ODD Evolvability; Operational Claim.

Subject Population: The population of persons or entities that an AI system's decisions act upon, distinct from the user population that operates the system. Examples include patients in a medical decision support system, pedestrians and cyclists in a vehicle perception system, or obstacle classes in an autonomous platform. (Framework-defined, AI-1.0) See also User Population.

User Population: The set of human operators, controllers, supervisors, or other authorized personnel who interact with the AI system to direct, monitor, or oversee its operation. Distinct from Subject Population, which refers to entities the AI system acts upon. (Framework-defined; see Subject Population.)

DATA TERMS

Ground Truth: Value of the target variable for a particular item of labelled input data. (ISO/IEC 22989:2022 3.2.7)

Input Data: Data for which an AI system calculates a predicted output or inference. (ISO/IEC 22989:2022 3.2.9)

Label: Target variable assigned to a sample. (ISO/IEC 22989:2022 3.2.10)

Production Data: Data acquired during the operation phase of a deployed AI system for which a deployed system calculates a predicted output or inference. (ISO/IEC 22989:2022 3.2.12)

Test Data: Data used to assess the performance of a final model. (ISO/IEC 22989:2022 3.2.14)

Training Data: Data used to train a machine learning model. (ISO/IEC 22989:2022 3.3.16)

Validation Data: Data used to compare the performance of different candidate models. (ISO/IEC 22989:2022 3.2.15)

RISK AND FAILURE MODE TERMS

Bias: Systematic difference in treatment of certain objects, people, or groups in comparison to others. (ISO/IEC 22989:2022 3.5.4)

Concept Drift: Change in the relationship between input data and the target variable over time (i.e., the decision boundary moves), requiring relabeling of training data and model retraining. (ISO/IEC 5338:2023 6.4.14 / ISO/IEC 22989:2022 5.11.9, tailored) See also Continuous Validation.

Data Drift: Change in the statistical characteristics of production data over time compared to the training data distribution, which can degrade model prediction accuracy. (ISO/IEC 5338:2023 6.4.14 / ISO/IEC 22989:2022 5.11.9, tailored) See also Continuous Validation.

Hallucination: The production of confidently stated but erroneous or false content. (NIST AI 600-1 §2.2 (Confabulation))

Model Drift: Degradation in model performance over time due to data drift, concept drift, or other factors affecting the relationship between inputs and outputs. (Tailored)

Out-of-Distribution: Input data that falls outside the statistical distribution of the training data, for which model predictions may be unreliable or undefined. (Tailored from ISO/IEC 23053:2022) See ISO/IEC 23053:2022 §8.7 (Operation), which discusses data drift, concept drift, and generalization affecting model performance.

Overfitting: Creating a model which fits the training data too precisely and fails to generalize. (ISO/IEC 23053:2022 3.2)

Performance Drift: Degradation of AI model performance metrics over operational time, typically arising from data drift or concept drift. Continuous monitoring per AI-4.4 and AI-4.6 detects performance drift; periodic revalidation per AI-4.5

quantifies it. (Framework-defined; complements Data Drift and Concept Drift.) See also Continuous Validation.

VERIFICATION AND VALIDATION TERMS

Bounded Evidence: Evidence that demonstrates a property under a declared budget (a perturbation size, an operating point, a scenario set). It holds at that budget and nowhere else; the budget is recorded with the result. One of the four evidence classes (Section 4.1). (Framework-defined)

Certified Configuration: The set of elements declared by the applicant and recorded by Safety Critical Labs on the certificate, to which certification attaches: the determination boundary and classification, the Operational Design Domain, the threshold values, the attack pattern sets declared under AI-7.1 and AI-7.2, and the foundational cybersecurity baseline. A change to any element is a change to the certified configuration. (Framework-defined, Section 1.6) See also Conformity Basis; Operational Design Domain.

Conformity Basis: The requirement statements of this framework on which the certification determination rests: each [R.AI-x.y] requirement statement verified per its paired [V.AI-x.y] verification statement, together with the procedural requirements of Sections 1 and 4. No document outside the framework forms part of the conformity basis unless a requirement statement names it and states the extent of the invocation. (Framework-defined, Section 1.6) See also Certified Configuration.

Continuous Validation: Ongoing monitoring process to verify that AI models continue performing satisfactorily during operation, including detection of data drift, concept drift, and performance degradation. (ISO/IEC 5338:2023 6.4.14, tailored) See also Data Drift; Concept Drift; Performance Drift.

Control Point: A defined verification or authorization gate in the AI system lifecycle through which an artifact, decision, or change must pass before proceeding to the next stage. Used in cross-references between AI-4.3 (Model Maintenance Criteria), AI-4.7 (Deployment Format Validation), and AI-7.4 (Model Integrity). (Framework-defined.)

Declared Evaluation Population: The population, partition or set on which a rate, score or coverage figure is measured, declared with its derivation basis, size and strata before the verification is executed and recorded in the certified configuration (Section 1.7, Part 2 C.8). (Framework-defined)

Deployment-Format Model: An AI model that has been transformed from its trained form for execution on operational hardware through processes such as

quantization, pruning, knowledge distillation, or framework conversion. Subject to revalidation per AI-4.7. (Framework-defined.)

Derivation Basis: The record that accompanies every declared threshold value: the hazard it mitigates, the worst credible consequence, the acceptable failure probability or rate adopted, the value, the sample size and confidence that support it, and the sensitivity of the determination to it (Section 1.5). (Framework-defined)

Deterministic Evidence: Evidence that establishes a property that is true or false of a named artifact or configuration, checked by computation or by inspection of the artifact itself. It closes its requirement. One of the four evidence classes (Section 4.1). (Framework-defined)

Evidence Class: The category that states what a verification's evidence can establish: deterministic, statistical, bounded or surveillance. Every verification statement names the class of its governing criterion (Section 4.1). (Framework-defined)

Fairness Metrics: Quantitative measures used to assess whether an AI system produces equitable outputs across different groups, operational contexts, and mission scenarios. Includes metrics for disparate impact, equal opportunity, demographic parity, and performance consistency across operational conditions. (Tailored from ISO/IEC TR 24027)

Insufficient Evidence: A verification outcome: the interval on a statistical measurement contains the criterion, the declared population cannot support the claimed rate at the stated confidence, or a bounded demonstration's budget is below the one the hazard analysis requires. Its effect on the determination is stated by tier in Section 1.6. (Framework-defined)

Residual: What a verification's evidence did not close: an insufficient-evidence outcome carried under Section 1.6, a bounded demonstration's budget, or a surveillance obligation outstanding. The certificate carries the residual list (Section 4.1). (Framework-defined)

Statistical Evidence: Evidence that estimates a rate or score on a declared population. It supports its requirement at a stated confidence with a stated residual and never closes it. One of the four evidence classes (Section 4.1). (Framework-defined)

Sufficiency Statement: The applicant's statement, one per applicable requirement set in the structure of Part 2 C.7, of how the sub-requirement evidence supports the parent outcome for the declared Operational Design Domain, with residuals and their compensations (Section 1.6). (Framework-defined)

Surveillance Evidence: Evidence produced by an obligation discharged continuously or periodically in service. It accumulates over time and never closes its requirement. One of the four evidence classes (Section 4.1). (Framework-defined)

System Safety Process: The system-level safety analysis and assessment activity through which a project identifies hazards and produces the hazard analyses that software-level standards consume, conducted per applicable domain practice (e.g., SAE ARP4754/ARP4761 safety assessment in civil aviation; ISO 26262-3 hazard analysis and risk assessment in automotive; ISO 14971 risk management for medical devices; MIL-STD-882E system safety in defense; the NASA safety program hazard analysis process, from which NASA-STD-8739.8 derives software safety criticality). Distinct from the software standard this framework supplements (e.g., DO-178C, NPR 7150.2D, ISO 26262-6), which consumes system safety outputs to determine software criticality but does not itself conduct the system-level hazard analysis. (Framework-defined, AI-1.5 and Section 1.2.5)

Test: Execution of the system or component under controlled conditions with defined inputs, producing quantitative measurements evaluated against pass/fail criteria.

Validation: Confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled. (ISO/IEC 22989:2022 3.5.18)

Verification: Confirmation, through the provision of objective evidence, that specified requirements have been fulfilled. (ISO/IEC 22989:2022 3.5.17)

OPERATIONAL TERMS

Assured Fallback: The element a runtime assurance monitor invokes in place of the learned component, assured under the applicable conventional standard at the level the hazard analysis assigns, and bringing the system to a declared safe or minimal-risk condition (AI-14.2). (Framework-defined; concept aligned with ASTM F3269 and the simplex architecture)

Autonomous Action: System action taken without requiring human approval or intervention at the time of execution. (Tailored)

Controllability: Property of an AI system that allows a human or another external agent to intervene in the system's functioning. (ISO/IEC 22989:2022 3.5.6)

Deployer Obligation: A requirement that binds the operating organization rather than the certified item, recorded as such on the certificate where the determination boundary excludes that organization (AI-9.7 to AI-9.9). (Framework-defined)

Fallback Behavior: Predefined safe system response invoked when AI outputs are unreliable, unavailable, or flagged as out-of-distribution. (Tailored)

Human Override: Capability for human operators to supersede or reject AI recommendations or autonomous actions, with documented rationale. (Tailored)

Reliability: Property of consistent intended behavior and results. (ISO/IEC 22989:2022 3.5.9)

Resilience: Ability of a system to recover operational condition quickly following an incident. (ISO/IEC 22989:2022 3.5.10)

Robustness: Ability of a system to maintain its level of performance under any circumstances. (ISO/IEC 22989:2022 3.5.12)

Runtime Assurance: An architecture in which a learned component is bounded in operation by an explicitly specified, conventionally verifiable monitor and an assured fallback, with declared switching conditions and timing; required at Safety-Critical (AI-14). (Framework-defined; concept aligned with ASTM F3269)

Runtime Assurance Monitor: The explicitly specified, non-learned element that observes the learned component's inputs, outputs or state against the hazards assigned to it and invokes the assured fallback when its switching conditions are met (AI-14.1). (Framework-defined)

SECURITY TERMS

Adversarial Attack: Deliberate manipulation of AI system inputs, training data, or model parameters intended to cause incorrect or harmful outputs. (ISO/IEC 5338:2023 6.4.3)

Adversarial Example: An input crafted by perturbing a legitimate input with imperceptible or low-magnitude modifications designed to cause AI model misclassification or incorrect output. Distinct from Adversarial Attack, which refers to the broader class of attack techniques. Per AI-7 and AI-12.6.

Adversarial Robustness: Ability of an AI system to maintain correct and safe operation despite adversarial inputs or attacks. (ISO/IEC 22989:2022 3.5.12, tailored)

Data Poisoning: Injection of malicious data into training datasets to influence model behavior in attacker-controlled ways. (ISO/IEC 5338:2023 6.4.3)

Input Manipulation: Crafting of inputs designed to cause model misclassification or incorrect outputs while appearing valid to human observers. (ISO/IEC 5338:2023 6.4.3)

Model Inversion: Attack technique that reconstructs sensitive training data by analyzing model outputs. (ISO/IEC 5338:2023 6.4.3)

Model Theft: Extraction of model parameters or behavior through systematic querying to replicate proprietary AI capabilities. (ISO/IEC 5338:2023 6.4.3)

EXPLAINABILITY AND TRUST TERMS

Confidence: Quantified measure of certainty or probability associated with an AI output or prediction. (Tailored)

Explainability: Property of an AI system to express important factors influencing the AI system results in a way that humans can understand. (ISO/IEC 22989:2022 3.5.7)

Graceful Degradation: System capability to maintain reduced functionality when AI is impaired. (Tailored)

Predictability: Property of an AI system that enables reliable assumptions by stakeholders about the output. (ISO/IEC 22989:2022 3.5.8)

Transparency: Property of a system that appropriate information about the system is made available to relevant stakeholders. (ISO/IEC 22989:2022 3.5.15)

Trust Calibration: Alignment between operator trust in AI and actual AI reliability. (Human factors)

PRIVACY AND DATA PROTECTION TERMS

Anonymization: The process of rendering personal data such that the data subject is no longer identifiable, by any means reasonably likely to be used. Regulatory standards for anonymization vary across jurisdictions, including GDPR Recital 26, HIPAA Safe Harbor at 45 CFR 164.514(b)(2), and HIPAA Expert Determination at 45 CFR 164.514(b)(1). (Framework-defined; aligned with GDPR Recital 26.) See also De-identification.

Data Controller: The natural or legal person, public authority, agency, or other body which, alone or jointly with others, determines the purposes and means of the processing of personal data. (GDPR Article 4(7))

Data Minimization: The principle and practice of collecting, processing, and retaining only the personal data strictly required for the AI system's operational task, with the minimum sufficient subset kept and the remainder discarded or replaced with derived features. Per AI-10 and Section 5.4.1. (Aligned with GDPR Article 5(1)(c) and ISO/IEC 27559.)

Data Processor: A natural or legal person, public authority, agency, or other body which processes personal data on behalf of the controller. (GDPR Article 4(8))

Data Subject: The identified or identifiable natural person to whom personal data relates. (GDPR Article 4(1))

De-identification: The process of modifying personal data to reduce the likelihood of associating it with specific individuals, encompassing techniques such as masking, generalization, perturbation, and aggregation. Distinct from Anonymization in regulatory frameworks (HIPAA, CCPA) where de-identification implies a quantified risk threshold. Per AI-10 and Section 5.4.2. See also Anonymization; Re-identification Risk.

Differential Privacy: A mathematical framework for quantifying the privacy risk of a computation over a dataset, characterized by a privacy budget parameter (epsilon) that bounds the contribution of any individual record to the output. (Framework-defined; see Dwork et al. 2006 foundational literature) Full citation: Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). “Calibrating Noise to Sensitivity in Private Data Analysis,” in Theory of Cryptography Conference (TCC 2006), Lecture Notes in Computer Science, vol. 3876, Springer, pp. 265-284.

DPIA (Data Protection Impact Assessment): A structured assessment of the privacy risks of a data processing activity, required under GDPR Article 35 and analogous regulations for high-risk processing. For AI systems processing personal data, DPIA is typically performed at design time and updated when the data processing materially changes. See also Privacy Impact Assessment (PIA).

Lawful Basis: The legal justification for processing personal data under an applicable privacy framework. Examples include the six bases enumerated in GDPR Article 6, HIPAA permitted uses and disclosures under 45 CFR 164.502, and CCPA business purposes. (Aligned with GDPR Article 6.)

Membership Inference: An attack class in which an adversary determines whether a specific data record was included in the training set of an AI model, typically by exploiting model confidence patterns or output behavior. Per AI-10 and Section 5.4.5.

Personal Data: Any information relating to an identified or identifiable natural person. The precise scope varies by regulatory framework. Examples include GDPR Article 4(1), HIPAA Protected Health Information under 45 CFR 160.103, and CCPA “personal information” under Cal. Civ. Code 1798.140. (Aligned with GDPR Article 4(1).)

Privacy Impact Assessment (PIA): Systematic assessment of the privacy risks associated with a processing activity and the mitigation measures applied. Under GDPR Article 35 and for high-risk processing, referred to as a Data Protection Impact Assessment (DPIA). (Framework-defined, AI-10.6; GDPR Article 35)

Privacy-Enhancing Technique (PET): Technical method designed to reduce privacy risk through data or computation transformation, including anonymization, pseudonymization, differential privacy, federated learning, secure multi-party computation, and homomorphic encryption. (Framework-defined, AI-10.4) See ISO/IEC 27559:2022, "Information security, cybersecurity and privacy protection - Privacy enhancing data de-identification framework."

Pseudonymization: The processing of personal data in such a manner that the data can no longer be attributed to a specific data subject without the use of additional information, where that additional information is kept separately and subject to technical and organizational measures. (GDPR Article 4(5))

Re-identification Risk: The likelihood that personal data which has been de-identified or anonymized can be associated back to specific individuals through techniques such as auxiliary data linkage, attribute inference, or longitudinal tracking. Per AI-10 and Section 5.4.2.

Special Categories of Personal Data: Categories of personal data subject to additional protection under GDPR Article 9 (racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data for identification, health data, sex life or sexual orientation) or analogous categories under other frameworks. Examples include HIPAA PHI and CCPA "sensitive personal information." (Aligned with GDPR Article 9.)

DOCUMENT REVISION HISTORY

Version numbering and change management. This document is identified as SCL-ARF-001 independent of its version. Version increments are defined by conformity impact, not by the size of the edit: a MAJOR increment (X) when the meaning of an existing requirement or verification changes or identifiers renumber, so that existing evidence and traceability break and transition arrangements are required; a MINOR increment (X.Y) when normative content is added with every existing identifier stable, so that existing evidence remains valid and new evidence is needed only for the additions; and a PATCH increment (X.Y.Z) for editorial changes with zero change to normative meaning. Each row of the table below records the change class of the release: Editorial, Normative Addition, or Normative Modification. This policy was adopted at version 3.7; the classes shown for earlier

versions are applied retrospectively for the reader and do not alter the numbers those versions were issued under. The framework is cited at its concept DOI, 10.5281/zenodo.19024420, which resolves to the latest version. Certificates cite the document identifier, the version assessed against, and that version's own DOI, never the concept DOI, so that the conformity basis of an issued certificate does not change when a new version is published. Transition arrangements between versions are defined in the assessment methodology.

The table below gives one line per version. The full description of each version is the Revision Record in Part 2 (SCL-ARF-002) and, from version 3.4, the version's change log.

Version	Date	Author	Change Class	Description
1.0	February 2025	Safety Critical Labs	Initial release	Initial framework development.
2.0	October 2025	Safety Critical Labs	Normative Addition	Added AI-4.6 Operational Validation Without Ground Truth, AI-9.6 Operational Role Verification and the risk score methodology.
2.1	January 2026	Safety Critical Labs	Normative Addition	Added AI-1.0 Operational Design Domain with the Operational Claim and eight attribute categories, AI-10 Privacy and Data Protection, the ODD procedural gate (Section 1.2.3, Table 1-4) and the domain standards citation convention (Section 1.3.1).
2.1.1	April 2026	Safety Critical Labs	Normative Addition	Added AI-9.7 to AI-9.9 (operator qualification, workload management, training) and AI-8.5 Public Disclosure Support.
2.1.3	05/04/2026	Safety Critical Labs	Normative Addition	Added AI-4.7 Deployment Format Validation, the Section 4 overview, Section 4.4 continuous and periodic verification and the cadence column of the matrix; editorial pass on Sections 1 to 3.
3.0	05/05/20	Safety	Normative	Major structural revision: Section 3

Version	Date	Author	Change Class	Description
	26	Critical Labs	Modificati on	Architecture and Paradigm Requirements with AI-11, AI-12 and AI-13; verification became Section 4 and implementation patterns Section 5.
3.1	06/11/202 6	Safety Critical Labs	Editorial	Editorial and formatting revision: verification statement formatting, title case, house style, typos, Table 1-3 duplication.
3.2	06/14/202 6	Safety Critical Labs	Editorial	Section 3 verification blocks reformatted to the Section 2 pattern; AI-4.7 phrasing aligned.
3.3	06/14/202 6	Safety Critical Labs	Editorial	Corrected the Test, Analysis and Demonstration definitions in Section 4.1; Test added to the glossary; IEEE 1012 citation updated to 2024; stale cross-reference corrected.
3.4	07/08/202 6	Safety Critical Labs	Normative Addition; Normative Modificati on	Applicability gate redesigned around the anchor criterion (Section 1.2.1, Table 1-2b); decision impact moved to classification; AI-1.9 AI Hazard Analysis Integration added; Appendix B.1 rewritten as a decision sequence.
3.5	07/29/202 6	Safety Critical Labs	Normative Modificati on	The undefined term host standard replaced by the project's system safety process in AI-1.9; System Safety Process defined; Section 1.2.5 names AI-1.9 as the single interface to the system safety process.
3.6	08/08/20 26	Safety Critical Labs	Normative Addition	The applicability determination made verifiable: correctness against explicitly written specification as the anchor criterion, the determination boundary as the unit

Version	Date	Author	Change Class	Description
				of determination, the deletion test; glossary gained Learned Artifact and Learned Component.
3.7	09/15/2026	Safety Critical Labs	Normative Addition; Normative Modification (Appendix D.5)	Section 1.6 Conformity Basis added; normative and informative references partitioned (Tables 1-9 and 1-10); declared attack pattern sets in AI-7.1 and AI-7.2; AI-7.7 Artifact Load Safety added; document identifier SCL-ARF-001.
4.0	2026-10 (public comment draft; deposit follows the comment window)	Safety Critical Labs	Normative Modification	Major revision: evidence classes and the insufficient-evidence outcome (Sections 1.6 and 4.1) with the residual list on the certificate; AI-14 Runtime Assurance mandatory at Safety-Critical; declared evaluation populations (Section 1.7) and derivation bases with adequacy criteria (Section 1.5); Section 4.6 monitoring clause; AI-15 Regulated Information Handling; AI-5.5 Hallucination Audit (audited miss rate) and detector recall in Section 4.6.1; AI-1, AI-2, AI-5 and AI-7 renumbered, with the identifier map at the end of this section; Part 2 (SCL-ARF-002) created for the classification determination, the evidence templates and the revision record. Full record: Part 2 Annex and the 4.0 change log.

Identifier changes in version 4.0. The table below maps every identifier of version 3.7 that was withdrawn, renumbered or newly created to this version; every identifier not listed is unchanged. Set numbers AI-1 through AI-13 are unchanged; AI-14 and AI-15 are new sets. The 4.0 change log carries the full disposition of every identifier.

Identifier Map, Version 3.7 to Version 4.0

Version 3.7 identifier	Version 4.0 identifier or disposition	Note
AI-1.5	AI-15.1	Content carried to AI-15.1 (AI-15, new set); identifier retired
AI-1.6	AI-15.2	Content carried to AI-15.2 (AI-15, new set); identifier retired
AI-1.7	AI-15.3	Content carried to AI-15.3 (AI-15, new set); identifier retired
AI-1.8	AI-15.4	Content carried to AI-15.4 (AI-15, new set); identifier retired
AI-1.9	AI-1.5	Renumbered; requirement modified as the change log records
AI-1.10	AI-1.6	Renumbered; requirement modified as the change log records
AI-2.6	Section 4.6.1	Content carried to Section 4.6.1; identifier retired
AI-2.7	Section 4.6.2	Content carried to Section 4.6.2; identifier retired
AI-2.8	AI-2.6	Renumbered; requirement modified as the change log records
AI-2.9	Section 4.6.4	Content carried to Section 4.6.4; identifier retired
AI-2.10	AI-2.7	Renumbered; requirement modified as the change log records
AI-2.11	AI-2.8	Renumbered; requirement modified as the change log records
AI-2.12	AI-2.9	Renumbered; requirement modified as the change log records
AI-2.13	AI-2.10	Renumbered; requirement modified as the change log records
AI-5.4	Section 4.6.4	Content carried to Section 4.6.4; identifier retired
AI-5.5	AI-5.4	Renumbered; requirement modified as the change log records
AI-6.4	Section 4.6.4	Content carried to Section 4.6.4; identifier retired

Version 3.7 identifier	Version 4.0 identifier or disposition	Note
AI-7.5	Section 4.6.4	Content carried to Section 4.6.4; identifier retired
AI-7.6	AI-7.5	Renumbered; requirement modified as the change log records
AI-7.7	AI-7.6	Renumbered; requirement modified as the change log records
(none)	AI-5.5	New in 4.0
(none)	AI-14.1	New in 4.0
(none)	AI-14.2	New in 4.0
(none)	AI-14.3	New in 4.0
(none)	AI-14.4	New in 4.0
(none)	AI-14.5	New in 4.0
(none)	AI-15.1	New in 4.0 (content of AI-1.5 in versions 3.x)
(none)	AI-15.2	New in 4.0 (content of AI-1.6 in versions 3.x)
(none)	AI-15.3	New in 4.0 (content of AI-1.7 in versions 3.x)
(none)	AI-15.4	New in 4.0 (content of AI-1.8 in versions 3.x)